SuperResearcher

The 100-Year Life

How longer lives reshape work, money, health, and society through 2050 and beyond

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Humanity is growing older in a way that has no precedent and no reverse gear. By 2050 some 1.6 billion people will be aged 65 or over — roughly one in six of everyone alive — the global median age will have climbed from 30 toward 36, and the population aged 80 and above will have doubled its share in the developed world and multiplied several-fold beyond it. This report examines what that transformation means, tracing it from the demographic transition itself through health, long-term care, working lives, pensions, income and wealth, housing, inequality and the emerging longevity economy, and ending with the projection machinery on which fiscal planning for an older world depends. Two facts frame everything else: the ageing of populations is driven more by falling fertility than by rising longevity, and it is proceeding fastest — compressing into two or three decades transitions that took Europe a century — precisely where incomes, institutions and care systems are least prepared. The evidence assembled here supports a consistent set of conclusions. How much of the longevity gain is lived in good health matters more than the gain itself: it decides whether longer lives are working and spending years or care-consuming ones, and it swings projected public spending more than any demographic variable. Rising participation converts most of the apparent dependency burden into a manageable one — older-worker employment stands at record levels, retirement ages are rising in most advanced economies, and the gap between demographic and economic dependency ratios measures a genuine policy space. What decides whether old age is lived in poverty is not average replacement rates but floors: indexation rules, eligibility conditions and the coverage of women, carers and non-standard workers, whose interrupted careers already cost them a fifth or more of their pensions. Care emerges as the binding constraint, underfunded and undervalued almost everywhere, while old-age inequality turns out to be manufactured across the whole life course — by education, employment and health shocks decades before retirement — and therefore only partly correctable at retirement age. None of this is destiny. The projections surveyed in these pages are conditional arithmetic, and the spread between their scenarios — healthy ageing against expansion of morbidity, participation against withdrawal, coverage convergence against family default — is the honest measure of what societies still control. The hundred-year life is best understood as a dividend: already earned by a century of progress in survival, but collected only by countries willing to rebuild institutions designed for shorter lives — investing early across the life course, protecting income floors, professionalising care, designing cities and transport for the people who will actually live in them, and measuring their progress by the condition of their worst-off older citizens rather than their averages.

The demographic transition: how long we will live

Population ageing worldwide

Population ageing — the shift of a society's age structure toward older ages — is the demographic signature of the era this report examines. Its mechanism is simple and, once fertility falls, relentless: "[a]s fertility rates drop during demographic transition, the shift in the age structure manifests initially as a swelling share of adults in the working and reproductive ages and falling proportions of children and youth. If fertility remains at lower levels, the population begins to age and eventually the share of older people starts to rise. Successive cohorts tend to live longer" [1]. Every country on earth is somewhere on this path; what differs — dramatically — is the speed.

The defining fact of contemporary ageing is that it is now fastest where institutions are least prepared. "Population ageing is progressing more rapidly in developing countries than it did historically in more developed countries": in most developed regions the share of the population aged 65 or above took "anywhere from 40 to 120 years" to rise from 7 to 14 per cent, and a further 20 to 50 years to go from 14 to 21 per cent, whereas "[b]y contrast, most developing countries will likely see a doubling in the share of older persons from 7 to 14 per cent in 15 to 35 years and a further increase from 14 to 21 per cent in just 10 to 30 years" [1]. The compressed timetable is the central policy problem of the hundred-year life: pension systems, health services and care arrangements that Europe built across a century of gradual ageing must be built elsewhere in a generation.

The median age — the line dividing a population's younger and older halves — tells the same story in a single number. "Globally, the median age is expected to climb from 30 years in 2021 to 36 years in 2050," with Latin America

and the Caribbean projected to rise fastest, "from 30 years in 2021 to 40 years in 2050," Europe and Northern America moving "from 40 to 46 years," and 16 countries expected to reach "a median age of 50 years or higher" by mid-century [1]. The oldest old are the fastest-growing stratum within this shift: "the greater share over age 80… is expected to increase from 2 to around 5 per cent of the global population between 2021 and 2050" [1] — the age band where care needs concentrate, and the subject of this report's later chapters on health and long-term care.

Ageing also rewrites the arithmetic of dependency, though not yet catastrophically. In 2002 "more than half of the global population was between 20 and 64 years old (54 per cent), meaning 8 or 9 individuals potentially depended on every 10 workers," and of those dependents "only 15 per cent were older people"; by 2021 the working-age share "had increased to 57 per cent" and "[t]his share will remain relatively stable through 2050, keeping the dependency ratio below 1" [1]. What changes is the composition: children remain the largest dependent group globally — 33 per cent of world population in 2021, falling to "around 28 per cent in 2050" — but three regions (Australia and New Zealand, Europe and Northern America, and Eastern and South-Eastern Asia) "will complete the transition from a high share of youth dependents to a high share of older dependents before 2050," with Latin America and the Caribbean following by 2060 and Central and Southern Asia by 2080 [1]. The sequencing of working-age peaks marks the transition's geography: the working-age population "peaked in Europe and Northern America in 2013," peaks in Eastern and South-Eastern Asia in 2027, in Latin America and the Caribbean in 2043 and in Central and Southern Asia in 2058, while "[a]ll other regions are expected to experience continued growth in the working-age population through the end of the century" [1].

The frontier of the transition lies in the least developed countries, where ageing arrives before wealth. Most sub-Saharan African LDCs "are still in the early stages of the transition," facing "persistently high levels of fertility and rapid population growth" and pressure "to accommodate their rapidly expanding school-age and working-age populations," while some Asia-Pacific LDCs "already have a large number of older persons, but they lack well-established social protection programmes and health care services" [2]. All LDCs "are expected to experience a significant rise in both the proportion and absolute number of older persons between now and 2050 and an accelerated pace of population ageing beyond 2050," and the strategic imperative is to harvest the demographic dividend first — human capital, girls' education, decent jobs — so as to avoid "getting old before getting rich" [2]. The chapters that follow take up each element of this worldwide transformation in turn: how long lives will be, in what health they will be lived, who will provide and pay for care, how work and pensions adapt, and how the gains and risks of a longer-lived world are shared.

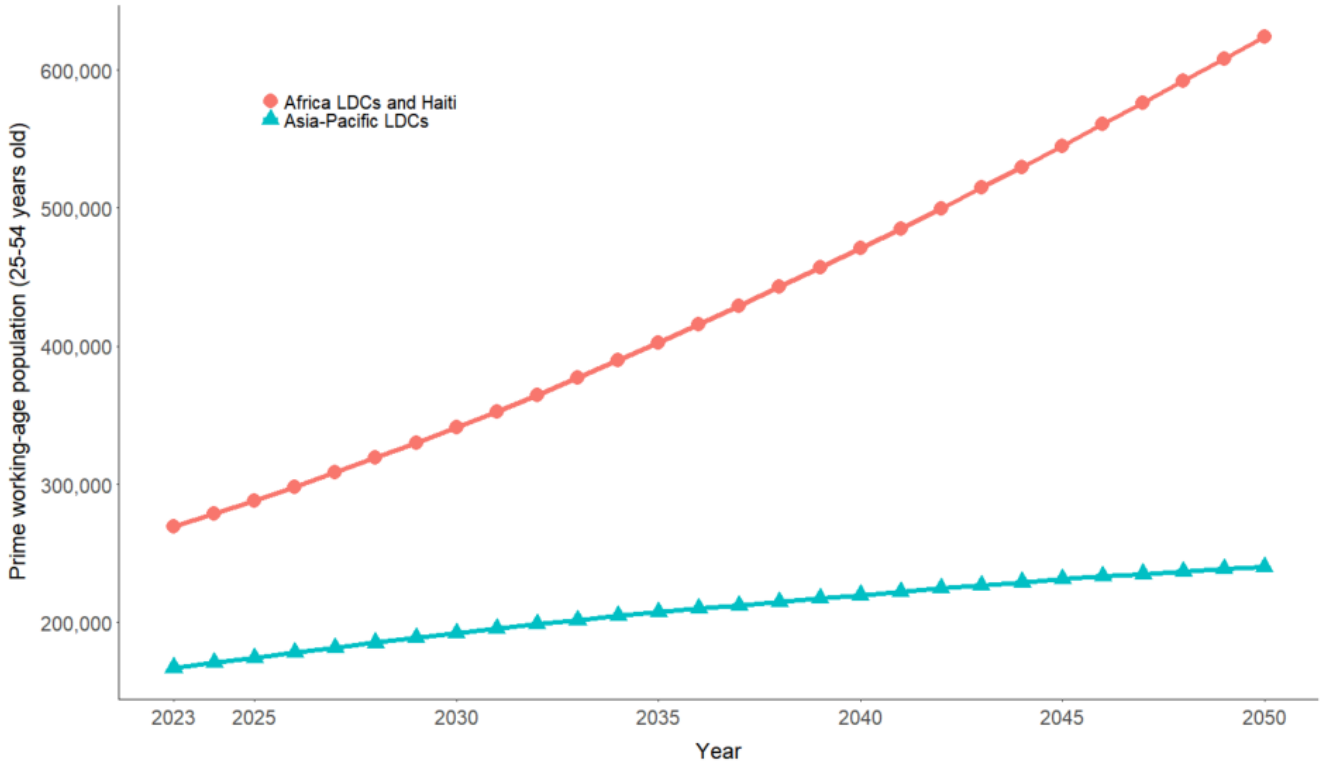


Exhibit. Projected number of working-age population in Africa LDCs and Haiti and Asia-Pacific LDCs, 2023–2050 Source: World Population Prospects 2022 (United Nations, 2022b). Source: S0005.

Life expectancy developments

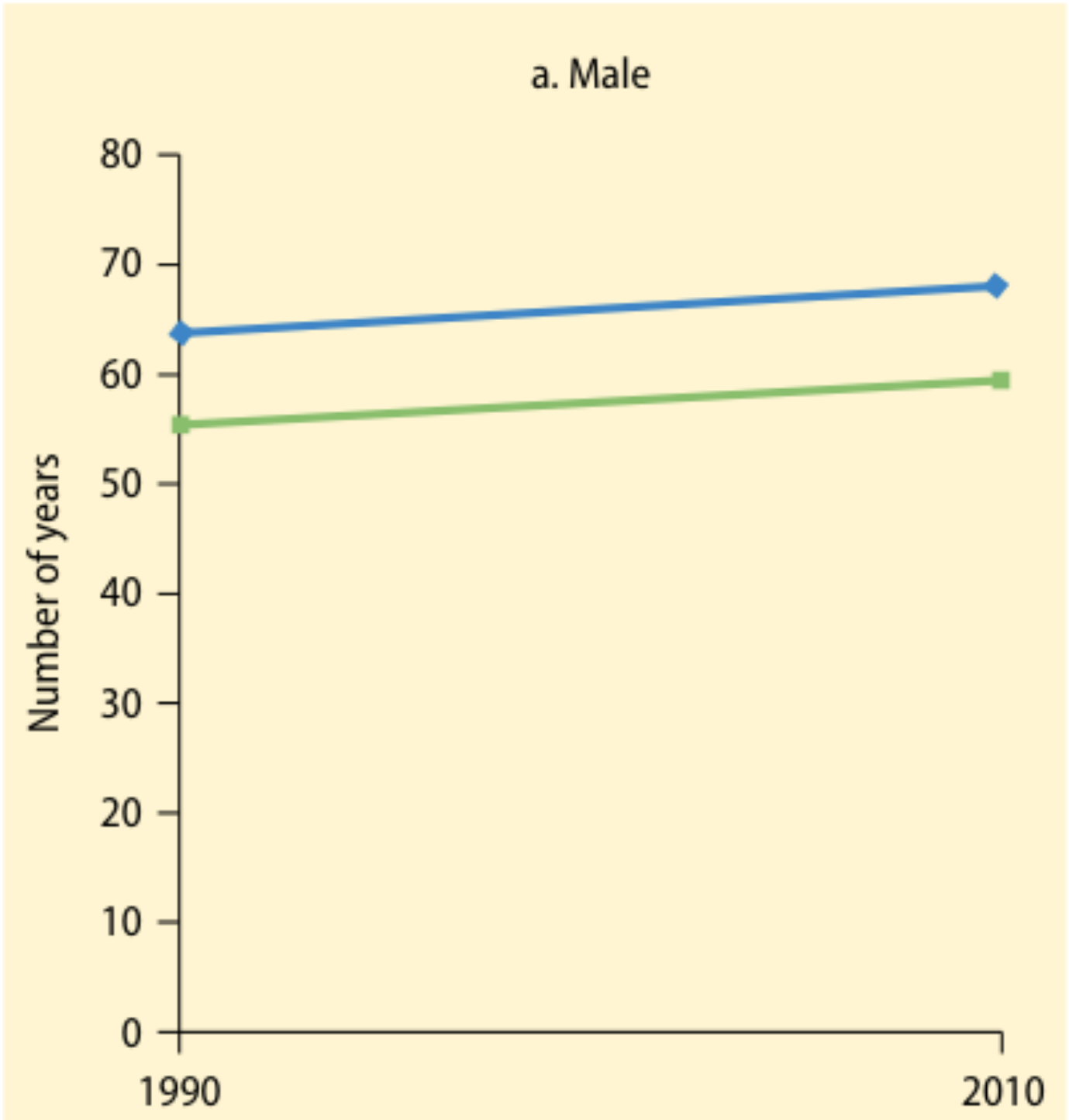
If population ageing is the era's structure, rising life expectancy is its engine. "Health improvements have been tremendous since the second half of the 19th century and life expectancy has increased for each successive cohort, apart from temporary mortality spikes due to the world wars" [3]. Longevity has become so reliable that survival to old age is now the statistical norm: "[o]n average across countries, 80% of males born 70 years ago have reached age 70 and 93% of males born today are expected to do so; for women, the respective shares are 89% and 96%," and "[t]he probability of surviving to age 85 will increase from 50% today to 75% by 2100 for males born today, and from 64% to 83%, respectively, for females" [3]. The hundred-year life is not a speculation about outliers; it is the central expectation for cohorts now being born.

The gains, however, arrive at very different speeds. "In OECD countries, life expectancy at birth rose by an average of 7.2 years between 1980 and 2015; by contrast, it increased by 20.5 years in India and by 11.3 years in Brazil over the same period. In Russia, however, the increase was only 2.4 years and South Africa lost 1.3 years of life expectancy" [3]. Catch-up is therefore real but not automatic — it depends on the mortality regimes a country manages to build. The demographic consequences compound: "[i]n 2015, old-age dependency ratios in Brazil and China were half the OECD average; by 2050 both countries will be rapidly closing in on the average OECD country and by 2075 they will be older than the average OECD country" [3].

Within the OECD, the league table of old age is stark. "In 1980, there were 20 persons aged 65 and over for every 100 people of working age (20-64 years) on average across the OECD; by 2015 this number had risen to 28… and with sharp acceleration is projected to almost double between 2015 and 2050" [3]. "Japan is by far the oldest OECD country: its old-age dependency ratio increased by 32 points between 1980 and 2015," ageing was also rapid in Finland, Greece, Italy and Portugal, and "Japan will remain the oldest country in 2050. Korea, however, is ageing the fastest, followed by Greece, Italy, Portugal and Spain" [3]. Meanwhile "life expectancy at age 65 ranges from 19 to 22 years in 2010-15" across OECD countries [3] — the two decades after the traditional retirement age that pension and care systems must now finance.

National averages conceal as much as they reveal. "On a national level, the share of the older population ranged from 6% in Mexico to 26% in Japan in 2014, but… [a]t the regional (TL3) level… [t]he share of the population aged 65+ ranged from 2% in the Chilean region of Antártica Chilena to 33% in Greece's Evrytania region," with "[u]rban centres draw[ing] younger and more mobile workers while rural areas are often already much older," and internal spreads of "about 20-percentage point differences between the youngest and oldest regions" in Spain, Mexico and Australia against "around 5 percentage points" in Central and Eastern Europe and Ireland [3]. Ageing is thus a local condition before it is a national statistic: services must be planned where old people actually are.

Two measurement cautions frame everything that follows. First, the headline figures understate what today's newborns will experience: "average cohort life expectancy is estimated to be equal to period life expectancy at birth plus 9.6 years, due to declining mortality at the same age across cohorts. In other words, the 2015 birth cohort will live about ten years longer than period life expectancy at birth in 2015 suggests" [3]. Second, the averages hide a steep social gradient that this report returns to in its inequality chapter: "[a] 25-year old university-educated man can expect to live 7.5 years longer than his low-educated peer, on average across countries; for women the difference is much lower at 4.6 years," narrowing by age 65 to gaps of 3.5 and 2.4 years but remaining, relative to remaining life, even larger [3]. Life expectancy is rising nearly everywhere; the open questions are how fast, for whom, and — the subject of the next sections — on what assumptions about fertility, mortality and migration the trajectory rests.


a. Male
80
70
60
50
40
30
20
10
0
Number of years
1990
2010

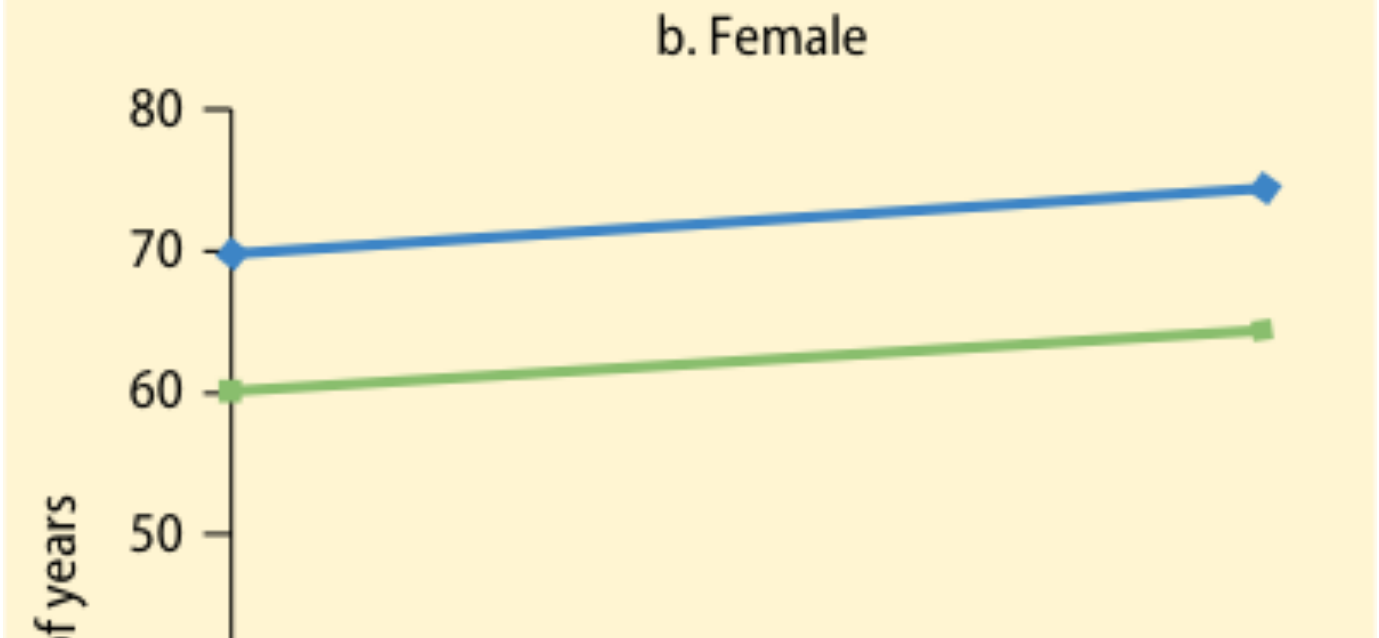

b. Female
80
70
60
50
f years

Exhibit. Between 1990 and 2010, East Asia and the Pacific gained years of both life expectancy and health-adjusted life expectancy for men and women. Source: S0029.

Fertility, mortality and demographic projections

Every number in this report's later chapters — pension costs, care needs, labour supply — descends from projections built on three assumptions: how many children will be born, how long people will live, and how many will move. The European Commission's Ageing Report states the machinery plainly: "[p]rojected changes in the size and age profile of the population are determined by assumptions regarding fertility rates, mortality rates and migration" [4]. Understanding those assumptions — and their fragility — is essential to reading the projections honestly.

The European Union offers the most fully documented worked example. "The total EU population is expected to rise from 449 million people in 2022 to a peak of 453 million people in 2026. After that, it would fall to 432 million in 2070, a decline by 4% compared to 2022," even though "in 13 Member States, the population is projected to increase" — while "an upward shift in the age distribution is expected in all Member States" [4]. The old-age dependency ratio "will rise sharply in all Member States": "[f]rom about 29% in 2010 in the EU, it rose to 36% in 2022 and would rise further to 59% in 2070, with most of the increase expected already by 2045. Put differently, the EU goes from having nearly thirty people aged 20 to 64 for every ten people aged over 65 years in 2022, to having less than twenty people by 2045" [4].

The three assumption dials are set as follows. On fertility: "[t]he total fertility rate in the EU is projected to rise from 1.5 live births per woman in 2022 to 1.6 by 2070," an increase that "reflects the assumption that, in the very long run, fertility rises to the highest rate currently observed across the EU Member States," while "[f]ertility rates would stay below the natural replacement rate of 2.1 in all countries" [4]. On mortality: "[l]ife expectancy at birth for males is expected to increase by 7.7 years in the EU, from 78.4 in 2022 to 86.1 in 2070. Female life expectancy at birth would rise by 6.4 years, from 84.0 in 2022 to 90.4 in 2070," with remaining life expectancy at 65 rising "about 5 years for both" and the projections resting on "the hypothesis of a partial upward convergence in longevity between Member States" [4]. On migration: "[n]et migration is assumed to return to pre-2022 levels in the coming years, at around 1 million people annually (0.2% of the EU population)," against "an exceptionally high inflow of around 1% of the EU population in 2022 because of the Russian war of aggression against Ukraine," averaging 0.3 per cent over 2022–2070 [4]. Even with participation rising — "by around 3 pps" for the working-age population "and by 10 pps for people aged 55-64" — "the number of people in the labour force is projected to decrease by 12% (25 million people) between 2022 and 2070 in the EU" [4].

East Asia and the Pacific show the same machinery at a different transition stage. Regional fertility fell from 5.51 to 3.03 children per woman, but unevenly: "[t]he older East Asian countries have extremely low total fertility rates (TFRs), averaging 1.28 children per woman… in 2010," younger countries stood at 3.45, and the intermediate group at 1.93 — "2010 TFRs had fallen to 1.66 in China, 1.41 in Thailand, and 1.75 in Vietnam" [5]. Mortality improvements were the fastest anywhere: "[d]ramatic gains in life expectancy at birth in the region, from less than 45 years in 1950 to 74 years today, mirror the decline in mortality rates" [5]. Yet the decomposition is unambiguous about what drives ageing: comparing counterfactuals, "the proportion of individuals 0 to 5 years of age in East Asia and Pacific would have been 7.0 percentage points higher in 2005 had fertility rates remained at 1960 levels rather than declined. This contrasts with a 0.1 percentage point decline in the same population had age-specific mortality rates remained at 1960 levels" [5]. Fertility decline, not longevity, does the ageing — longevity determines how long the aged state lasts.

Both exercises carry a shared warning label. The convergence assumptions embedded in projection models do quiet but decisive work: the UN's projected fertility recovery in the oldest East Asian countries "is far from certain and is driven to a significant extent by the convergence assumption used in the UN models" [5]. Projections are best read as the current settings of three dials — fertility edging up but below replacement, mortality improving with partial convergence, migration reverting to trend — applied to populations whose momentum is already locked in: "the low fertility and mortality rates that evolved between 1960 and 2005 will continue to drive population aging as those birth cohorts move through the age distribution, even as the future rates themselves are projected to stabilize" [5]. The next section turns from the world to its regions: where ageing is fastest, and how far the oldest age groups will grow by 2050.

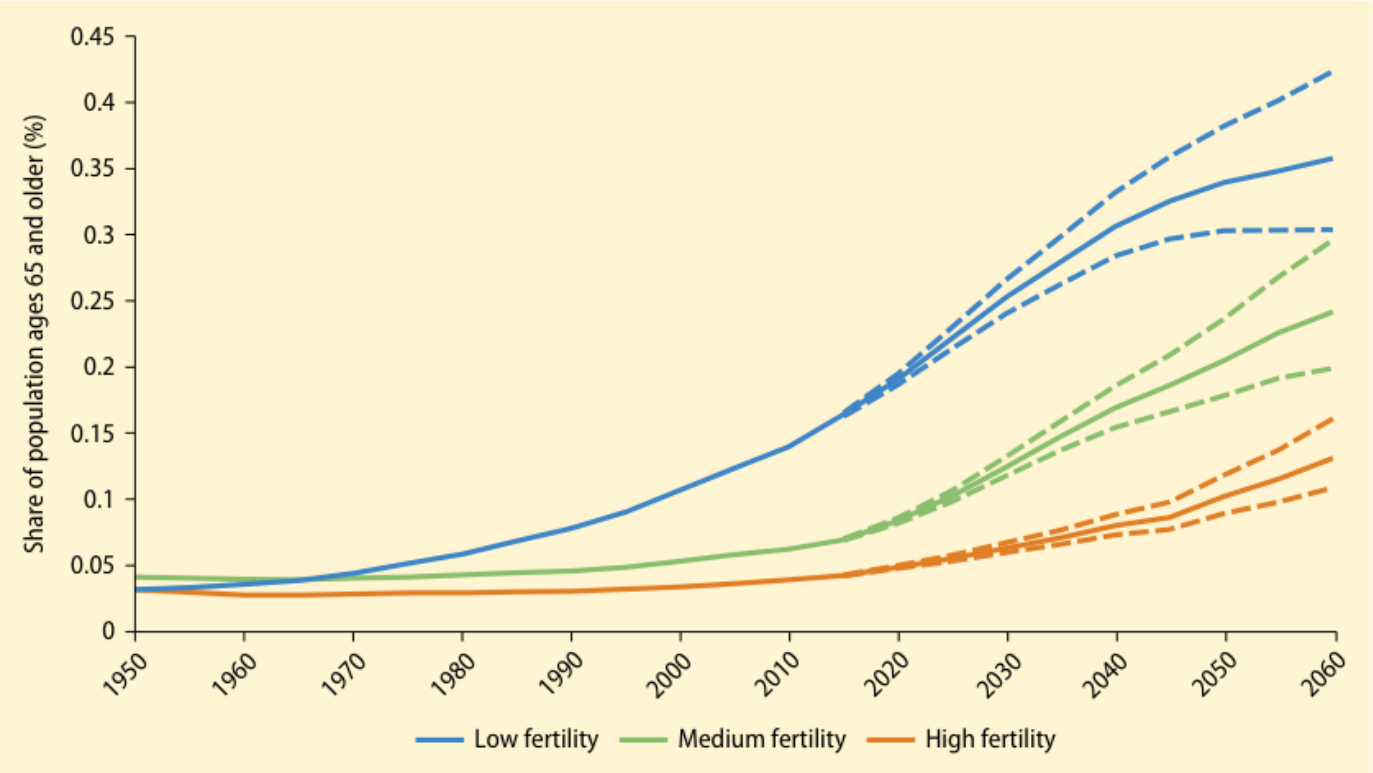


Exhibit. Trajectories of the share of people aged 65 years and older by country group under low, medium and high total fertility rate assumptions, showing how alternative fertility paths reshape the pace of population ageing over the long run. Source: S0029.

Rapid population ageing across regions: East Asia and Pacific, Europe

Ageing is a global process with a distinctly regional geography. Nowhere is the contrast between Europe's long transition and Asia's compressed one sharper than in East Asia and the Pacific, the region that now holds the world's largest older population in absolute terms: "[i]n 2010, East Asian and Pacific economies were home to around 187 million people ages 65 and above (around 150 million of them in developing countries). This number represents around 36 percent of the global population in this age group. China is responsible for the bulk of this elderly population, with nearly 114 million people ages 65 and older" [5]. East Asia, led by China, has more old people than any other region — while its average older share, "around 7 percent of the total population in 2010," was similar to Latin America and the Caribbean (6 per cent) and only modestly above the Middle East and North Africa and South Asia (5 per cent) [5]. Scale and share tell different stories, and policy must answer both.

Within the region, three distinct ageing patterns organise the diversity. The high-income "red" economies — Hong Kong SAR, Japan, the Republic of Korea and Singapore — are already aged societies, with people 65 and older representing "14 percent of the total population of this group in 2010" [5]. The "orange" group — China, Indonesia, Malaysia, Mongolia, Thailand and Vietnam — "are aging very quickly," while the "green" group of Cambodia, Lao PDR, Myanmar, Papua New Guinea, the Philippines, Timor-Leste and the Pacific Island countries "are still young—just 4 percent of the population was age 65 or older in 2010—but they will begin to age quickly in coming decades" [5]. At the young extreme, "only 3 percent of the populations of Papua New Guinea and Timor-Leste are 65 years old and above—the same as the average for Sub-Saharan Africa" [5].

The speed differential against Europe is the section's central fact. "East Asia and Pacific is also aging more quickly than any other region in history. The population ages 65 and over in many countries in the region will increase from 7 percent to 14 percent of the total population in just two or three decades—a change that took 45 years in the United Kingdom, 69 years in the United States, and 115 years in France" [5]. The country timings make the compression concrete: Vietnam crosses from ageing to aged in about 15 years; Lao PDR, Malaysia, Indonesia, Thailand and Korea in about 20; China, Singapore, Japan, Mongolia, Myanmar, Cambodia and Timor-Leste in about 25 [5]. "In terms of speed of aging, the typical East Asian and Pacific economy... moves from aging to aged in around 25 years, but countries such as Indonesia, Korea, and Vietnam are notable for their even more accelerated demographic transitions" [5]. Europe's pension and care institutions were built across the long transitions these figures describe; East Asia must build equivalents in a single generation, at lower income levels.

The consequences arrive in sequence. Working-age populations diverge first: “[b]etween 2010 and 2040, the share of the working-age population will fall by more than 15 percent in Korea and by at least 10 percent in China, Japan, and Thailand,” while “in green and some orange countries, the working-age share of the population is not expected to shrink until after 2040,” with “[t]he Philippines and Indonesia… account[ing] for the largest share of the regional increase” [5]. The oldest old follow: “[t]he population of those ages 80 and above—the ‘oldest old’—will also grow faster in East Asia and Pacific than in any other region” [5]. And the endpoint converges with Europe’s present: “by 2060, the average share of population ages 65 and over in East Asia (22 percent) will approach that of the OECD countries (24 percent) and Eastern Europe and Central Asia (26 percent),” and “[w]hereas in 2010 only 1 of the 25 ‘oldest’ economies… was in East Asia (Japan), by 2060 five East Asian economies are projected to be among the 25” [5].

The regional literature closes with two cautions that apply to every projection in this report. First, the record is biased in a known direction: “[t]he record of demographic projections in the region is mixed… Fertility decline, for example, has tended to consistently outpace projections in the region; assumptions about future reversals in fertility for richer East Asian and Pacific economies risk extrapolating from unstable short-run dynamics” [5]. Second, the yardsticks themselves are ageing: “[t]he static definition of the working-age population as people ages 15 to 64… is increasingly being questioned. Alternative dependency measures that have been proposed (such as years of remaining life or ratios based on age profiles of actual labor force participation) may be more relevant for policy purposes as people live longer” [5]. The next section quantifies the destination of these trajectories: the shares of the population aged 65+ and 80+ on the road to 2050.

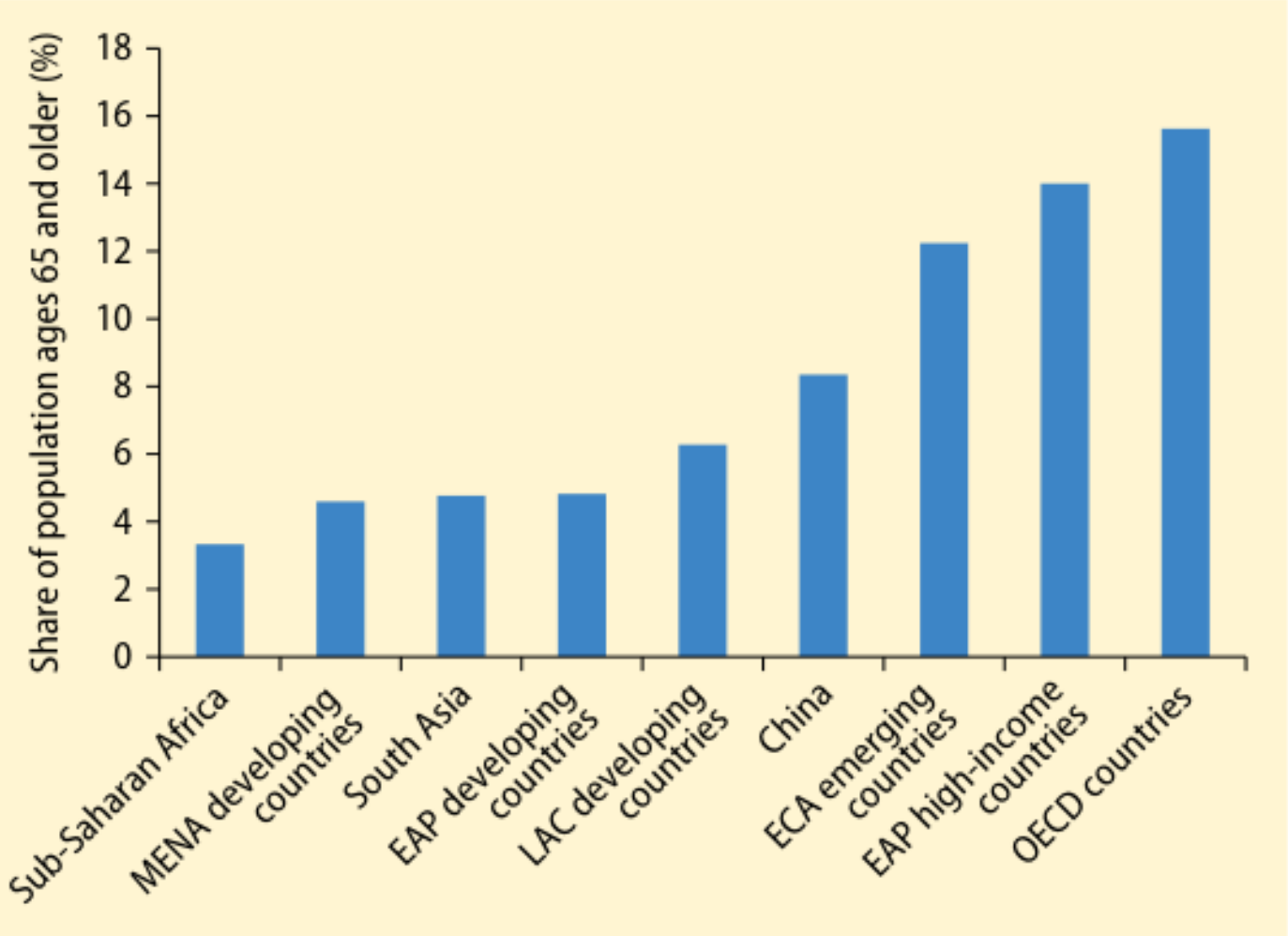


Exhibit. Share of the population aged 65 and above in 2010 across selected world regions and East Asia and Pacific subregions, benchmarking the region’s ageing against South Asia, Sub-Saharan Africa, MENA and developed countries. Source: S0029.

The older population: shares aged 65+ and 80+ to 2050

The destination of the trajectories traced so far can be stated in two numbers. “The number of people aged 65 years or older worldwide is projected to more than double, rising from 761 million in 2021 to 1.6 billion in 2050. The

number of people aged 80 years or older is growing even faster" [1]. In proportional terms, "[b]etween 2021 and 2050, the global share of the older population, defined as people aged 65 years or over, is projected to increase from less than 10 to around 17 per cent" — "[i]n 2021, 1 in 10 people worldwide were aged 65 or above. In 2050, this age group is projected to account for 1 in 6 people globally" [1]. The UN's assessment is categorical: "[p]opulation ageing is an irreversible global trend. It is the inevitable result of the demographic transition – the trend towards longer lives and smaller families – that is taking place even in countries with relatively youthful populations" [1].

The OECD record shows what the mature stage looks like. "In recent decades, the share of the population aged 65 and over has doubled on average across OECD countries, increasing from less than 9% in 1960 to 18% in 2021," and "[a]cross the 38 OECD member countries, more than 242 million people were aged 65 and over in 2021, including more than 64 million who were at least 80 years old" [6]. The projection continues upward: the 65+ share is "projected to continue increasing in the coming decades, rising from 18% in 2021 to 27% by 2050," with "[i]n five countries (Korea, Japan, Italy, Greece and Portugal), the share of the population aged 65 and over will exceed one-third by 2050," while "[a]t the other end of the spectrum, the population aged 65 and over in Israel, Mexico, Australia and Colombia will represent less than one-fifth of the population in 2050, owing to higher fertility and migration rates" [6].

The oldest old are the steepest curve. "Between 2021 and 2050, the share of the population aged 80 and over is predicted to double on average across OECD member countries, from 4.8% to 9.8%. At least one in ten people may be 80 and over in nearly half (18) of these countries by 2050, while in five countries (Korea, Japan, Italy, Greece and Portugal), more than one in eight people may be 80 and over" [6]. Korea is the extreme case of speed: its 80+ share undergoes "nearly quintupling – from below the OECD average in 2021 (3.9% versus 4.8%) to well above it (16.5% versus 9.8%) by 2050," making Korea "the most rapid population ageing among OECD member countries" in the coming years [6]. Outside the membership, the giant transitions dominate the global arithmetic: "[i]n China, the share of the population aged 65 and over will increase much more rapidly than in OECD member countries – more than doubling from 12.6% in 2021 to 30.1% in 2050. The share of the Chinese population aged 80 and over will rise even more quickly, increasing more than four-fold from 2.3% in 2021 to 10.3% in 2050," while Brazil, "whose share of the population aged 65 and over was only around half the OECD average in 2021," reaches "nearly 22%… by 2050" [6].

Two compositional facts complete the picture. First, old age is disproportionately female, though less so than before: "women represented 56% of the population aged 65 and over in 2021, a slight decrease from 59% in 2000," exceeding 65 per cent in Latvia, Lithuania and Estonia and falling to "just 52%… in Iceland" [6] — and the UN expects "small but noticeable reductions in the sex gap in life expectancy and in the female share of the population at older ages… in coming decades" as male survival converges toward female [1]. Second, the growth in numbers is fastest where shares are still lowest: Northern Africa and Western Asia and sub-Saharan Africa "are expected to experience the fastest growth in the number of older people over the next three decades," while "[t]oday, Europe and Northern America combined have the highest share of older persons" [1].

These shares translate directly into service demand. "[B]etween 2015 and 2030, the number of older people in need of care around the world is projected to increase by 100 million," yet "the supply of LTC workers per 100 elderly people (aged 65 and over) has stagnated in most countries since 2011" [6]. The chapters that follow take up the implications in order — the case for action, the health of longer lives, and the care systems that the 80+ curve makes unavoidable.

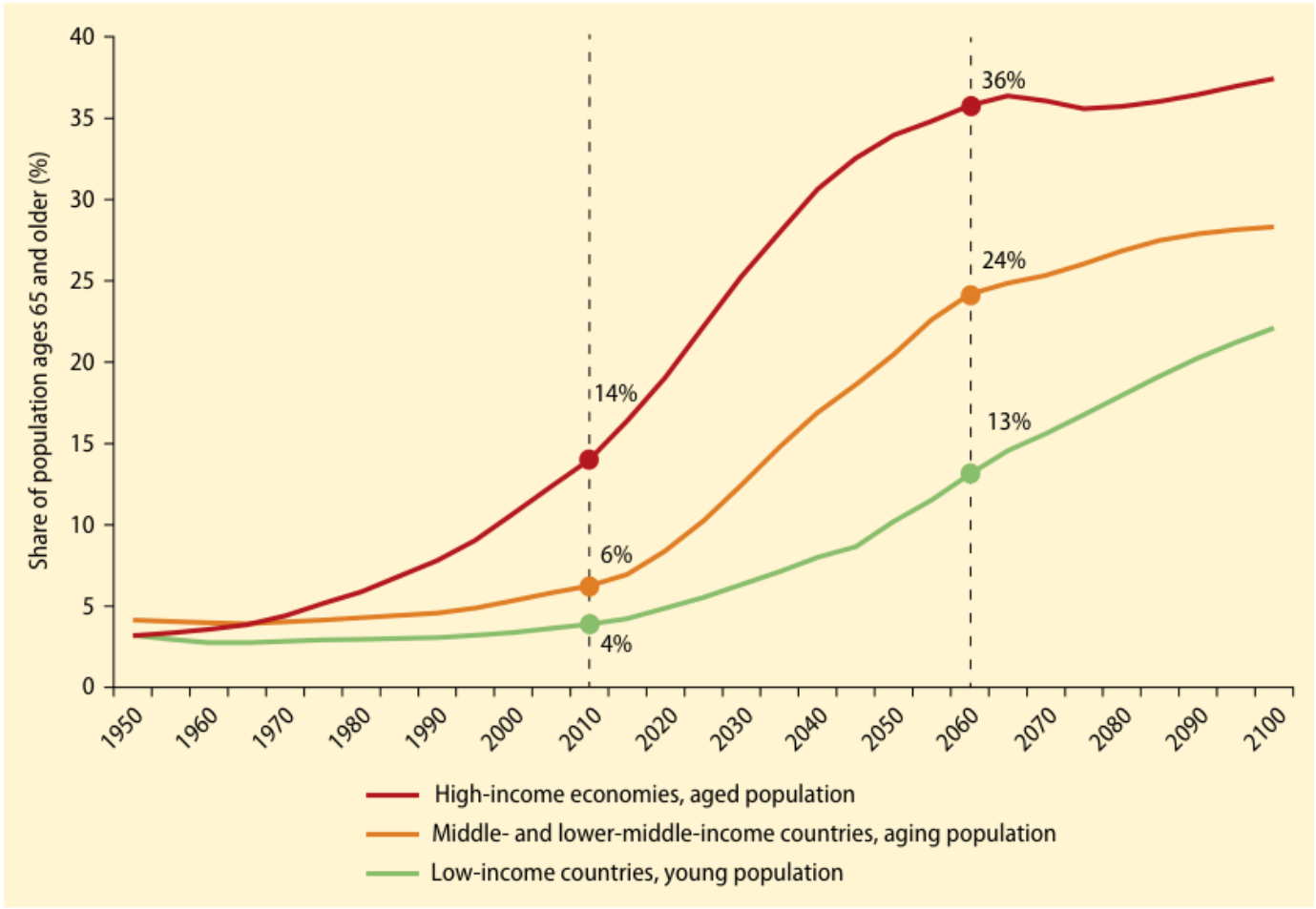


Exhibit. Three distinct ageing patterns among East Asia and Pacific economies, comparing the share of the population aged 65 years and older across high-income economies with already aged populations and their younger regional counterparts. Source: S0029.

The case for action

The case for acting on population ageing begins with a reframing: longevity is an achievement, not a misfortune. "The rise in human longevity is a success story caused by improved sanitation and medical therapies, greater access to education and family planning, and strides towards gender equality and women's empowerment" [1]. But the achievement outruns the institutions built for shorter lives. The shift toward an older population — "gradual and largely irreversible" — "brings into question current arrangements of old-age support in countries both young and old," and "[a]s the health and wealth of societies continue to improve, traditional notions of dependency at older ages are becoming less and less valid. Yet, effective systems of old-age support will continue to be needed, as will the intergenerational solidarity required to sustain them" [1]. Action is required not because ageing is a crisis, but because the default settings of work, retirement, health and care were calibrated to a demography that no longer exists.

The sharpest reason to act early is distributional. "Not everyone has benefitted to the same extent from the social and economic improvements that drive longevity," and "[h]igh and rising levels of inequality threaten to become a defining feature of present and future generations... Without swift and bold policy action to counter this trend, future cohorts of older persons may be even more economically unequal than those alive today. But rising inequality is not inevitable, and policy makers can influence the future direction of inequality as populations continue to live longer" [1]. The window for influence is before people reach old age: "[p]olicies to promote healthy ageing, prevent poverty and foster employment and opportunities for decent work before people reach older ages are vital to enhance economic security and reduce inequality among older persons," and mobilising "the skills, expertise and knowledge of older persons, women, and other groups that have traditionally been excluded from or disadvantaged in the labour market can go a long way towards promoting equity and ensuring that old-age support systems are fiscally sustainable" [1]. Hence the UN's timing verdict: "[q]uestions of intergenerational equity in an ageing world need to take centre stage... Now is the time to plan for the long term, to prepare for the challenges ahead and take advantage of the new opportunities these demographic shifts bring" [1].

The opportunity side of the case is now argued in market terms. The World Economic Forum's longevity-economy work urges "a fundamental reengineering of markets and systems from a first-principles perspective – one that prioritizes advancing healthspans and wealthspans over merely extending lifespans. This shift reframes ageing from a 'liability model' to an 'asset model of longevity'" [7]. The asset base is material: "[w]ith older consumers becoming the fastest-growing economic force, pension funds and their $28 trillion in assets can finance green growth, ensuring a healthier planet and population" [7]. The same source insists the longevity and climate agendas are inseparable: "[b]y 2050, the interplay between climate change and longevity will shape both human well-being and economic development. Climate change directly impacts health and longevity through rising temperatures, extreme weather events and the spread of infectious diseases, which disproportionately affect vulnerable populations, including older people" [7].

Lived experience supplies the urgency that aggregate numbers cannot. In the United States, "19% of the rural population is already 65 or older, compared to 15% in urban areas," yet rural communities face "limited healthcare access, economic disinvestment and infrastructure gaps" — "[t]he longevity economy cannot thrive by 2050 if rural communities are left behind" [7]. In Japan, "the impacts of ageing are evident everywhere, from labour shortages to working-age individuals juggling jobs and caregiving. Remarkably, even people in their 40s are still considered 'young'… serious intergenerational dialogue is urgently needed" where "the burden is unfairly shifted to the young" [7]. In Nigeria, the foundation itself is missing: "over 80 million youth and 53.40% unemployed" and an "underfunded public health framework" mean that "without health, employment or satisfaction, a thriving longevity economy is unattainable" [7].

Underneath these cases sits a structural change in the shape of a life. "[T]raditional career paths are becoming more varied. Individuals are navigating longer, more complex careers and multistage lives, with more fluid transitions between education, work, caring responsibilities and retirement," so "enabling financial resilience during those key transitions is essential not only for individual well-being but also for sustained economic growth" [7]. The demographic chapters have established that the hundred-year life is coming; this chapter's conclusion is that systems designed for the three-stage life — education, work, retirement — must be rebuilt for the multistage one, and that the rebuilding is cheapest and fairest when it starts now. The report turns next to the foundation on which everything else depends: the health of longer lives.



Exhibit. Key life transitions — including career changes, caring responsibilities and retirement — at which financial resilience is tested as economies adapt to longer lives, and where support determines both individual well-being and sustained economic growth. Source: S0010.

Health in a longer life

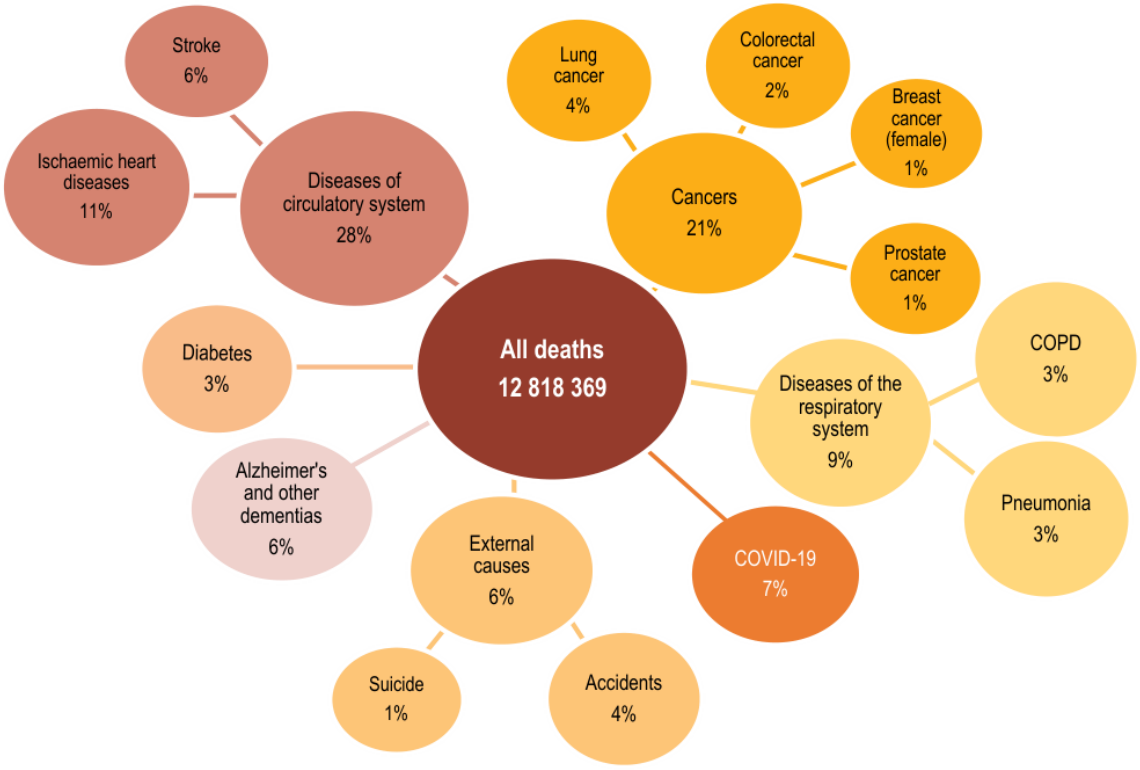
Health status, disability and mortality trends

Whether the hundred-year life is a blessing or a burden turns on a question demography alone cannot answer: are the added years lived in good health? The mechanisms cut both ways. “In the past 100 to 150 years, medical technologies have been effective at saving lives from a range of fatal diseases. Often, however, this success has meant that the life years saved are spent in poor health and characterized by chronic illness and disability” [5]. Three hypotheses organise the possibilities. Under the expansion of morbidity hypothesis (Gruenberg 1977; Olshansky et al. 1991; Verbrugge 1984), “the decline in mortality is largely due to a reduction in disease fatality rather than a reduction in prevalence or incidence of those diseases. Consequently, falling mortality is accompanied by an increase in morbidity and disability” [5]. Under the compression of morbidity hypothesis (Fries 1980, 1989), “the incidence and prevalence of chronic illnesses and morbidity are compressed toward the later period of life at a faster pace than reductions in mortality. People do not just live longer; they live longer in better health” [5]. Under dynamic equilibrium (Manton 1982), “the effects of compression and expansion of morbidity offset one another, with, on one hand, decreasing prevalence or incidence of chronic illness, and on the other hand, decreasing fatality rates of diseases, leading to longer prevalence of disability” [5].

The empirical record refuses to pick a single winner — because different diseases move in different directions. "[H]igher levels of some disabling conditions (dementia and musculoskeletal diseases) go along with falling rates of prevalence of other conditions (cardiovascular and chronic respiratory diseases). The growing consensus from Organisation for Economic Co-operation and Development countries is as follows: the prevalence of severe disability has been reduced substantially, the prevalence of mild to moderate disability has increased, and overall, health status has improved (Christensen et al. 2009)" [5] — a pattern closest to dynamic equilibrium with a favourable tilt. Measured disability also moves for non-medical reasons: "[t]rends in disability may reflect underlying trends in other domains, such as the rising use of assistive technology, improved accessibility at home and in public spaces, and changes in social perceptions of disability," as well as "improvements in living conditions, not only of those currently elderly, but also of their childhood living conditions" [5].

Healthy life expectancy (HALE) data from East Asia and the Pacific illustrate both the gains and the gaps. Life expectancy and health-adjusted life expectancy in the region both increased between 1990 and 2010, with healthy life expectancy at birth rising by 11 months in the period covered, "[t]he biggest improvements in healthy life expectancy in the region… in low-income countries, and the slowest improvements… in middle-income countries (IHME and World Bank 2013)" [5]. But "[a] significant gap exists between total and healthy life expectancy in the region": "[a]t the age of 50, men and women in Japan can expect to live free of disability approximately 10 years more than their counterparts in Papua New Guinea," "[y]ears spent in good health are considerably lower in Mongolia, Papua New Guinea, and some Pacific Islands compared with not only high-income countries such as Australia and Japan but also middle-income countries such as Indonesia and China," and "[t]he gender gap also is significant in all East Asian and Pacific countries, with women spending more years free of disability than men" [5]. The difference between life expectancy and HALE — years of healthy life lost to poor health — is the quantity health systems must shrink.

The measurement toolkit has broadened accordingly. Alongside chronological indicators (shares above 65, median age, old-age ratios), analysts now use prospective measures based on "the expected time until death (i.e., the prospective old-age dependency ratio)," economic ratios of consumers to workers, and health-based measures — "the disability-adjusted life years (DALYs) or the healthy life expectancy (HALE) at birth" — which "try to capture the diversity and heterogeneity of the older population, including differences in their health, functional ability and socioeconomic status" [8]. The frontier cases show why health-adjusted thinking matters: in countries whose populations have already peaked, "[t]he number of persons aged 65 or older… is projected to reach 409 million by 2027, surpassing the number of children under age 18," while "[t]he number of persons aged 80 or older in the same group is increasing even faster and is likely to triple, rising from 85 million in 2024 to around 267 million by 2100" [8]. And no inflow rescues the arithmetic: "[i]nternational migration alone… cannot offset population decline or population ageing in the long term and should not be seen as a 'solution' to what is ultimately a universal and irreversible process: the demographic transition" [8]. The policy conclusion follows directly: since the years are coming regardless, the contest is over their quality — which is why the next sections turn to what health systems spend, who staffs them, and whom they actually reach.



Note: Other causes of death not shown in the figure represent 21% of all deaths.
Source: OECD Health Statistics 2023.

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Figure 3.6. Main causes of mortality by country, 2021 (or nearest year)

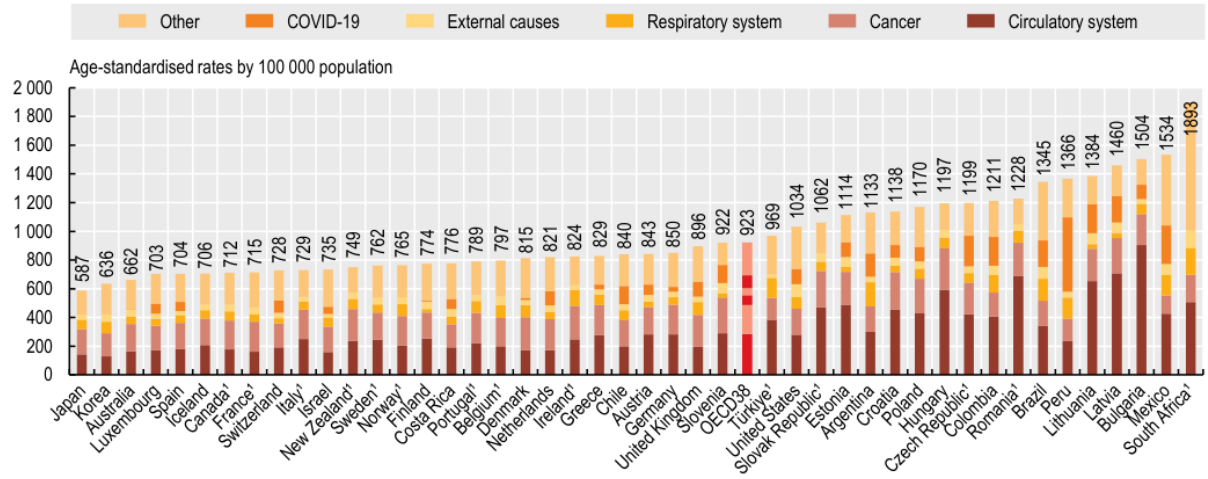


Exhibit. Main causes of mortality across OECD countries in 2021 (or nearest year), with diseases of the circulatory system and cancers accounting for the largest shares of deaths, followed by respiratory diseases and other causes. Source: S0026.

Health expenditure and health expenditure in relation to GDP

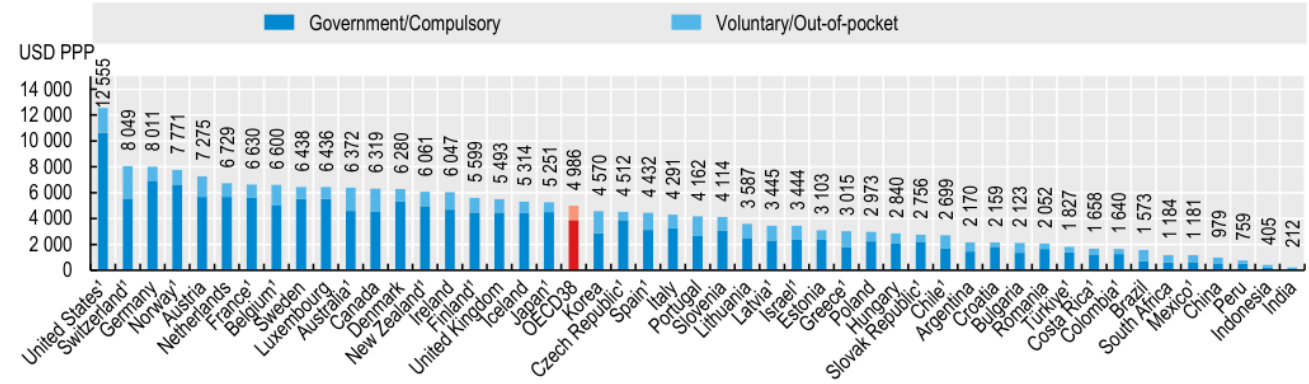
Health spending is where the demography of the previous chapters converts into budgets. “In 2022, average per capita health spending in OECD countries (when adjusted for differences in purchasing power) was estimated to have reached nearly USD 5 000. In the United States, it reached the equivalent of USD 12 555 for every US citizen. Switzerland and Germany were the next highest spenders in the OECD, but at around USD 8 000 this was still less than two-thirds of the level in the United States,” with Norway, Austria and a group of western European countries plus Australia, Canada and New Zealand at USD 6–7,000, and “spending in Mexico (USD 1 181) at around a quarter of the OECD average” [6]. The financing structure is predominantly collective: “[o]n average across OECD countries, about three-quarters of all health spending is financed through government or compulsory insurance schemes” [6] — which is why ageing-driven health costs land, in the main, on public budgets and mandatory contributions.

Age structure explains part, but only part, of the level differences. “[A] population with an older demographic structure can expect higher mortality rates, greater incidence and prevalence of certain diseases, and thus higher demands for healthcare and, by consequence, higher spending on health” — yet standardising for age reverses some familiar rankings: “countries such as Israel and Ireland, and some of the Latin American OECD member countries could expect higher health spending relative to the OECD average if a standard population structure was applied, whereas those countries with older populations (e.g. Japan, Germany and Italy) could expect lower spending” [6]. Japan, Germany and Italy spend less than their age structures would predict; demography is a driver, not a destiny — prices, volumes and system design mediate it.

The growth record shows both a trend and a shock. “In the years leading up to the COVID-19 pandemic, annual average per capita spending on healthcare grew by an average of 2.6% across OECD countries,” with Latvia, Lithuania and Korea growing “between 6 and 8%” in 2015–2019 while “in most Nordic countries and France, growth was much more moderate at less than 1% on average” [6]. Then “[t]he emergence of COVID-19 in 2020 led to sharp increases in health spending, particularly from governments… Between 2019 and 2022, average per capita spending growth in the OECD accelerated to 3.3% per year, with a peak reached in 2021 before contracting in the most recent year” [6]. Around two-thirds of OECD countries saw faster growth during the pandemic than before it, “[s]ome countries – Latvia and Türkiye, have seen double-digit growth in health spending between 2019 and 2022,” Korea and New Zealand grew more than 8 per cent, and “only Mexico is expected to have seen overall negative growth during the most recent three-year period” [6].

For projections, the European Commission’s Ageing Working Group turns the previous section’s morbidity hypotheses into explicit spending scenarios — the cleanest illustration in the literature of how health-status assumptions move fiscal numbers. The baseline reference scenario combines “the demographic impact of population ageing,” a “‘neutral’ assumption on the evolution of health status, by assuming that half of the extra years of life gained through higher life expectancy are spent in good health,” unit costs evolving with GDP per capita, and an income effect “modelled through a cost sensitivity of 1.1 in the base year converging to unity by 2070” [4]. Around it sit the diagnostic alternatives: a demographic scenario isolating “the pure ageing effect”; a ‘no healthy ageing’ scenario implementing expansion of morbidity, in which “all the gains in life expectancy are assumed to be spent in poor health. The number of years spent in good health remains constant” [4]; a ‘healthy ageing’ scenario based on “the relative compression of morbidity hypothesis,” in which “the number of years spent in poor health during a lifetime remains constant… all future gains in life expectancy are assumed to be spent in good health” [4]; and a labour intensity scenario recognising that health care “is and will remain a highly labour-intensive sector,” with unit costs tied to productivity per hour — the Baumol mechanism by which care gets relatively more expensive even with no demographic change at all [4].

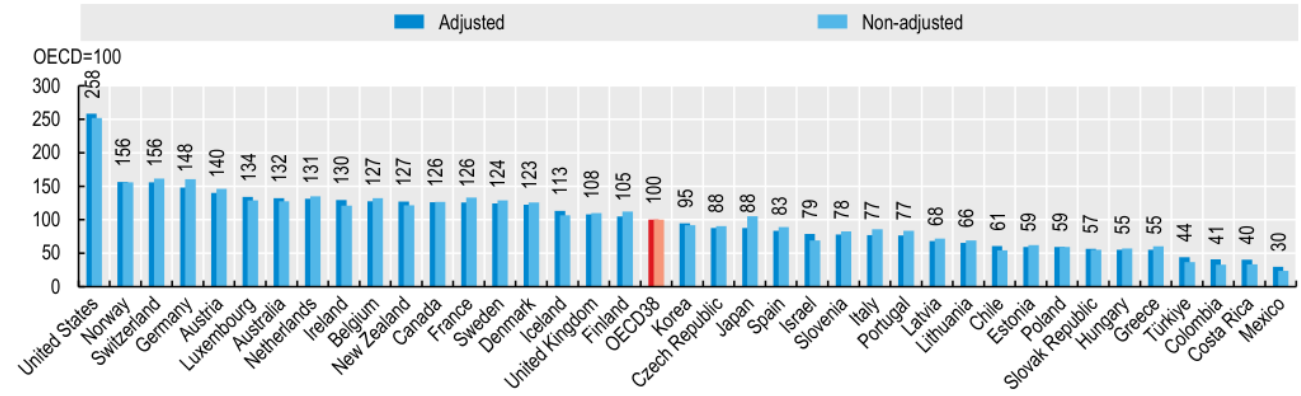
The lesson for readers of any health-spending projection is that the spread between these scenarios is the honest output: ageing pushes spending up mechanically, but how much depends on whether morbidity compresses or expands and on whether care’s labour intensity keeps raising its relative price. The next section turns from money to people: the health and social care workforce that spending must buy.



OECD estimates.
ource: OECD Health Statistics 2023; WHO Global Health Expenditure Database.

StatLink https://stat.link/m6pzqb

igure 7.5. Impact of age-adjusting health expenditure per capita, 2022



ote: Health spending relative to OECD average (OECD=100) after indirect standardisation based on a derived OECD age-spending profile.
ource: Calculations based on OECD (forthcoming[1]), "Understanding international measures of health spending: Age-adjusting expenditure on health".

StatLink https://stat.link/giu21p

igure 7.6. Average annual growth in per capita health expenditure (real terms), 2015-19 and 2019-22

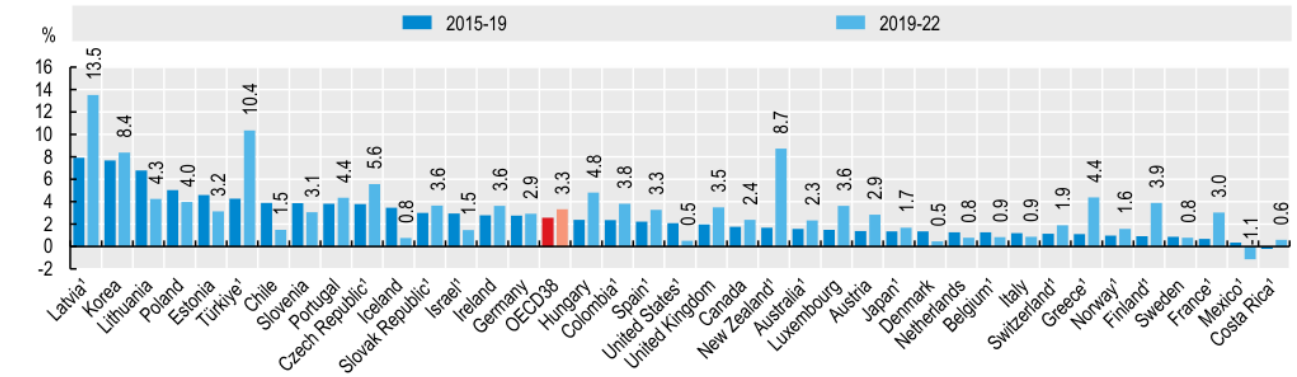


Exhibit. Health expenditure per capita in 2022 (or nearest year), in US dollars at purchasing power parity, split between government and compulsory financing schemes and voluntary or out-of-pocket payments across OECD countries. Source: S0026.

Health workforce and the health and social care workforce

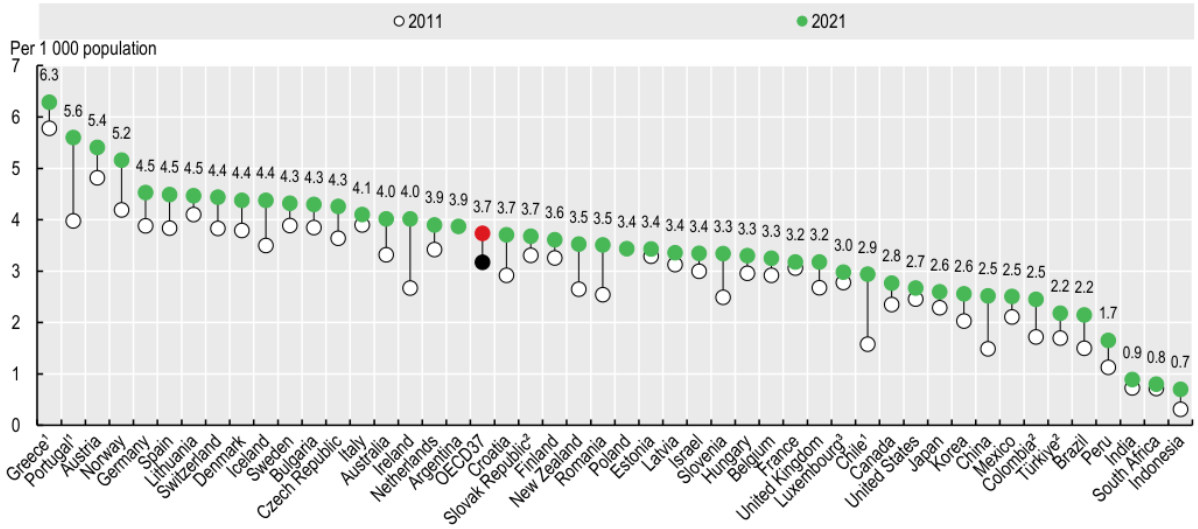
Health and social care is now one of the largest employers in the developed world — and still cannot hire fast enough. "In OECD countries, health and social care systems employ more workers now than at any other time in history. In 2021, more than one in every ten jobs (10.5%) was in health or social care, up from 9.5% in 2011," and "[i]n Nordic countries and the Netherlands, more than 16% of all jobs were in health and social work" [6]. The sector's growth dwarfs the rest of the economy: "employment in health and social work increased by 24% between 2011 and 2021 – over twice the rate of overall employment growth" [6]. Its composition is distinctive: "[i]n most OECD countries, over 75% of workers in the health and social care sector are women," with women's jobs "concentrated more in

lower-skilled and lower-paid occupations" even as "half of all doctors on average across OECD countries in 2021 were female" [6]. "Nurses make up the most numerous category of health and social care workers in most OECD countries, accounting for approximately 20-25% of all workers," while "[p]ersonal care workers... also account for a relatively large share, sometimes exceeding the number of nurses" [6].

The workforce proved to be a survival variable, not just an input: "[d]uring the COVID-19 pandemic, higher national numbers of health and social care workers were significantly associated with lower mortality across OECD countries, based either on registered COVID-19 deaths or the broader measure of excess mortality" [6]. Yet the same period exposed the retention fragility underneath the growth. Nurses numbered "9.2 practicing nurses per 1 000 population on average across OECD member countries, up from 8.2 in 2011," but the density "ranged from fewer than 3 per 1 000 population in Colombia, Türkiye and Mexico to over 18 per 1 000 in Finland, Switzerland and Norway," with partner countries South Africa, India and Indonesia below 2 per 1,000 and China rising rapidly "from 1.7 in 2011 to 2.9 in 2021" [6]. Numbers even fell slightly in Latvia, the Slovak Republic and Sweden, and growth has often been bought by diluting skill mix — in Switzerland "the number of associate professional nurses increased almost three times faster than the number of professional nurses," lifting their share "from 30% in 2011 to 36% in 2021" [6]. Training pipelines alone do not solve it: "[i]t is not enough to train more nurses: there is also a need to retain nurses in the profession once they have completed their studies," and "the dropout rate continues to be high, especially among nurses working in long-term care," despite measures such as Norway's Competence Lift programmes [6]. The pandemic turned attrition into a headline risk: "[c]oncerns about a 'great resignation' of nurses and other health workers emerged in early 2021 in the United States, and a little later in 2021 and 2022 in the United Kingdom," where "the number of nurses who left the NHS in England reached an all-time high in 2021/22, and more than half of leavers were aged under 40" [6].

Migration fills part of the gap, shifting the shortage across borders: "[n]early half of all nurses in Ireland are foreign-trained, and over a quarter in New Zealand and Switzerland obtained their first nursing degree in another country" [6]. Pay dynamics complicate recruitment further: while real health-sector wages saw "large increases in most Central and Eastern European countries since 2011," they were "stagnant or declining" in Finland, Italy, Portugal, Spain and the United Kingdom, and "[h]igh inflation has eroded health sector wages recently in some countries, making it harder to attract and retain health professionals" [6].

The forward outlook is unambiguous. "Population ageing, technological change and rising incomes are expected to continue to boost demand for health workers in the coming years and decades," with US projections showing healthcare and social assistance "projected to create about 45% of all new jobs between 2022 and 2032" and Canada forecasting its largest employment increases in health between 2022 and 2031 [6]. Long-term care is the tightest segment: "growth in demand for long-term care workers over the next decade across OECD countries will be much higher than the actual recorded increase over the past decade," and "[f]urther efforts will be required to increase the attractiveness of the profession and recruitment and retention of workers in the long-term care sector to avoid a sharp increase in unmet needs and workforce shortages" [6]. The next sections ask who actually gets care from this workforce — coverage, affordability and the unmet needs that persist despite the spending.



1. Refer to all doctors licensed to practise, resulting in a large overestimation of the number of practising doctors (e.g. around 30% in Portugal). 2. Includes not only doctors providing direct care to patients but also those working in the health sector as managers, educators, researchers, etc. (adding another 5-10% of doctors). 3. Latest available data 2017.

Exhibit. Practising doctors per 1,000 population in OECD countries in 2011 and 2021 (or nearest year), showing how physician density has changed over the decade. Source: S0026.

Population coverage for healthcare: affordability, availability and use of services

Coverage has three dimensions: who is entitled, which services are included, and what share of the cost is paid collectively. "In addition to the share of the population entitled to core health services, the extent of healthcare coverage is defined by the range of services included in a publicly defined benefit package and the proportion of costs covered," with differences arising from services "included or excluded in the publicly defined benefit package," from "different cost-sharing arrangements," and from services "only being covered for specific population groups in a country (such as dental treatment)" [6]. Universal entitlement on paper can thus coexist with shallow protection in practice — the distinction that decides affordability for older people, who use services most.

The depth gradient runs from hospital care down to medicines and teeth. "On average across OECD countries, around three-quarters of all healthcare costs were covered by government or compulsory health insurance schemes in 2021… In nearly all OECD countries, inpatient services in hospitals were more comprehensively covered than any other type of care, with 90% of all costs borne by government or compulsory insurance schemes in 2021," reaching near 100 per cent in Sweden, Norway, Iceland and Estonia, but "only around two-thirds of total costs" in Australia, Greece and Korea [6]. Outpatient care follows: "[n]early four out of every five dollars spent (79%) on outpatient medical care in OECD countries were paid by government and compulsory insurance schemes," ranging "from under 60% in Portugal and Korea to over 90% in the Czech Republic, the Slovak Republic, Denmark and the United Kingdom" [6].

Below that, protection thins sharply. "Public coverage for the cost of dental care is far more limited across OECD countries due to restricted service packages (frequently limited to children) and higher levels of cost-sharing. On average, less than one-third of dental care costs were borne by government schemes or compulsory insurance," with more than half covered "[i]n only three OECD countries (Japan, Germany and France)" and "very low" compulsory coverage in Greece, Spain and Israel [6]. Pharmaceuticals sit between: "across OECD countries, 56% of pharmaceutical costs were financed by government or compulsory insurance schemes. The most generous coverage was found in France (83%), Ireland (82%) and Germany (82%)," while "[i]n Canada and Poland, over one-third of all pharmaceutical spending was financed via voluntary health insurance schemes, while in Chile, out-of-pocket payments financed nearly 80% of pharmaceutical spending" [6]. For older people managing multiple chronic conditions, these shallow zones — dental and medicines — are precisely where daily costs accumulate.

Coverage breadth showed its value under stress. "[C]ountries where the entire population had health coverage for a key set of health services experienced better health outcomes" during the pandemic, and governments improvised extensions where gaps existed: "[i]n Poland, for example, the National Health Fund covered uninsured as well as insured people for health services combatting COVID-19," and "in Ireland all COVID-19-related treatment, testing and remote GP assessments during the pandemic were available to all residents free of charge" [6].

Even so, entitlement does not guarantee receipt. "On average across 26 OECD countries with comparable data, only 2.3% of the population reported that they had unmet medical care needs due to cost, distance or waiting times in 2021," but "over 5% of the population reported unmet care needs in Estonia (8.1%) and Greece (6.4%), while in Germany, the Netherlands, Austria and the Czech Republic, fewer than 0.5% of the population reported unmet needs" [6]. The distribution is the policy problem: "[s]ocio-economic disparities are significant: people in the lowest income quintile were three times more likely to report unmet medical care needs than those in the highest quintile in 2021," a gradient present "in all analysed countries" and largest "in Greece, Latvia and Türkiye… with a difference of over 6 percentage points" [6]. The next section examines unmet needs in full — their composition by service, their reasons, and their trends since 2011.

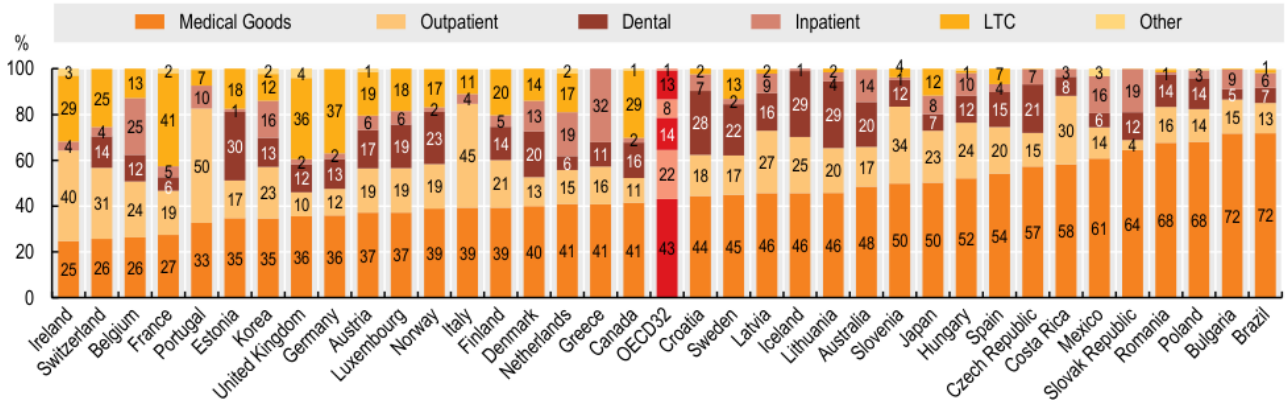


Exhibit. Composition of out-of-pocket spending on health by type of service in 2021 (or nearest year), showing the shares accounted for by medical goods, outpatient care, dental care, inpatient care, long-term care and other services. Source: S0026.

Unmet needs for healthcare

Unmet need is where coverage statistics meet lived experience: care that people judge they needed but did not get, because of cost, distance or waiting. Its level is low on average across the OECD but its distribution is anything but random — and its composition reveals exactly where the coverage architecture is thin. The clearest mirror is dental care, the least publicly covered major service. "Reported unmet needs are generally larger for dental care than for medical care… This reflects the fact that dental care is less well covered by public schemes than medical care in most OECD countries, so people often must pay out of pocket or purchase additional private health insurance" [6]. "[M]ore than 7% of people in Portugal, Latvia, Iceland and Greece reported unmet dental care needs in 2021, compared to fewer than 0.5% in the Netherlands, Germany and Austria," and "[i]n all analysed countries, the burden of unmet needs for dental care falls disproportionately on people on lower incomes. This was most evident in Portugal and Latvia, where more than 16% of people in the lowest income quintile reported forgoing needed dental care in 2021, compared to fewer than 2% in the highest quintile" [6]. Portugal's response — "creating dental health offices within public primary healthcare facilities" [6] — treats the gap as the coverage failure it is. For medical care, too, the burden concentrates: "[i]n Greece and Estonia, more than one in ten people in the lowest income quintile reported unmet medical care needs" [6].

The reasons people give are shifting from price to queue. "The main reason cited for unmet needs for medical care was typically waiting times, with 1.4% of people reporting this issue in 2021, on average across 26 OECD countries. In Estonia, Slovenia and Finland, more than 4% of the population cited waiting times as a barrier. Cost was also cited as an important barrier to access, and was the main reason for unmet needs in Greece, Iceland, Türkiye and Latvia, and accession country Romania. Distance to travel was also cited as a barrier, but less frequently than waiting times or cost" [6]. The trend since 2011 confirms the rotation: "[u]nmet medical care needs due to cost have generally fallen in most countries since 2011 (other than in Portugal, Luxembourg and Denmark). However, unmet medical care needs due to waiting times have often increased since 2011 – particularly in Slovenia, Estonia, Ireland and the Slovak Republic" [6]. Systems have been getting better at affordability and worse at timeliness — a trade that hits older patients hardest, since their needs are least postponable. Estonia's countermeasure is administrative: its national e-booking system "now includes a function where patients can select a treatment service and the system automatically searches for an appointment time that matches their preferences," helping government "track which health services have longer waiting lists and analyse the reasons" [6].

Nowhere is the queue more consequential than in elective surgery — the procedures that restore mobility to older bodies. The underlying demand is vast: osteoarthritis, the main indication for joint replacement, affects "about 528 million people" worldwide with symptomatic disease, "an increase of 113% since 1990," and "[a]ge is the strongest predictor of the development and progression of osteoarthritis" [6]; OECD averages run at "172 per 100 000 population for hip replacement, and 119 per 100 000 for knee replacement" [6]. Before the pandemic, waiting was already extreme in places: for cataract surgery, "over 60% of patients remained on the waiting list… for more than three months in Costa Rica, Norway and Finland," and "[t]he median number of days a person waited was almost a year in Poland (336 days), and over 100 days in Costa Rica, Slovenia and Ireland" [6]. For hip replacement, the share waiting over three months in 2019 "ranged from around 30% in Sweden and Italy, to almost 90% in Chile and over 70% in Costa Rica and Norway," with "[t]he median number of days a person waited was 663 days in Poland" [6]. COVID-19 then "massively exacerbated" the problem: cataract median waits "more than doubled in Costa Rica, Hungary, Spain

and Chile," hip waits "more than doubled in Chile and England (United Kingdom)," and although "[i]nitial data for 2022 indicate an improved situation in most countries," hip waiting times remained "generally still worse than in 2019, particularly in terms of median waiting times" [6]. The human meaning of the queue is stated plainly: "[b]y postponing the expected benefits of treatment, it means patients continue living with pain and disability for longer than they need to, and may worsen health outcomes for patients after the intervention" [6]. For a report on longer lives, that is the sharpest formulation of the access problem: years added, but spent waiting in pain for the procedures that make them livable. The next sections turn to two system responses — digital health, and prevention.

Digital health and digital health readiness

Digital health is the health system's most plausible lever against the workforce and waiting constraints documented in this chapter — and its least mature capability. The OECD defines the field broadly: digital health is "[t]he field of knowledge and practice associated with the development and use of health data and digital technologies to improve health," expanding eHealth "to include digital consumers, with a wider range of smart devices, connected equipment, and digital therapeutics," plus "the Internet of things, artificial intelligence, big data and robotics, and predictive and prescriptive analytics" [6]. "Digital health readiness" — the foundation for using that stack well — has four dimensions: "**analytic readiness** (for responsible analytics); **health data readiness** (for integrated health data); **technology readiness** (for robust technology); and **human factor readiness** (for capacity, co-operation, and oversight)," which "need to be designed to work together to optimise health outcomes while minimising harms" [6].

The starting point is backwardness, memorably quantified: "health systems continue to rely on fax machines, with 75% of global fax traffic used for medical services," and "it can take 17 years for published leading practice to become common practice," while "health providers express concern over their new digital burden while not receiving benefits from modern technologies" and "the public cannot engage meaningfully in their care without access to their own health records" [6]. Digital transformation "has been described as a determinant of health, as digital technologies, access, and literacy increasingly influence health, well-being and health transformations" [6] — meaning unreadiness is itself a health inequality, falling hardest on the older, poorer and more rural patients this report centres.

COVID-19 compressed a decade of adoption into two years. Digital health enabled countries to scale specific functions at speed: Canada, Latvia, Spain, the United Kingdom and the United States scaled "remote disease management and monitoring"; Costa Rica, the Czech Republic, Finland, Latvia, Spain and the United States improved "care co-ordination and integration"; Australia, Austria, the Czech Republic, Luxembourg and Spain improved "electronic prescribing" [6]. The infrastructure numbers moved correspondingly: "[i]n 2021, on average over 93% of primary care practices used EMRs, an increase from 70% in 2012… In 13 OECD countries, all primary care practices used EMRs, whereas in some countries such as Poland, Mexico, Switzerland and Japan, around 40% or fewer had EMR availability" [6]. Patient-side engagement shows what maturity looks like: "more than half of citizens in Finland reported personally accessing their EHRs regularly online in 2019. They interacted with their records to renew prescriptions, update consent, post living wills, and record organ donation testaments" [6]. But the substrate remains uneven: mobile is the dominant access route globally, subscriptions run at "almost two… for every individual" in Japan and Estonia but "less than one per person in 12 OECD countries" [6].

The returns case is strong — "[t]here is an opportunity for investments in digital strategy to generate potential returns of USD 3 for every USD 1 of investment," through "improved health outcomes, reduction of waste, and minimised duplication, while also supporting more resilient health systems" [6] — but so are the risks, and readiness means governing both. Artificial intelligence offers gains "from automating administrative processes to aiding health professionals in diagnosis, powering medical devices for improved treatment, virtually testing millions of antibiotics for superbugs," alongside "risks… including biased algorithms that exacerbate inequities, lack of clinical validation that risks patient safety, and potential for privacy breaches" [6]. Security is the binding constraint: "the cost of cyberattacks (across all industries) may reach USD 10.5 trillion by 2025," and "[h]ealth is a prime target for cybercrime given the sprawl of health technologies, the value of health data, and the risk of disruption in health services from technical outages" [6]. Uncoordinated private offerings add a structural risk: subscription care models that, "without suitable regulation, create data silos," leaving "health systems being unaware of inequities" [6]. For ageing societies, the stakes are concrete: digital tools are how scarce clinicians reach dispersed older patients, how chronic disease is managed at home, and how the data for every projection in this report get collected. The next section turns to the oldest digital-free intervention of all — prevention, beginning with vaccination.

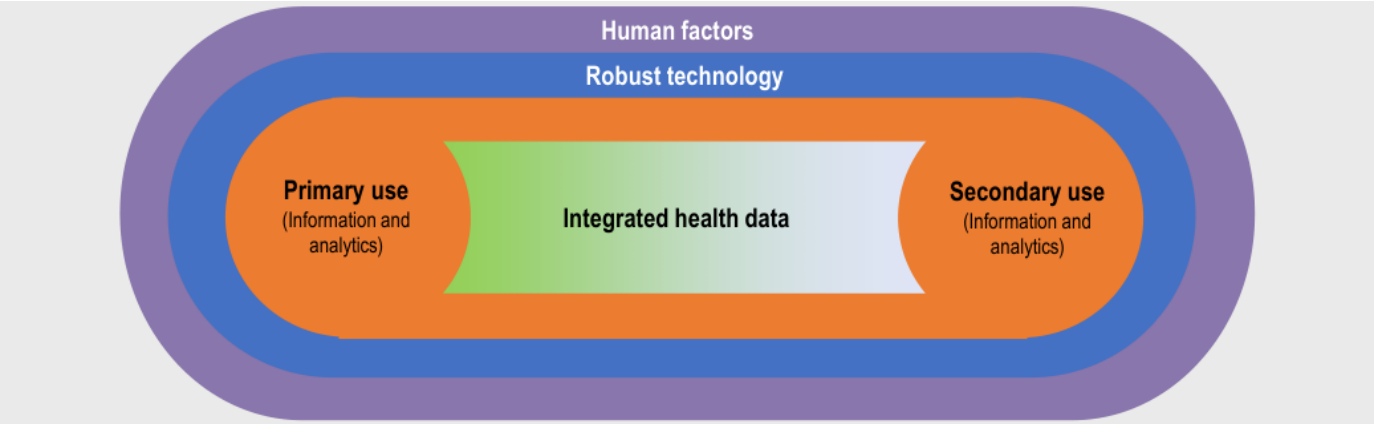


Exhibit. Figure 2.1. Integrated Digital Health Ecosystem Human factors Robust technology Primary use Secondary use (Information and Integrated health data (Information and analytics) analytics). Source: S0026.

Routine vaccinations and prevention in older age

Prevention is the cheapest capacity a health system can buy, and vaccination is its most cost-effective instrument: "[v]accines are a cost-effective tool for protecting against infectious diseases," and mass vaccination is "the most effective way to defeat infectious diseases such as diphtheria, tetanus and pertussis (DTP), measles, hepatitis B, and influenza" [6]. For older people the pivotal programme is influenza. "Influenza is a common infectious disease, responsible for 3-5 million severe cases worldwide each year, along with up to 650 000 deaths (WHO, 2019)," and "[s]ince older people are at greater risk of developing serious complications from influenza – such as pneumonia and sepsis, which can result in serious illness or death – WHO recommends that 75% of people aged 65 and over should be vaccinated against seasonal influenza" [6].

Performance against that target is poor. "The WHO target of 75% was only attained in the United Kingdom (81%), Korea (80%) and Denmark and Ireland (75%) in 2021, whereas the influenza immunisation rate among this vulnerable group was very low at below 15% in Latvia, Poland and the Slovak Republic" [6]. The direction, however, is encouraging — and unusual: "[u]nlike childhood immunisation, influenza vaccination rates for people aged 65 and over increased to 55% on average across OECD countries in 2021, from 48% in 2019, reflecting increasing public confidence in the flu vaccination in recent years," with "[t]he increase in vaccination rates… most notable in Denmark and Norway, with around a 20 percentage point increase since 2019," though rates fell by around 5 points in Costa Rica, Latvia, Korea and Mexico [6]. The pandemic, which disrupted so much else, appears to have taught older cohorts the value of the annual shot.

The childhood end of the schedule matters to this report because prevention is a life-course asset: healthy ageing "begins in childhood, when access to healthcare services from newborn screening to childhood vaccinations, and programs to provide adequate nutrition are essential to ensure that children are launched on a healthy trajectory for life" [9]. Childhood coverage remains high but is slipping at the margin: "around 93% of children on average receiving the recommended DTP and measles vaccinations in 2022," yet "nearly half of countries fall short of the minimum immunisation levels (95%) recommended by the World Health Organization (WHO) to prevent the spread of the disease," with measles coverage "particularly low in Poland (71%) and Estonia (78%)" and DTP below the 90 per cent WHO floor in almost one in five OECD countries, Mexico at 83 per cent [6]. The driver is confidence: children's rates "declined slightly (by less than 2 percentage points) between 2019 and 2022… [which] may reflect wavering public confidence in vaccination during the pandemic," with perceived importance of childhood vaccines down "by more than a third in Korea (reaching 48%) and Japan (54%) by 2022" compared with 2015, and EU confidence in MMR vaccination at 85 per cent in 2022, down from 88 per cent in 2020 [6]. The OECD's warning applies to both ends of life: "[h]igh national coverage rates… may not be sufficient to stop disease spreading if vaccination rates are uneven within the country, or if take-up is low in specific population groups. Hence, government action to garner trust and public confidence in the safety and efficacy of vaccination across all population groups is essential" [6].

Vaccination sits inside a broader prevention doctrine that the longevity literature states explicitly. The Stanford Center on Longevity's New Map of Life argues that breakthrough therapeutics alone will not deliver healthy ageing: "[t]he full promise of geroscience can only be realized… if this individualized approach to therapies is complemented by a much-expanded investment in population-level delivery of prevention and health promotion by public health and the public health system. Otherwise, the discoveries to come through geroscience will widen health disparities, excluding those who cannot pay" [9]. Its principle — "healthy aging requires investments in health at every

life stage" [9] — extends from childhood immunisation through midlife risk-factor management to older-age vaccination. The gap between the 75 per cent influenza target and the 55 per cent OECD average is thus best read as a down payment still owed on the compression of morbidity: every prevented case of influenza, pneumonia or sepsis in an 80-year-old is a year of healthy life the care system does not have to buy back at hospital prices. The next section turns to whether the care that is delivered achieves its outcomes — quality and outcomes of care.

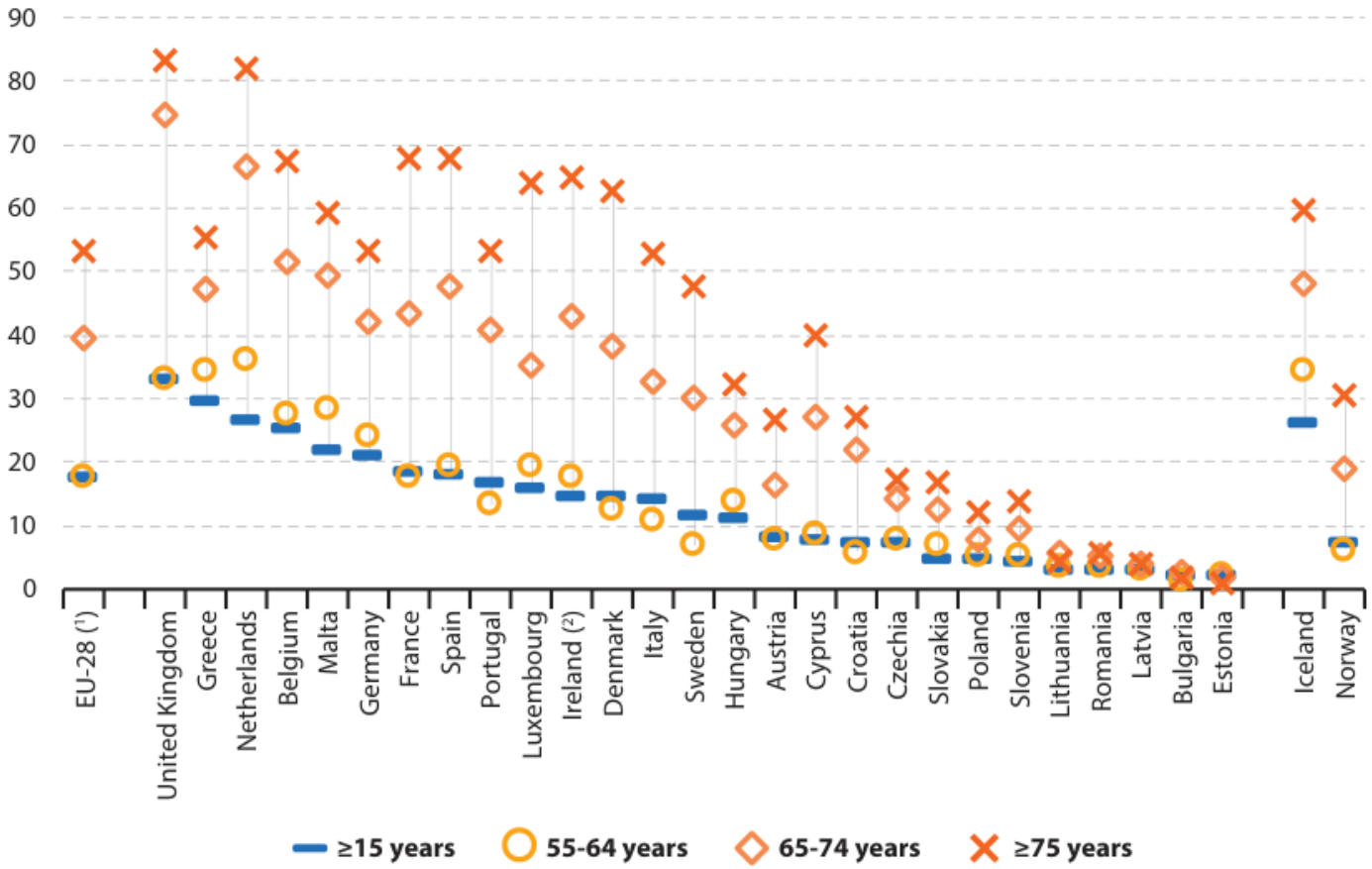


Exhibit. Self-reported vaccination against influenza by age class in 2014, comparing uptake among people aged 55–64, 65–74 and 75 and over with the population aged 15 and over across EU countries. Source: S0014.

Quality and outcomes of care

Spending, staffing and coverage measure what health systems deploy; quality indicators measure what patients get. The OECD's current assessment is one of genuine improvement shadowed by strain. On safety: "[p]atient safety indicators show encouraging results: for example, safe prescribing in primary care has improved in most countries over time, with reductions in the average volume of antibiotics, opioids and long-term prescriptions of anticoagulants. Still, patient safety remains a concern, with 57% of hospital physicians and nurses perceiving staff levels and work pace to be unsafe" [6]. The workforce shortages of the previous sections are not merely an access problem; the people delivering care judge their own conditions unsafe — a quality risk that compounds as demand rises.

On effectiveness, two trend lines matter for older patients. First, primary care is doing more of its preventive job: "[a]voidable hospital admissions have fallen in most OECD countries over the past decade, with large reductions observed in Lithuania, Mexico, Poland and the Slovak Republic. This is an indication that primary care is helping to keep people well and treating uncomplicated cases" [6] — admissions for asthma, COPD, heart failure and diabetes that good community care should prevent, falling as populations age. Second, acute care keeps getting better at its fundamental task: "[i]n almost every OECD country, 30-day mortality rates following a heart attack or stroke are lower than ten years ago. However, these mortality rates slightly increased between 2019 and 2021 on average, due to treatment delays during the pandemic" [6] — the same pandemic that lengthened the elective queues also cost lives in emergency care, a debt still being repaid.

The most important measurement shift is toward what patients themselves report. "A deeper understanding of quality of care requires measuring what matters to people. Patient-reported outcomes show, for example, average qual-

ity of life 6-12 months after hip surgery improved in all countries, reaching a score equivalent to 80% or higher, up from scores equivalent to 35-50% pre-surgery (based on the Oxford Hip Score)" [6]. For this report's readership the point is direct: the joint replacements whose waiting times were documented earlier are among the highest-value interventions in medicine when they happen — doubling measured quality of life. The queue and the outcome are two views of one system.

How care is organised is itself a quality intervention. Ambulatory (same-day) surgery has diffused unevenly: it "accounts for 90% or more of all cataract surgeries in around three-quarters of OECD countries with available data," with nearly all cataract surgery as day cases in several countries, "however, the rate is relatively low in Lithuania and Mexico, with fewer than 65% of surgeries performed as ambulatory cases" and under 50 per cent in Bulgaria and Romania [6]. The catch-up can be rapid where incentives align: "[i]n Austria, the share of cataract surgeries performed as day cases increased from only 46% in 2011 to 91% in 2021; in Hungary, it increased from 35% to 77%," and Denmark, France and the United Kingdom have used payment design — DRG adjustments and day-case incentives — to push the model [6]. Day surgery "[frees] up capacity within hospitals to focus on more complex cases or to reduce waiting lists," though the OECD cautions that lower unit costs "may be offset by overall growth in the volume of procedures performed" plus post-acute and community care costs [6].

Finally, quality increasingly depends on the data layer beneath care. "While 90% of OECD countries have an electronic health portal in place, only 42% reported that the public could both access and interact with all their data through the portal," and "[a]round one-third (38%) of countries have no clinical standards or vendor certification of electronic health record systems, limiting the interoperability of health data" [6]. A hundred-year life generates a century-long record; systems that cannot move that record with the patient will deliver fragmented care to exactly the multimorbid older patients who most need coordination. The health chapter closes with the policy frame that binds these findings — the strategies governments have adopted for ageing populations.

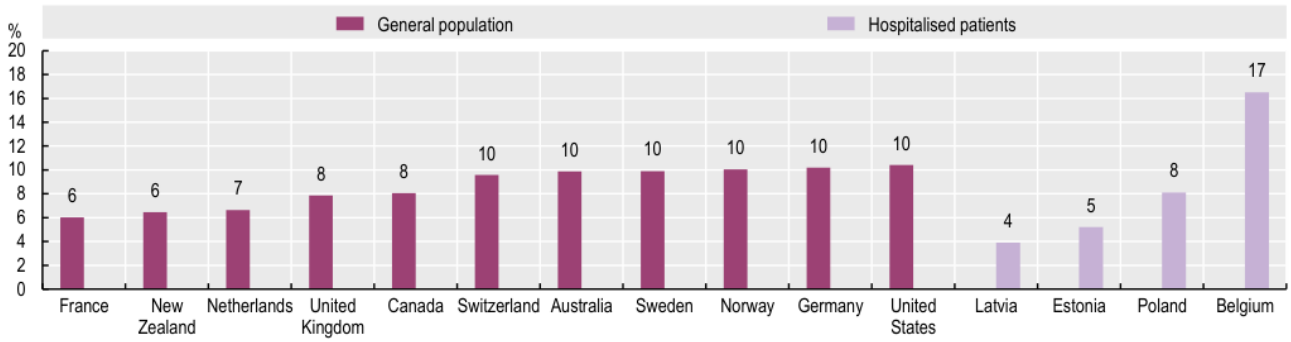


Exhibit. Share of patients reporting that a patient safety incident occurred during their treatment or care in 2020 (or nearest year), comparing the general population with hospitalised patients. Source: S0026.

Health policies and strategies for an ageing population

The policy architecture for ageing was built in layers over four decades. "At the First World Assembly on Ageing convened in Vienna, Austria, in 1982, population ageing was identified for the first time as a global issue with vital connections to development. The Second World Assembly on Ageing, held 20 years later in Madrid, concluded with the adoption of a Political Declaration and the Madrid International Plan of Action on Ageing, a landmark agreement in which Governments committed to 'building a society for all ages'" [1]. The Madrid Plan's three priorities — "older persons and development, advancing health and well-being into old age, and ensuring enabling and supportive environments" [1] — still organise the field. The 1994 Cairo Programme of Action "recognized population ageing as an irreversible trend and called on Governments... to take advantage of the associated opportunities by helping older people to maintain their economic and social independence, and by providing health care and support through family and other formal and informal safety nets," with specific protection for older women, "the majority of older persons globally" [1]. After the Millennium Development Goals "only marginally included consideration of older persons," the 2030 Agenda restored universality — "leaving no one behind... for all segments of society and at all ages" — and "[i]n December 2020, the United Nations General Assembly proclaimed 2021-2030 as the United Nations Decade of Healthy Ageing," with member states acknowledging ageing's "major impacts on health systems and many other aspects of societies, including labour and financial markets and services for education, housing, long-term care and social protection" [1].

Within that frame, the UN's strategy guidance is balanced between opportunity and protection. Early in the transition, countries hold "a window of opportunity to benefit from a demographic dividend… Expanding opportunities for decent work and investing in quality education and health-care services help maximize the potential benefits of the dividend and steer economic growth that is inclusive" [1]. But fiscal anxiety can corrupt the response: "[i]n most cases, concerns have emerged around the fiscal sustainability of programmes to respond to ageing. Unfortunately, some measures focused primarily on balancing the budget may also give short shrift to essential forms of social protection and, in doing so, intensify poverty and inequality among older people" [1] — including shifting investment risk onto individual pensioners, which "leaves individual beneficiaries bearing the uncertainties of investment performance, which may result in insufficient income later in life" [1]. The optimistic scenario is conditional: "future cohorts of working-age and older persons are likely to be healthier and better educated, and therefore more productive… Whether these potentially positive effects materialize depends on countries maintaining or increasing public investments in health care and education for all, including lifelong learning" [1].

The operational healthy-ageing agenda, as the OECD frames it, starts from an uncomfortable fact: "[a]ll OECD countries, despite the common endorsed principle of adequate access to health care for all people, have disparities in access to health services, and many people report not being able to meet health care needs, largely among low income groups," with cost the main reported reason, and with healthcare innovations that "are likely to have benefited groups with higher levels of education and income more" [3]. The prescriptions follow directly: "[c]ost effective prevention, primary care and screening services should thus be provided for low or no cost to prevent diseases (e.g. vaccination against influenza among elderly people) and encourage early detection of cancer and other diseases," and "[m]any OECD countries have introduced mechanisms to facilitate health care access for low-income patients through co-payment reductions or exemptions" [3]. Geography needs its own instruments — mobile facilities and telehealth such as "mobile diabetes counselling in Germany" and "the Humber Digital Health Community in the United Kingdom" — with the caveat that "telehealth should not be seen as a replacement for more traditional face-to-face consultations, particularly among older populations who may be less able to use new mobile health (mHealth) applications" [3].

The delivery model must change shape: "health care should be better integrated across various disciplines towards a patient-centred approach; physician and nurse specialisation in geriatric care should also be further developed," delivered by "interdisciplinary teams who can provide a seamless care between health and social care," requiring "changes in education and training to promote inter-disciplinary education" [3]. And prevention targeted at middle and older ages demonstrably narrows inequality: Australia's "'Living Longer Living Stronger' physical activity intervention improved health outcomes for adults ages 50 and over"; the US Senior Farmers' Market Nutrition Program "targets coupons at low-income, older adults encourages healthy eating, is cost-effective, and improves health outcomes"; Spain's PUMP programme "encourages physical activity by providing pedometers and a support network targeting the elderly and people with disabilities, including those in remote areas… [and] has been replicated in other countries" [3]. Health literacy completes the toolkit: "[i]ncreasing health literacy, particularly among lower educated and disadvantaged groups, should be a high priority to improve or at least prevent a worsening of health conditions" [3]. With health systems thus reframed, the report turns to the domain where ageing bites longest and least formally: long-term care and the people — paid and unpaid — who provide it.

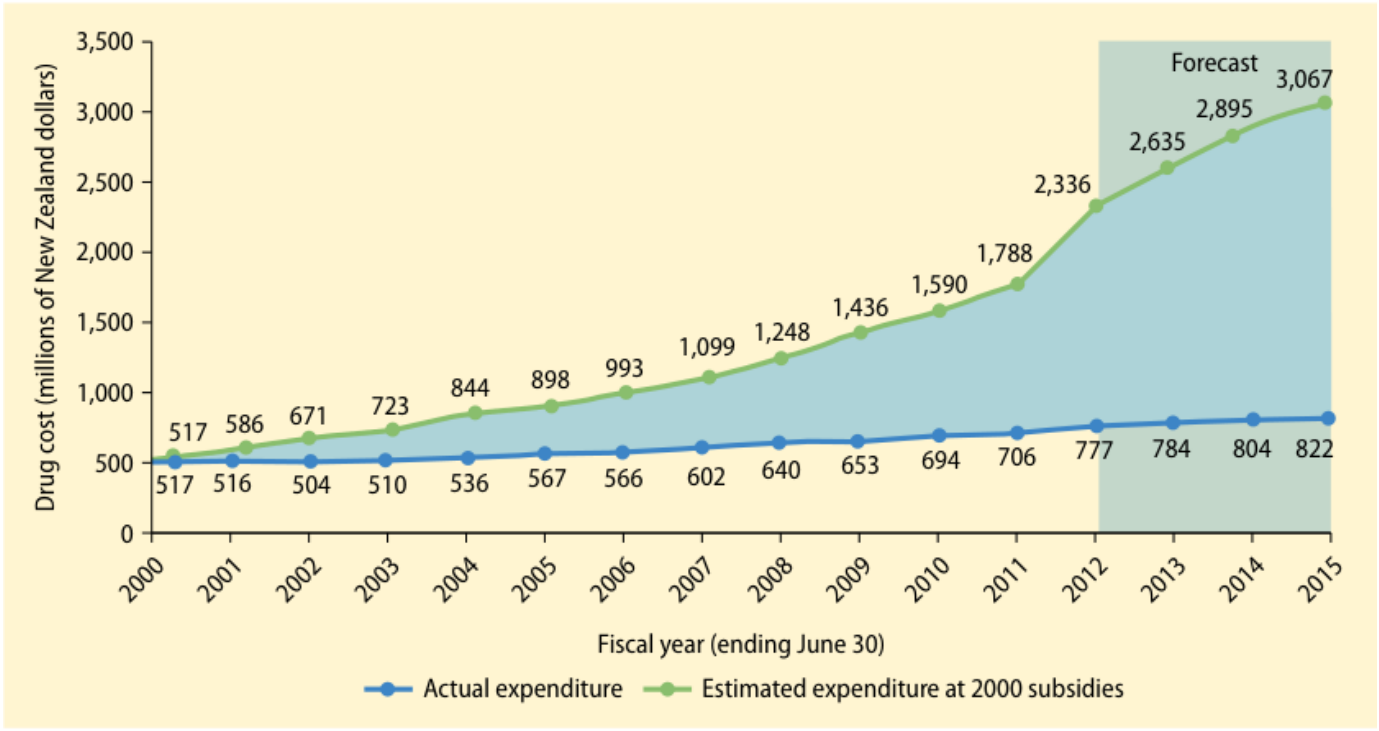


Exhibit. Actual pharmaceutical spending in New Zealand against spending projected at 2000 subsidy levels over 2000–2015, showing how PHARMAC's purchasing arrangements held drug expenditure below the projected path. Source: S0029.

Long-term care and caregiving

Ageing and long-term care systems

Long-term care is where population ageing stops being a statistic and becomes a daily human transaction — help with dressing, bathing, eating, mobility and medication, delivered for months or years. Yet it is the least built-out part of the welfare state. On spending, the world divides into tiers. "In 2019-2020, the Netherlands and Scandinavian countries (Denmark, Norway and Sweden) spent the most by far on long-term care, at around 2.5 to 3 per cent of GDP. Elevated spending mirrors the more developed formal long-term care systems in these countries. A second group of high-income countries, including Finland, France, Germany, Japan, Switzerland and the United Kingdom, allocate between 1 and 2 per cent of GDP to long-term care" [1]. Below that, provision thins quickly: "average public expenditure on long-term care is low, remaining below 1 per cent of GDP from 2006 to 2010… Little has changed over the past decade. Among OECD countries, average public expenditure for formal long-term care was just above 1 per cent of GDP in 2016 and 2020" [1]. Where formal systems are absent, "[p]eople with long-term care needs rely to a greater extent on unpaid family members" [1] — the default LTC system of most of humanity.

The governance gap is as striking as the spending gap. "[M]any countries across regions lack a specific policy, plan or strategic framework for long-term care. The situation is particularly critical in South-Eastern Asia where population ageing is occurring rapidly" [1]. The consequences are concrete: "[m]issing or inadequate regulations on long-term care and their enforcement can impair quality, put prospects for decent work at risk for care workers, and increase the vulnerability of older persons to abuse. A lack of legislation on labour standards, for instance, has left care workers without guarantees of minimum daily and weekly hours and little employment protection or assistance in case of unemployment… driv[ing] rising inequality in working conditions and therefore high turnover and low morale" [1]. And formal policy on paper is not delivery: "[e]ven in countries that do have care provision policies, poor implementation has often left caregivers with benefit packages that do not meet minimum requirements. Older persons, especially the most vulnerable, are stranded amid fragmented, confusing and inadequate care options (Scheil-Adlung, 2015)" [1].

The system's economics rest on undervaluation — the UN's verdict is in its heading: care work is "undervalued, underpaid, undertrained." The unpaid foundation is immense: "[t]he actual value of unpaid care work is massive, with 16.4 billion hours spent on it every day, equivalent to 2 billion people working eight hours per day with no remuneration," Argentina's economy ministry estimating unpaid care and domestic work at "15.9 per cent of GDP in 2020," and in the United States in 2015, "15.9 million family members and friends provided 18.1 billion hours of unpaid care

to people with Alzheimer's and other dementias, with an estimated economic value of $221.3 billion" [1]. Paid care carries a wage penalty: "[a]cross the 27 members of the European Union, non-residential long-term care workers make 80 per cent of the average national hourly wage," and "[c]aregivers tend to be underpaid even compared to other occupations where workers have similar skills, education and experience; this is referred to as the care penalty" [1].

The workforce is increasingly migrant and correspondingly vulnerable: "[m]igrant caregivers are more likely to hold short-term temporary visas… limiting their rights in the host country and amplifying their vulnerability to exploitative practices," "[f]oreign-born care workers in China, Taiwan, Province of China… only receive 86 per cent of the minimum wage of their native-born peers," and their departure leaves "a growing gap in care provision in their countries of origin" [1]. Training lags need: care requires skills from personal assistance to geriatrics, yet "[i]n OECD countries, about 70 per cent of formal care workers are personal care workers with no mandatory standard or minimum qualifications; the remaining 30 per cent are nurses with a minimum number of years of training," and family caregivers "typically have little care literacy" [1].

Support for the supporters is patchy. "Paid leave entitlements as well as flexible working arrangements for family or other informal caregivers… can protect the economic security and mental health of caregivers. Yet even among wealthier OECD countries, a third do not offer paid leave benefits to care for an ill family member (non-child)," qualifying relatives are "in most cases limited to partners/spouses, parents and sometimes siblings only," and "[t]emporary respite care… is not available in all countries, especially in developing ones" [1]. Unsurprisingly, "[m]any carers choose to remain in part-time jobs with lower salaries," caregiving brings extra costs from "medical supplies, assistive technologies and physical adaptations to homes," and the strains "can spark family tensions and caregiver-recipient confrontations, including elder abuse" [1]. The next section prices this economy of care in full — what caregiving costs, who bears it, and what societies get back.



Exhibit. Rising numbers of individuals taking on caring roles for ageing relatives and children, and the growing pressure this places on families, communities and public long-term care systems. Source: S0010.

Economics of caregiving and long-term care

Long-term care is the largest unfunded liability of the ageing transition, and most of it never appears in a budget. Globally, "average public expenditure on long-term care is low, remaining below 1 per cent of GDP from 2006 to 2010," and "[l]ittle has changed over the past decade. Among OECD countries, average public expenditure for formal long-term care was just above 1 per cent of GDP in 2016 and 2020" [1]. The spending league table measures system formality more than need: "[i]n 2019-2020, the Netherlands and Scandinavian countries (Denmark, Norway and Sweden) spent the most by far on long-term care, at around 2.5 to 3 per cent of GDP," a second group (Finland, France, Germany, Japan, Switzerland and the United Kingdom) allocates "between 1 and 2 per cent of GDP," while in parts of South-Eastern Europe and Latin America "formal provision of care is less comprehensive" and "[p]eople with long-term care needs rely to a greater extent on unpaid family members" [1].

The unpaid economy that fills the gap is taxed invisibly. "An individual's unpaid care responsibilities may limit their time for paid work. Many carers choose to remain in part-time jobs with lower salaries that are easier to combine with care responsibilities," and "[f]amily caregiving is also associated with additional expenditure, including purchasing medical supplies, assistive technologies and physical adaptations to homes" — strains that "can spark family tensions and caregiver-recipient confrontations, including elder abuse" [1]. Public support for these caregivers is patchy even where it exists: "[e]ven among wealthier OECD countries, a third do not offer paid leave benefits to care for an ill family member (non-child)," and among those that do, qualifying family members are "in most cases limited to partners/spouses, parents and sometimes siblings only," while "[t]emporary respite care… is not available in all countries, especially in developing ones" [1].

Above the household sits a policy vacuum. “Many countries across regions lack a specific policy, plan or strategic framework for long-term care,” a situation “particularly critical in South-Eastern Asia where population ageing is occurring rapidly”; “[m]issing or inadequate regulations on long-term care and their enforcement can impair quality, put prospects for decent work at risk for care workers, and increase the vulnerability of older persons to abuse,” with care workers often lacking “guarantees of minimum daily and weekly hours and little employment protection” — driving “high turnover and low morale” [1]. Even where frameworks exist, “poor implementation has often left caregivers with benefit packages that do not meet minimum requirements,” leaving older persons “stranded amid fragmented, confusing and inadequate care options” [1]. The economics of care, as this chapter assembles them, describe a three-way transfer that policy has barely begun to price: from families (overwhelmingly women) who supply the labour unpaid or underpaid, from care workers whose conditions reproduce the shortage, and from the future, as deferred system-building converts a plannable cost into a crisis expense.

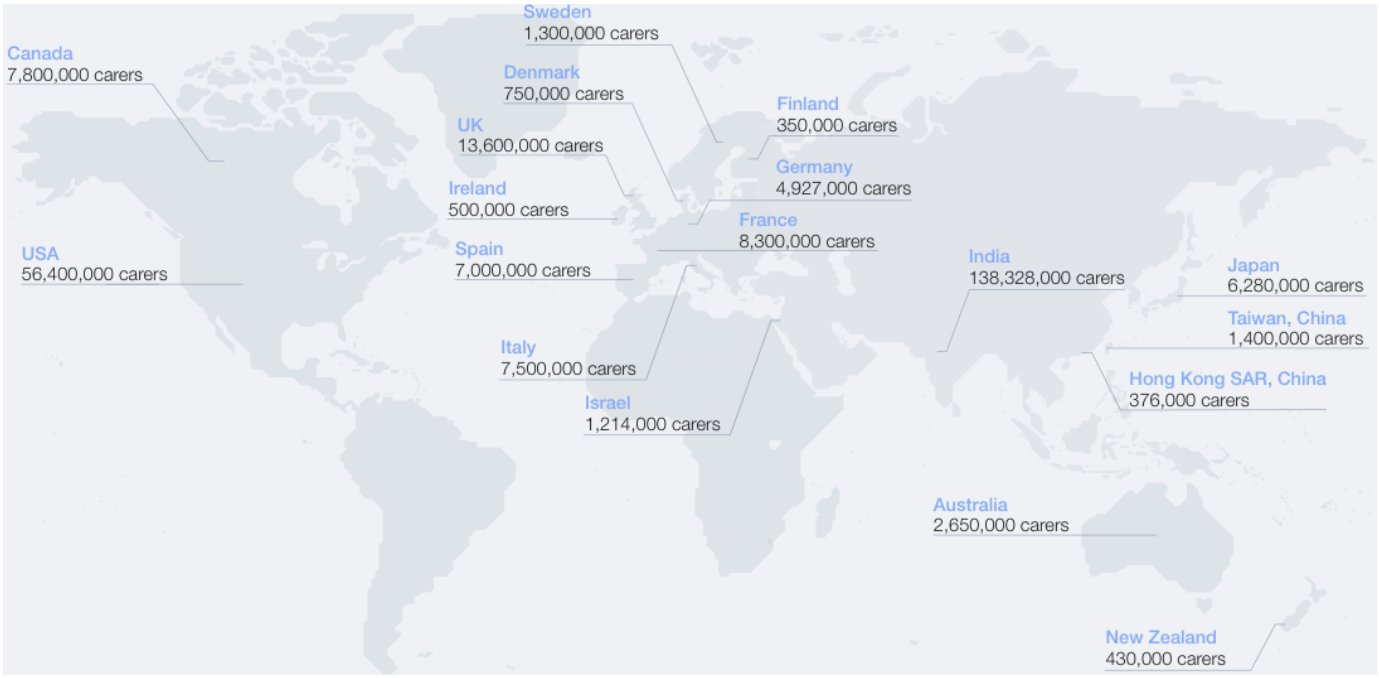


Exhibit. Estimated numbers of informal carers in selected countries around the world, ranging from hundreds of thousands in smaller economies to many millions in the largest, illustrating the scale of unpaid care provision. Source: S0010.

Universal social protection and access to long-term care

The rights-based case for universal social protection starts from the changing shape of disease. Among people over 50, the leading causes of disability are now “cancer, chronic kidney conditions, hearing impairment, dementia and falls,” and as non-communicable diseases rise with age, “[d]isability accounts for a greater share of the global disease burden” — “[p]eople tend to suffer less from early death but more from long-term conditions,” with multiple comorbidities concentrating in older adults and NCDs making them “particularly vulnerable to certain impacts of climate change, such as rising temperatures” [10]. Much of this burden is tractable early: “[a] large proportion of NCDs can be prevented or limited through early detection and appropriate management,” risk factors are mitigable through diet, exercise and avoiding smoking and drinking, “[p]hysical exercise reduces the risk of falls for older persons – falls that are often responsible for rapid degradation of health status, resulting in earlier dependency and hence increased needs for LTC” [10]. Prevention, however, is socially conditioned — the behaviours and monitoring involved depend on “addressing the factors likely to stand in the way” of healthy lives, “[a]nd it is in this area that social protection policies can, and should, make an important contribution” [10].

The institutional vehicle is the social protection floor. “The ILO Social Protection Floors Recommendation, 2012 (No. 202) calls for the urgent establishment of national social protection floors accessible to all and guaranteeing that people have effective access to health care without hardship and income security through a life-cycle approach” [10]. Universal social protection, grounded in the Universal Declaration of Human Rights, the International Covenant on Economic, Social, and Cultural Rights, Convention No. 102 (1952) and Recommendation No. 202, has three core dimensions: “[u]niversal coverage… All should have effective access to social protection throughout the life cycle, if and when needed”; “[c]omprehensive protection,” spanning health care and income security across the nine classic contingencies — “medical care… sickness, unemployment, old age, employment injury, family respons-

ibilities, maternity, invalidity and survivorship" — and explicitly extending to "new and emerging risks, such as needs related to LTC"; and "[a]dequate protection," benefits set at a level that "effectively prevents poverty, vulnerability and social exclusion" and allows people "to lead healthy and dignified lives" [10].

The significance for long-term care is categorical. Under this framework LTC is not a discretionary welfare add-on but a named social risk inside the universal floor — the same status old-age income itself acquired a century ago. That reframing answers the previous chapter's economics directly: where care needs are treated as an insurable life-cycle contingency, the costs stop falling by default on unpaid family carers and under-protected care workers, and access stops depending on whether a country's family structure happens to absorb them. The floor approach is also deliberately a floor — nationally defined guarantees, progressively realised — which makes it the one framework in this report addressed to the countries ageing fastest with the least fiscal room, not only to those already spending 3 percent of GDP on care.

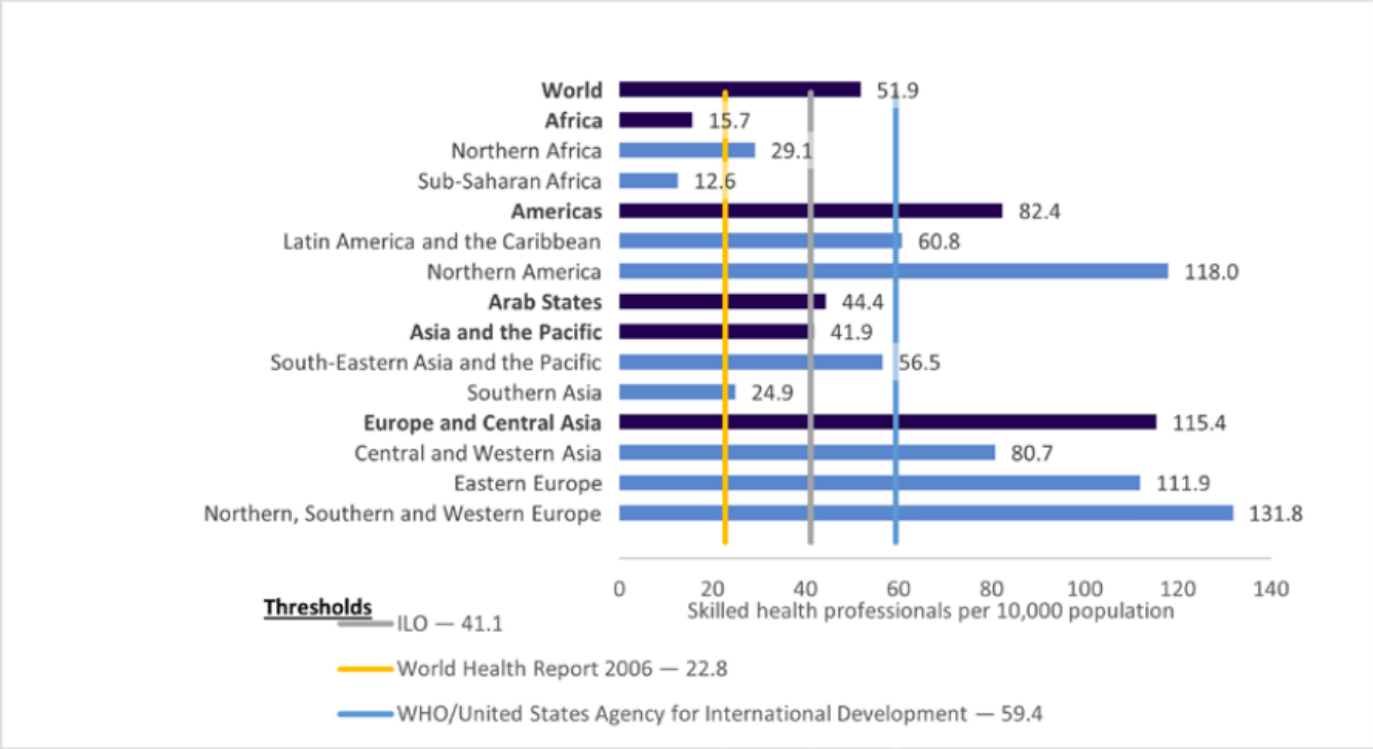


Exhibit. Skilled health staff density across world regions, benchmarked against three threshold levels relevant to delivering long-term care within universal social protection systems. Source: S0023.

Long-term care in ageing East Asia and Pacific

East Asia and the Pacific is where every thread of this chapter converges under maximum pressure: the fastest ageing in the world, family-care norms eroding under urbanisation and low fertility, fiscal space at its thinnest, and long-term care systems that are mostly still to be built. The World Bank's regional assessment frames LTC first as a fiscal risk: "[t]he biggest risk to aggregate savings rates in East Asia and Pacific will be through the impact on government saving caused by rising pension, health, and long-term care spending. This risk is of particular concern in developing East Asian and Pacific countries where projected fiscal space is very limited" [5]. The compounding problem is timing: "[t]he fiscal risks are compounded in many East Asian and Pacific countries because governments are pursuing rapid coverage expansion of health insurance and pensions and exploring more active public financing for long-term care" [5] — that is, the region must build the entitlement, the delivery system and the financing simultaneously, on the steepest demographic slope, rather than sequentially over generations as Europe did.

The pension projections in the same assessment indicate the scale of the competition for that fiscal space: IMF projections "point to a significant increase in pension spending in China and Malaysia, with more modest increases for Indonesia, the Philippines, and Thailand," with "[t]he increase in China of over 3 percent of GDP… particularly notable," while "Korea experiences a significant increase in pension spending approaching 4.5 percent in the 2010–30 period, reflecting the maturation of its relatively young pension system and the continuation of rapid aging," and Japan "exhibits a slight fall in pension spending as a share of GDP, reflecting its already aged population and cost-

containment measures" [5]. LTC spending must be layered on top of these trajectories — in most of the region, from a base of essentially zero formal provision.

Against that backdrop, the region's policy literature concentrates on the two social-insurance pioneers and the giant building from scratch. Japan's programme is treated explicitly as the transferable template — Campbell's assessment is titled "Japan's Long-Term Care Insurance Program as a Model for Middle-Income Nations," alongside the foundational accounts "Long-Term Care Insurance Comes to Japan" and "Japan's Radical Reform of Long-Term Care" [5]. Korea's parallel system is studied comparatively against Japan's — "The Present Situation and Problems of the Long-Term Care Insurance in South Korea: From Comparative Perspectives between South Korea and Japan" [5]. And China's challenge is framed in the literature as system construction under speed: "China's Rapidly Aging Population Creates Policy Challenges in Shaping a Viable Long-Term Care System," with institutional provision emerging market-first — "An Industry in the Making: The Emergence of Institutional Elder Care in Urban China" — and the state's problem defined as "Reining in Market Forces through Regulatory Oversight" [5]. The dedicated regional synthesis is Somanathan's World Bank background paper, "Long-Term Care in Aging East Asia and Pacific" [5].

The regional pattern that emerges is a three-way race. The red-group pioneers (Japan, Korea) prove social LTC insurance is feasible in East Asian fiscal and cultural conditions, and supply the design evidence — eligibility assessment, benefit packages, provider markets — for everyone behind them. The orange group, led by China, must build at continental scale while pension costs alone rise by over 3 per cent of GDP. The green group — Cambodia, Lao PDR, Myanmar, the Philippines and the Pacific islands — still has the demographic time the others have lost, but the least fiscal room to use it. For all three, the lesson of this chapter holds: care systems are cheapest to build before they are needed, and East Asia is the region where that window is closing fastest. The report now turns from care to the other great adjustment of longer lives — working longer.

Working longer: labour markets at older ages

Labour force projections and the participation rate of older workers

Whether longer lives translate into longer working lives is, in the official projections, a question about participation rates — and the machinery used to project them reveals how much of the future labour force is already locked into cohort behaviour. The EU's Ageing Report projections rest on the cohort simulation model: "[t]he cohort simulation model (CSM) as developed by the European Commission (DG ECFIN) has been used since the 2006 Ageing Report to project participation rates by gender and single age. The rationale for using the CSM is to reflect the substantial changes in labour market behaviour in recent decades across different cohorts and sexes. This methodology accounts, for instance, for the significant rise in female labour force participation in recent decades, as younger women, with a much stronger attachment to the labour force, gradually replace older women with relatively low participation rates" [4]. Mechanically, "[t]he central feature of the CSM is the estimation of labour market exit and entry rates for a 'synthetic' cohort… These probabilities are calculated on the basis of participation rates for the past ten years – 2013-2022 for the 2024 Ageing Report" [4], with the base year 2022 and the projection period 2023–2070 [4]. Two guards keep the machinery honest: a ten-year unweighted average so no single cyclical year dominates, and "[a] corrective mechanism for young cohorts (15-24)… in order to avoid that any increase in education enrolment rates (and the corresponding decline in participation rates) feeds into future declines of participation rates for prime-age workers" [4]. Hours are handled conservatively: "[a]lthough work rates and weekly hours worked are frozen per age group over the projection period, total hours worked changes due to compositional effects that mostly reflect the projected increase in female labour force participation, given the higher incidence of part-time work among women" [4].

Onto this behavioural baseline, pension reform is layered as the one deliberate policy lever: "[p]ension reforms are modelled through their estimated impact on the labour market exit rates of older workers (aged 51-74)… The estimation of the adjustment takes into account country-specific information about the relationship between retirement behaviour and the parameters of the pension system, as well as cross-country evidence. This framework for analysis is able to incorporate a broad typology of measures, inter alia, increases in the statutory or early retirement age, the convergence of lower female statutory retirement ages to that of men, the linking of the statutory retirement age to changes in life expectancy, and changes in financial incentives for early or delayed retirement" [4]. The legislated reforms captured in the 2024 round — with France, Slovakia and Sweden showing "the most significant new reforms since the 2021 Ageing Report" — include packages raising minimum early retirement ages to 63 with 42-year career requirements and statutory ages rising "from 65 years to 66 years in 2025 and to 67 years in 2030. A career of 45

years remains required for drawing a full pension" [4]. The lever also runs in reverse: one member state's "2017 reform... caps the increase in the statutory retirement age at 65, reversing the 2011 reform that increased retirement ages approximately in line with gains in life expectancy" [4] — a reminder that projected participation is a political variable, not a demographic destiny.

The headline result is that participation gains substantially offset population shrinkage at the employment margin: "Employment in the EU is projected to increase from around 75% of the working-age population in 2022 to around 79% in 2070. This change mostly reflects higher employment rates among older people and women in general, given higher participation among these groups and the assumption of declining unemployment over time" [4]. Three implications follow for the working-longer agenda. First, most of the projected gain is cohort replacement, not reform: tomorrow's 60-year-old workers will participate at higher rates mainly because today's 40-year-old women already do — the dividend arrives whether or not policy acts, but it is finite and largely exhausted by mid-century. Second, the residual lever is the exit margin at ages 51–74, which is precisely where pension parameters bite — retirement ages, career-length requirements and early-retirement penalties are labour-supply policy. Third, the projections assume away the intensive margin (hours frozen) and depend on unemployment declining as assumed; the binding constraints in practice are the subjects of the next sections — whether employers retain older talent, whether skills and health keep pace, and whether working conditions make longer careers choosable rather than merely required.

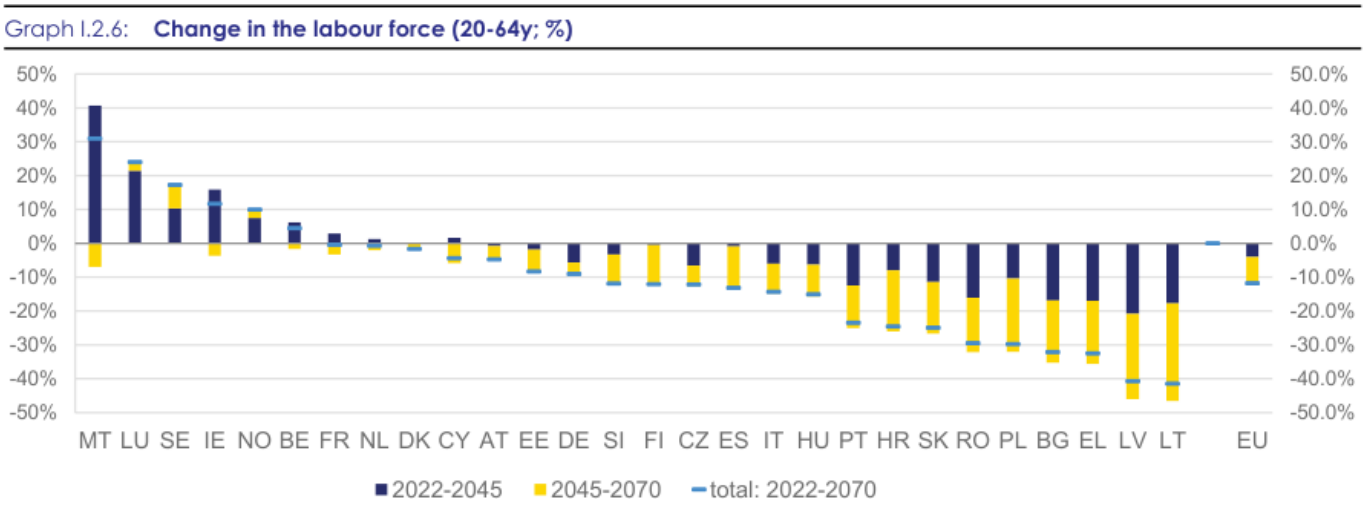


Exhibit. Projected change in the labour force aged 20–64, in percent, by Member State over the projection horizon, ranging from strong growth in countries such as Malta and Luxembourg to deep declines elsewhere. Source: S0015.

Retaining talent at all ages

The participation projections of the previous section describe what workers will do; this section is about what employers must do to keep them. The strategic context is scarcity: "[t]he share of 20-64 year-olds in the population is projected to fall from an average of 58% across OECD countries in 2021 to 53% in 2060, limiting the inflow of new talent into the labour market," while "[i]n 2019, prior to the pandemic, about 55% of employers in a survey of more than 40 000 employers across all industry sectors in 40 countries were reporting talent shortages, by 2022 this had risen to 75%" [11]. In a shrinking pipeline, retention stops being an HR metric and becomes the supply strategy itself — every experienced worker kept is a hire that does not have to be found.

The leakage pattern is age-specific. "Average job tenure fell by about 8% (nine months) in the past decade (for workers aged 15-64)," and "[l]ess than 50% of workers aged 55-59 will still be in the same job five years later. For workers aged 40-44, the share is more than 70%" [11]. But older workers rarely job-hop: only "4% of 55-64 year old workers change job every year compared with 8% of 30-54 year old workers," while "6% of 55-64 year old workers exit the labour market every year compared with 3% of 30-54 year old workers" [11]. The retention problem at older ages is therefore not poaching but exit — and exit is largely permanent, taking firm-specific experience with it, which "[d]amages the business talent pipeline and ability of firms to reap the benefits of a multigenerational workforce" [11].

Why do they leave? For early retirees the top reasons are proximity to retirement, health problems and caregiving; for job-switchers, low pay, feeling undervalued and lack of advancement [11]. Health is the largest involuntary channel: "[m]any people who leave their job voluntarily do so because of poor health," with ill-health accounting for 25%

of quits among workers aged 50–64 versus 20% among those aged 35–44 [11]. Pay operates with unusual force at the bottom: "[o]lder workers are 40% more likely to quit their job when working in low-paying firms," and across six countries "the quit rate is 50% higher in the 20% of firms paying the lowest wages compared with the 20% of firms paying the highest wages" — while "[m]ature workers aged (50-59) are much less likely to leave their firm for another one paying better wages compared to younger workers but are much more likely to exit the labour market altogether" [11]. Low pay does not redistribute older workers between firms; it ejects them from work entirely.

Flexibility is the second currency, and it is misallocated. Based on UK and Australian evidence, "the option to work at home on a regular basis is equivalent to just over 20% of their average annual salary" for women and 7% for men, while "having flexible start and finish times is worth in monetary terms about 48% of the average annual salary for women, and just under 30% for men" [11]. Yet "[e]vidence from the 2022 AARP Global Employee Survey in 12 countries shows that mature workers have much less flexibility in their jobs (such as being able to work from home or flexible hours) than younger workers" [11] — the group that values schedule control most, and needs it for health and caregiving, gets least of it. Management capability lags similarly: "[d]iscriminatory and negative attitudes towards older workers can damage the talent pipeline," yet "only 42% of employers offer their managers training on management practices for a multigenerational workforce" [11]. What works is known: "high performance work practices (HPWP) such as work flexibility and autonomy, benefits such as bonuses, and performance management have proved to promote better skills use within workplaces and retention" [11].

The proof case is Ford's Fiesta plant in Cologne. Closing an in-house plant affected 5,000 employees, "including 500, mainly older workers with acquired disabilities" who "could only carry out simple assembly jobs"; Ford and the IQPR institute formed "[a]n interdisciplinary integration team... which used software to match the demands of a job with the precise abilities of a particular worker," with ability profiles prepared by physicians with employee consent [11]. "Over 300 workers were re-integrated into production jobs which brought about huge economic advantages including saving recruitment costs of up to USD 9 million per year. Rates of sickness absence also fell from over 20% to about 7%," and the project shifted the firm "from a 'deficit-oriented approach', with the focus on what an employee can no longer do, to a 'resource-oriented approach'" [11]. The generalisable economics are favourable: "[a]ccommodation costs are often minimal, as it usually involves an increase in flexibility for employees rather than an increase in expenditure," and graded return-to-work lets employees come back sooner on lighter duties, preventing "the loss of skills and experience" [11]. Retention, in short, is mostly job design — which is why the next sections take up its components in turn: skills and training, then health, pay and working conditions.

Investing in workers' skills and training

Longer careers are only productive if skills keep pace, and the central failure of current systems is that training stops roughly when careers most need it to continue. "Workers who regularly maintain and upgrade their competencies throughout their working lives fare better in the labour market. Upskilling, reskilling, and a greater focus on lifelong learning will be especially critical to help older workers cope with rapidly evolving skill demands due to technological progress and globalisation. However, available data from OECD countries shows that older workers continue to receive less training than other population groups thereby compounding their disadvantage" [12]. The gap is large even where systems are strong: "[l]arge differences continue to persist even in some of the top performing OECD countries, including Germany Canada and France where they exceed 25 percentage points. They are the lowest in Israel, the Slovak Republic and the United States, where they are at or below 15 percentage points" [12] — younger adults (25–34) versus older adults (55–64) in job-related training, per the Survey of Adult Skills (PIAAC).

The causes form a self-reinforcing trap. "One key factor preventing older workers from closing the skill gap with younger employees lies in the fact that the employers usually do not see the benefits of investing in the training of their employees. Older people themselves may also see little value in training if they face poor prospects of remaining much longer in work. In addition, like other groups, lack of time is still the largest barrier, particularly so in Korea, Japan and Spain. In Japan, the inconvenience of training is also cited as a major barrier while cost appears to play a major role in the United States" [12]. Employers under-invest because expected tenure looks short; workers under-invest for the same reason; skills then obsolesce, tenure actually becomes short, and both sides' pessimism is confirmed. Digital change tightens the trap: "[t]he digital transformation is generating deep and rapid changes in the labour market. Many existing jobs will be overhauled and new ones will be created... Older workers may be potentially more exposed to the risk of skills obsolescence," compounded because "many older people exhibit lower levels of digital readiness than their children and grandchildren" [12]. Breaking the loop is precisely what the longer statutory working lives of the previous sections do from the other side: raise expected tenure, and training's payback period

lengthens for both parties — “longer working lives should strengthen the willingness of firms to train older workers” [12].

Policy attacks the three barriers — cost, motivation and information. On cost: “[t]o encourage employers to invest in training for older employees, some countries (e.g. Australia, Germany and the Netherlands) have embarked on initiatives to reduce the cost of training older workers relative to other employees,” and “[e]valuations of the WeGebAU programme in Germany, which targets to support the training of low-skilled and older workers show that such programmes can keep older workers at work for longer” [12]. On guidance: “[i]n mid-2019, Australia rolled out nationally the Career Transition Assistance program for job-seekers aged 45 and above, following a trial of the program in five regions. The program combines tailored career assistance and functional digital literacy training”; “[i]n the Netherlands, workers aged 45 and more can participate in subsidised career development guidance (Ontwikkeladvies)… Participants develop a personal development plan that describes the actions that will be taken to ensure employment until retirement age”; and “[i]n Korea, Job Hope Centres offer re-employment services for vulnerable individuals aged 40 and over, including counselling and guidance services… Almost 30 000 people benefited from this programme in 2017” [12]. Recognition completes the set: “many countries are deploying programmes that recognise existing skills through validation and certification to boost participation of older workers in training” [12] — certifying what experienced workers already know converts invisible competence into portable employability.

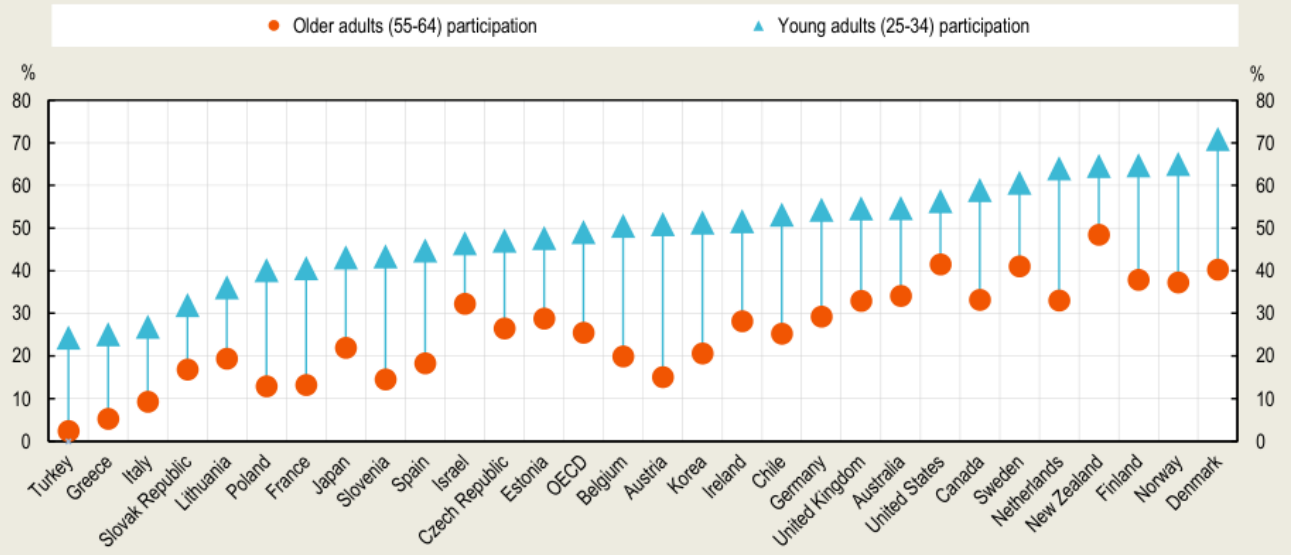
For those who do lose jobs late, training is also the re-entry mechanism — and design matters. “Job loss can be particularly damaging when it occurs at an older age. For older unemployed workers finding a job has remained difficult as demonstrated by their rising share in long-term unemployment. Recent evidence on the effectiveness of different types of active labour market programmes covering OECD and non-OECD countries suggests that a package of placement, training and counselling measures targeted at disadvantaged older workers may be more effective than wage subsidies alone,” with the OECD Faces of Joblessness studies reinforcing “the importance of tailoring the services offered to the circumstance of individual jobseekers” [12]. The same logic reaches the hardest cases: for workers in hazardous or arduous jobs, “[r]eskilling and upskilling aim to enable workers to remain productive and adapt to new job tasks… A professional-training and guidance framework has therefore to be put in place by governments and social partners to allow workers in hazardous or arduous jobs to make the needed extra effort into acquiring new skills, such that they can prolong their careers in different jobs. Not being able to continue working in one specific job should not be dealt with by early retirement” [13]. Skills policy is thus the hinge of the whole working-longer chapter: pension reform extends the horizon, retention keeps workers in firms, and training keeps them employable across the horizon — provided the next section’s conditions, health and pay, make staying worthwhile.

Did you know?

Longer working lives should strengthen the willingness of firms to train older workers. Yet, older adults across the OECD participate far less in training than their younger counterparts. Large differences continue to persist even in some of the top performing OECD countries, including Germany Canada and France where they exceed 25 percentage points. They are the lowest in Israel, the Slovak Republic and the United States, where they are at or below 15 percentage points.

Figure 5.2. Older adults participate in training far less than younger adults

Share of young adults and older adults in job-related training, 2011/12 or 2014/15



Note: Year 2014/15 for Chile, Greece, Israel, New Zealand, Slovenia and Turkey. OECD is an unweighted average of the countries in the chart.
Source: OECD calculations based on the Survey of Adult Skills (PIAAC) (2012), http://www.oecd.org/skills/piaac/.

StatLink https://doi.org/10.1787/888933991527

Exhibit. Shares of young adults aged 25–34 and older adults aged 55–64 participating in job-related training in 2011/12 or 2014/15, showing that older adults participate far less than younger adults. Source: S0019.

Improving employee health and stepping up pay and working conditions

Longer working lives are conditional. "By supporting longer working lives – where possible – and boosting the retention and productivity of older workers, governments can help ameliorate the possible negative effects of population ageing and longevity on economic growth and government budgets. However, working longer is not necessarily possible for all workers in the absence of improvements to job quality and health" [11]. The fiscal logic runs in both directions: premature exit is doubly expensive because "[o]lder people who are displaced or voluntarily leave the labour market often struggle to find jobs and face much larger earning losses relative to younger workers if they do re-enter employment… increased social welfare payments and reduced income tax receipts," and because "[t]he health status of older workers who leave the labour market prematurely often declines, with an increase in the incidence of depression, alcohol-related conditions and higher mortality" [11] — exit worsens the health that caused it, at public expense. With the OECD's 80+ population heading from 68 million in 2022 to 151 million by 2060 [11], the margin matters.

Health is the first condition. "Ill-health is a key driver of labour market turnover," and beyond exit itself, "health problems are also a key cause of sickness absence and presenteeism which if left untreated can have detrimental consequences for employees, employers and governments because of lost productivity and lower employee retention. On average, in 2019 the sickness absence rate was 1.3% among young workers, 2% among prime aged workers,

and 3.2% among mature workers" [11]. Workplace health programmes work only when serious: "[m]any large employers have put in place workplace health and wellbeing programmes but evidence suggests to be effective they must take a comprehensive and integrated approach to improve health and business outcomes," while for SMEs "governments and social partners are strengthening occupational health services, as in the United Kingdom, developing national accreditation for health and well-being providers, and by creating certified recognition programmes for employers, as in Japan," with subsidies for employer investment "also gained importance in recent years" [11].

Institutional design shapes incentives. "Most OECD countries have a combination of employer-paid sick leave and sickness benefits for workers on sickness absence, except Japan and the United States. Evidence from Switzerland and the Netherlands shows that stronger employer-paid sick leave can provide greater employer incentives to prevent ill-health, support rehabilitation and return-to-work" [11] — when firms bear the cost of absence, prevention becomes a business decision. "[G]radual return to work schemes following illness, as in Germany, can also be a win-win allowing workers to return sooner while preventing the loss of skills and experience," and "[m]aking accommodations for workers with health conditions, such as changes to the physical working environment, working time arrangements, or workplace responsibilities, can play an important role in allowing workers with health issues to continue to work" [11]. The gap is legal scope: "in most OECD countries, employers are obliged to make adjustments for workers with disabilities, but this does not usually extend to workers experiencing sickness, illness or injury" [11] — precisely the conditions that dominate older workers' exits. Prevention starts earlier still: "[p]olicies aimed at preventing avoidable long-term sickness need to start early in life, for example by preventing obesity, smoking and harmful use of alcohol" [11]. And the newest condition is digital: France, Greece, Italy, Luxembourg, the Slovak Republic and Spain "now have legislation in place giving workers the right to digitally disconnect" [11], a mental-health response to telework's always-on drift.

Pay is the second condition, and the policy question is whether to take the "high road." In "retail, hospitality, and other consumer facing services, the 'low road' practice of setting wages low and managing high turnover is all too common" [11] — a model whose turnover costs fall hardest on older workers, who exit rather than switch. The evidence on floors is reassuring: "minimum wage increases in many countries have raised the pay of workers at the bottom of the income distribution and have not substantially reduced employment or hours of work (Dube, 2019)," and "[m]inimum wages can also help reduce employee turnover… and reduce wage inequality" [11]. Calibration matters: "[o]n average across OECD countries minimum wages are set at around 50% of the median," but Dube's review argues "there is room for exploring a more ambitious minimum wage of between 60% to two-thirds of the median wage" [11]. Coverage is broad either way: "[c]urrently, 29 out of 38 OECD countries have statutory minimum wages. In the nine OECD countries without statutory minimum wages (Austria, Denmark, Finland, Iceland, Italy, Norway, Sweden, Switzerland and Costa Rica), a large part of the workforce is formally covered by wage floors specified in collective agreements" [11]. Bargaining adds what floors cannot: "[c]ollective agreements can often improve other aspects of job quality such as the design and implementation of occupational health programmes, management practices, intimidation and discrimination, and reskilling and training" [11], while performance-linked pay helps "motivate workers and increase their attachment to an organisation" [11]. Underpinning all of it is management quality: "[b]etter people management skills including career support, clear expectations about work tasks, coaching, consultation, a positive attitude and trust, have a substantial effect on staff retention" [11]. Health, pay and conditions are not welfare add-ons to the working-longer agenda; they are its feasibility constraints. The next section turns to the shape of the careers themselves — career length and the unemployment breaks that scar them.

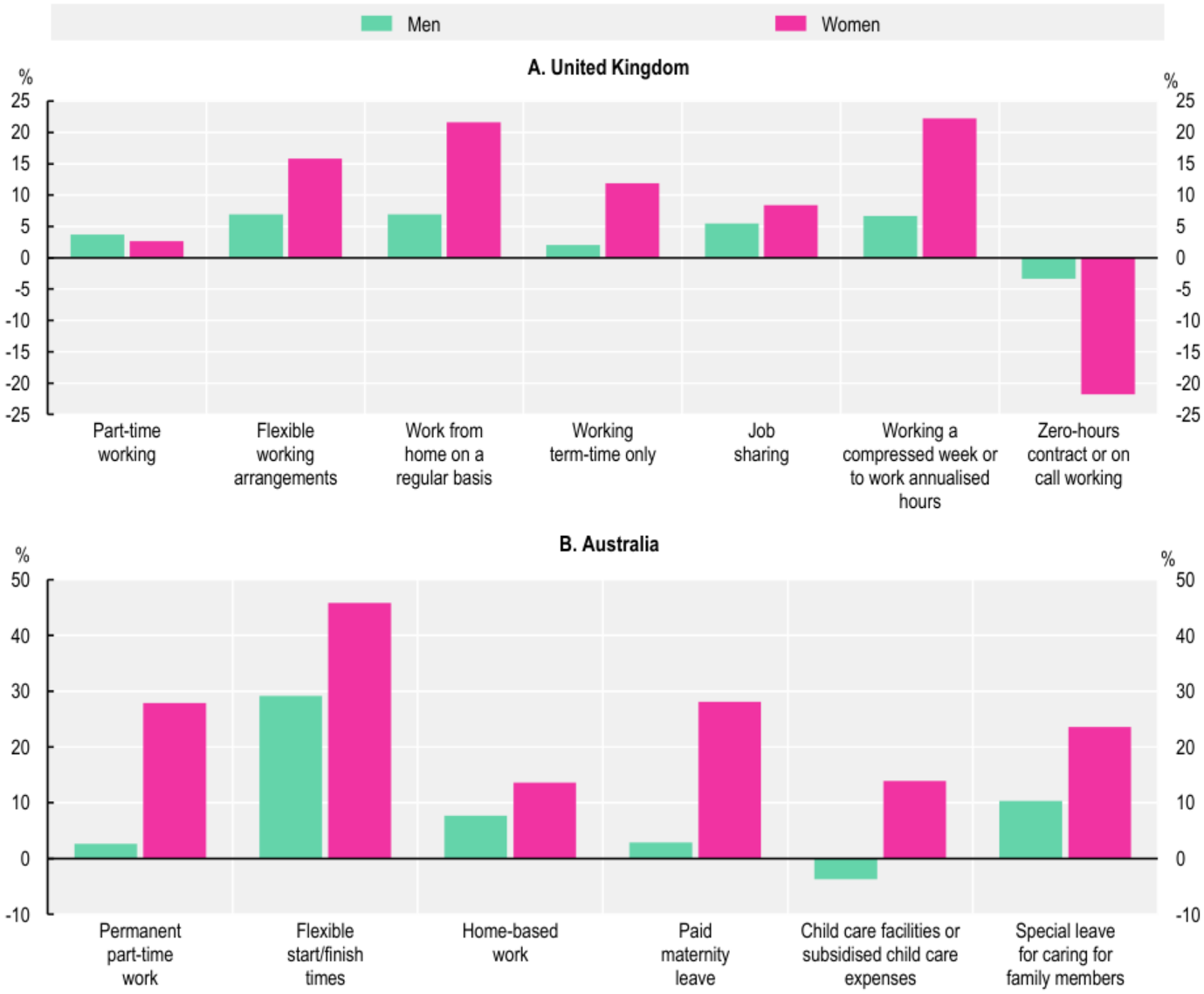


Exhibit. The value that men and women place on flexible working practices and certain family leave policies, expressed as a percentage of annual average income. Source: S0028.

Career length and the impact of unemployment breaks

Statutory retirement ages assume a career; actual careers are messier, and the pension consequences of the mess are large, unevenly distributed and only partly insured. "Working a full career is becoming increasingly uncertain in today's labour markets. Many countries currently experience very high levels of unemployment, particularly among the young. Workers with careers shortened by late entry or breaks will usually receive lower pensions, especially in systems where pension benefits and lifetime earnings are closely tied" [3]. The OECD's stress test combines the two commonest shocks: "pension entitlements are therefore calculated for male average earners who enter the labour market at 25 years old (rather than the standard 20) and spend ten years unemployed between the ages of 35 and 45," measured against a full career from age 20. "In the absence of any mechanism to offset late starts and career breaks, such an incomplete-career scenario would translate into a drop of about 35% in pension benefits. The OECD average projected fall based on legislated pension rules is 22% on average across countries... This means that redistributive/stabilisation devices would offset about 37% – that is (35-22)/35 – of the shortfall in this relatively extreme case" [3]. The late-start, broken-career worker retires on "on average, 78% of that of a 20-year-old starter with an unbroken career" [3].

System design decides who bears the loss. "Pensions in Ireland, New Zealand and the United Kingdom are unaffected by shorter careers as they are either residence-based basic in New Zealand, or contribution-based basic in Ireland and the United Kingdom. But these three countries are exceptions" [3] — flat basic pensions simply do not price career shape. At the other pole, "[t]he biggest drops – of 30% or more – would come in Hungary, Greece, Chile, Turkey, Mexico, Poland, Israel, Iceland and Italy" [3], tightly earnings-linked systems where every missing year is a missing accrual. Between the poles, design features mediate: "[i]n Australia, although there is a 15-year gap in

contributions to the defined contribution scheme, the Age Pension makes up for over two-thirds of the fall in pension levels, leading to a benefit equal to 92% of that of the full-career worker," while "the proportion is 70% in Israel, by contrast, where the pension system also has a strong DC component and no means-tested element" [3] — a means-tested floor under a DC pillar is the difference between a 8-point and a 30-point penalty. Not all breaks are priced alike, either: for childcare, "[a]verage-wage women with two children and taking five years out of the labour market to care for the children will have a pension equal to 95% of that for a full-career female worker" [13] — credits and care provisions protect the breaks societies have chosen to protect, leaving unemployment and late entry comparatively exposed.

Where systems do compensate, they often charge time instead of money. "Whilst the retirement age in most countries is relatively independent of contribution histories, workers who have gaps in their records need to work to a later age than full-career workers in six countries – France, Germany, Greece, Luxembourg, Slovenia and Spain. For example, late-start workers with unemployment gaps in their careers need to work two years longer in Germany and Spain, four in France, and five in Greece, Luxembourg and Slovenia to avoid being penalised for early withdrawal" [3]; in France the trade buys back most of the loss — "a latecomer with an unemployment history also qualifies for around 90% of a full-career worker's pension by working four years longer at older ages" [3]. The working-longer agenda thus cuts both ways for broken careers: later retirement ages raise the bar for everyone, but contribution-linked systems quietly demand the most extra years from those whose careers were interrupted — often involuntarily, and often by the same labour-market failures (youth unemployment, long-term joblessness) that policy elsewhere tries to cure.

This is the context for the OECD's 2015 Council Recommendation on Ageing and Employment, whose agenda spans the whole career: pension systems that "encourage and reward later retirement in line with increased life expectancy," restriction of "publicly funded early-retirement schemes which encourage workers to leave employment while they are still in good health," action to "[a]ddress age discrimination in recruitment, promotion, and retaining workers," pay-setting reviewed "to reflect productivity and competences, not age," and — directly on this section's theme — measures to "[l]imit the impact of job loss by providing effective employment assistance via a package of counselling and training measures targeted at those most at risk of long-term joblessness" [12]. The conditions agenda matters here too: "more than one in four [older workers] experience job strain as measured by the difference between job demands and job resources," and responses range from Denmark's "obligatory psychosocial risk assessment of working practices" to Sweden's mandate for "special measures to prevent forced early workforce exit due to heavy workload or inappropriate working postures in the health sector" [12]. Careers, in short, are pension policy written over forty years: every unemployment break is a future benefit cut unless the system deliberately decides otherwise. The next section examines the sharpest case — pension provisions for workers whose jobs are hazardous or arduous.

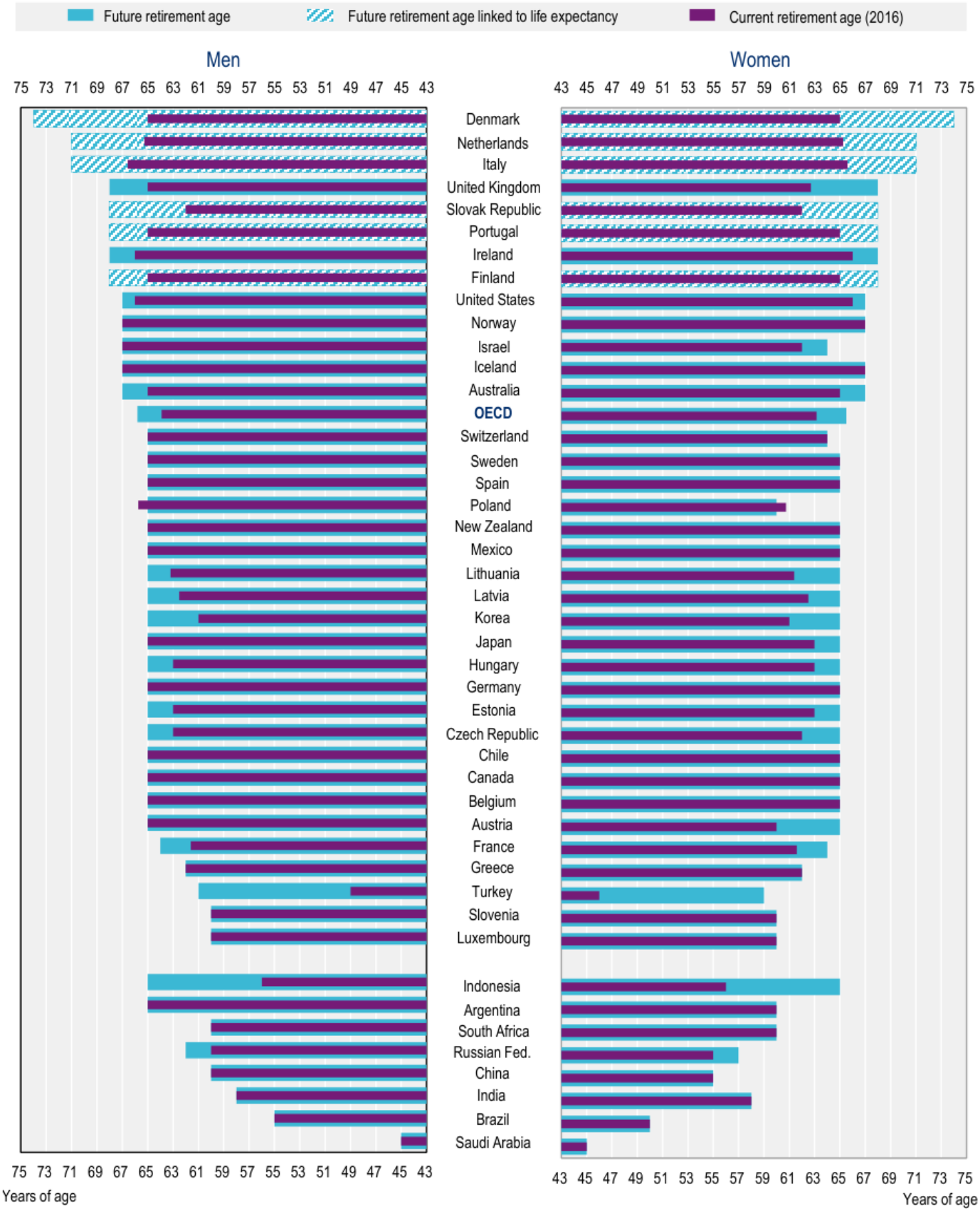


Exhibit. Current (2016) and future statutory retirement ages for men and women who entered or will enter the labour market at age 20, distinguishing countries whose future retirement age is linked to life expectancy. Source: S0019.

Pension provisions for workers in hazardous or arduous jobs

Every pension system carries exceptions for work that wears bodies out — miners, firefighters, dancers, seafarers — and the OECD's assessment of them is strikingly sceptical: the exceptions mostly treat the wrong problem, in the wrong institution, for the wrong people. "Hazardousness or arduousness of jobs thus raise first and foremost an issue for labour market policies. That is, the policy priority should be, when possible, to improve working conditions

for such jobs. Long-term exposure to related risk factors, such as physical strain, noise or uncommon working-time patterns, is likely to generate some delayed negative health consequences. Improving working conditions can be achieved through health and safety regulations to limit exposure to risky factors as well as by encouraging or incentivising social partners to take measures to limit hardship and health risks" [13]. The second instrument is information: "[c]ommunicating about the risks – backed by evidence… helps workers weigh different job opportunities and ask to be compensated for the risks to be taken, which in turn contributes to internalising the negative impact of those jobs, mainly by raising their cost and therefore the price of related goods and services" [13]. Where workers lack real choice and compensation fails, "policy efforts should focus on overcoming the labour market limitations that prevent the compensation mechanism from working" — not on pension carve-outs [13].

The factual basis for special treatment is real: occupational mortality gradients are large and persistent. "Higher mortality rates for manual workers are found for ages 20-59 in Finland, Japan, the United Kingdom and for ages 50-75 in Denmark, England, Finland, France and Sweden," and "[d]ifferences in mortality rates after age 30 among men result in workers in professional occupations living about 5 years more than manual workers in Italy" [13]. At retirement ages the gaps persist: "[t]he difference in life expectancy at age 65 between high-earnings and low-earnings (which are mainly low-skilled) occupations has been estimated between 2 and 5 years in Germany in 1998, France in early 2000s and in the United States in 1997-2014," and a 29-study European review concludes differences at 65 of "about 2-3 years in Germany, Norway, Sweden and almost 5 years in France" [13]. Within a single employer the gradient is stark: among French hospital workers, "life expectancy at age 65 of middle-level employees is shorter by 2.3 years for men and 1.4 years for women compared to executives while for blue-collar workers these differences increase to 3.7 and 2.2 years, respectively" [13]. Occupation itself contributes independently: Finnish evidence shows "factors related to occupation itself have an independent effect on the incidence of disability and mortality" beyond education and income, with physically strenuous work and chemical exposure driving the pattern [13].

Yet the schemes built on this evidence mis-target. "Remaining life expectancy of low-skilled and manual workers after retirement is relatively low, but workers covered by special pension schemes for hazardous or arduous work often do not belong to this group of workers" [13] — the shortest-lived are the low-skilled generally, those who never worked, and people with disabilities, while special schemes typically cover defined, often organised occupations. And the remedy itself is wasteful: "[t]he permanent withdrawal from the labour market in countries where special pension schemes exist, sometimes at very early ages, is an inefficient solution to address the decline in the ability to perform some specific tasks at older ages. Indeed, people in this case can most often successfully perform other jobs. In some countries, ballet dancers are covered by special pension provisions (but not sport professionals) which represents an extreme case in this context" [13]. Three further defects compound: "if too broadly defined, early retirement provisions for hazardous or arduous work provide obvious disincentives to continue working for people who can actually work"; "[o]ccupation-specific pension provisions can also reduce job mobility if changing jobs results in losing some pension entitlements"; and "special pension provisions, when publicly financed, lower the incentives by firms to reduce the incidence of hazardous or arduous work" [13] — the subsidy protects the hazard.

The disability system is an imperfect backstop: sickness and disability benefits "do not compensate for the health impacts that may become fully visible only after having retired," and "[e]ligibility to higher benefits provided by disability or injury insurance, requires establishing a causal link between work and the loss of work capacity. Even when existing, these causal links are often difficult to prove, in particular when prolonged exposure to hazardous or arduous conditions result in delayed diseases" [13]. The constructive alternative is transition, not exit — reskilling frameworks that let damaged workers prolong careers in different jobs — and it has working models: "Finland's skill development system is one of the most successful in the OECD… Public employment services purchase training, which is free for individual, from education providers and companies"; "Austria introduced a co-ordinated programme to enable adults to obtain basic competences and educational qualifications free of charge in 2012. Between 2012 and 2017 approximately 50 000 individuals participated"; and the Netherlands supported older unemployed with "mandatory free-of-charge jobsearch training, and the introduction of training vouchers" [13]. The conclusion closes the working-longer chapter with a discipline the pensions chapter will need: occupational harm is real, measurable in years of life — and is an argument for safer jobs, priced risks and mid-career transitions, with early retirement reserved as the exception it was meant to be.

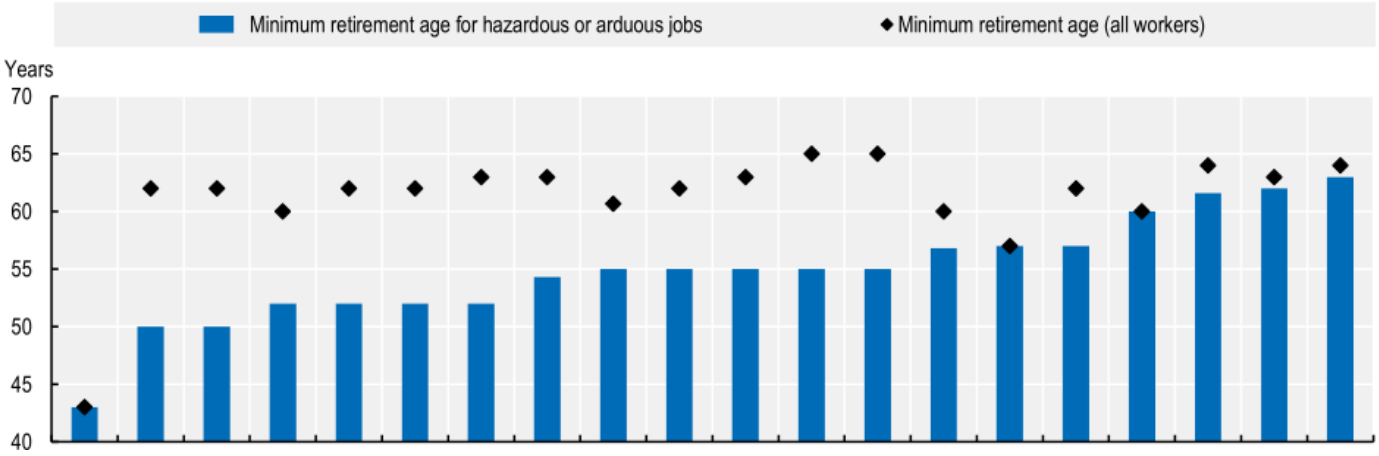


Exhibit. Minimum retirement ages in 2022 for workers in hazardous or arduous jobs compared with the minimum retirement age for all workers, excluding ballet dancers and artists. Source: S0002.

Labour market policies to address ageing

The working-longer agenda assembles into a single framework in the OECD's Council Recommendation on Ageing and Employment, and this closing section states it plainly. The precondition is that pension and benefit levers alone do not suffice: "[t]here are a number of policy levers to structure retirement pensions and other out-of-work benefits in a way that encourages and rewards later retirement. However, if working longer is to be an attractive and rewarding proposition for older workers, action on both the demand side and the supply side will need to be taken. Indeed, an increasing number of countries are also focusing on encouraging employers to retain and hire older workers. This includes: extending or abolishing mandatory retirement ages; fighting age discrimination; promoting further training over the whole working life; encouraging age-management approaches by employers; strengthening employment services for older job-seekers; and improving the health of older workers through good workplace practices throughout working careers" [12]. Each item maps to a section of this chapter: mandatory retirement and discrimination to retention, training to skills, age management and health to conditions, employment services to the broken-career problem.

The Recommendation organises the agenda in three areas: "i) strengthening **incentives** for workers to continue working at an older age; ii) ensuring there are good job **opportunities** for older workers by encouraging employers to retain and hire them; and iii) promoting the **employability** of workers throughout their working lives for better work choices and opportunities at an older age" [12]. To make the framework measurable, the OECD built composite indicators for each dimension: "[i]ncentives for older people to continue working at an older age are captured by focusing on i) a combination of current retirement ages (men), ii) the gross replacement rates and iii) the impact on benefits when working and deferring pensions. Job opportunities for older workers the focus are captured by i) retention rates and ii) the full-time earnings ratio 55-64/25-54. Finally, the employability of older workers is proxied by several indicators, including i) training, ii) job strain and iii) life expectancy at 65" [12].

The framework is not merely taxonomic — it predicts outcomes. "Higher country scores in each of these three areas are associated with better labour market outcomes for older people as summarised by either the employment rate of 55-64 year olds or the effective age of retirement... Stronger incentives to keep working such as higher retirement ages or less generous pension benefits are closely correlated with both higher employment rates and later exit from the labour force. But employer behaviour also matters as shown by the positive correlation with the opportunities index. Lastly, there is also a close association between a higher employability index and better outcomes of older workers" [12]. Hence the conclusion: "countries which provide better incentives, better opportunities and invest more in employability of workers via training and good quality jobs do better in promoting employment for older workers. For example, the Nordics such as Iceland, Sweden, Norway and Denmark as well as New Zealand and Canada, are among the best performing countries across the three policy dimensions and in terms of employment outcomes for older people. In contrast, Turkey, Greece and Luxembourg have relatively low employment rates for older people, reflecting relatively low retirement ages, weak incentives for working longer, low retention rates" [12] and weaker performance across the other dimensions.

Read against this chapter's evidence, the three pillars are complements, not substitutes. Incentives without opportunities produce the low-pay ejection documented earlier — workers priced out of firms rather than retained by them. Opportunities without employability produce the training gap's obsolescence trap. Employability without in-

centives produces skilled workers who retire on schedule anyway because the pension system pays them to. The hazardous-work analysis adds the boundary condition: where jobs themselves destroy health, the labour-market instruments — safer work, priced risk, mid-career transition — must do the work, since pension exceptions both miss the worst-off and subsidise the hazard. And the projections chapter supplies the stakes: cohort replacement will deliver much of the participation gain automatically, but the residual — the difference between the Nordics' outcomes and the laggards' — is exactly what these policies determine. With the labour-supply side established, the report turns to the systems that must finance the longer lives themselves: the architecture and design of national pension systems.

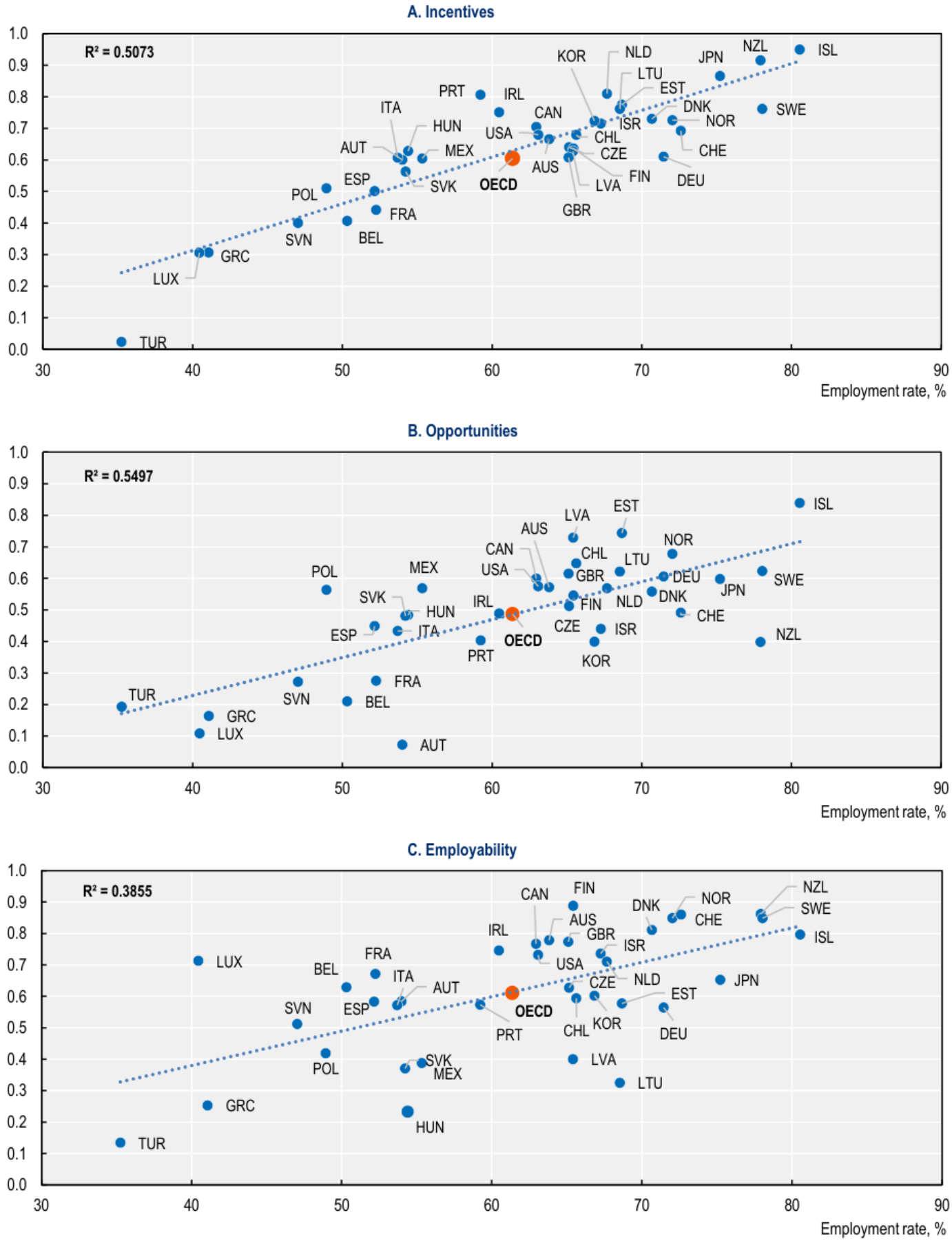


Exhibit. Further policy reforms that can aid higher employment of older workers, summarising where additional reform effort would most raise older workers' employment. Source: S0019.

Pensions: financing a hundred-year life

Architecture and design of national pension systems

Pension systems look bewilderingly diverse, but the OECD's taxonomy reduces them to a common skeleton. "Retirement-income regimes are diverse and often involve a number of different programmes. The taxonomy of pensions used here consists of two mandatory 'tiers'; the first generates retirement income independent of past earnings level with the second covering earnings-related components. Voluntary provision, be it personal or employer-provided, comprises the third tier" [13]. The division of labour is functional: "[t]he first tier comprises programmes offering the first layer of social protection in old age, and for which past earnings are irrelevant in the calculation of retirement income. Such schemes often target some minimum standard of living in retirement. Mandatory earnings-related components (second tier) contribute to smoothing consumption, and therefore standards of living, between working life and retirement" [13]. Everything in this chapter — and much of the inequality analysis later — follows from how countries weight protection against smoothing.

The first tier has three instruments. **Basic pensions** pay flat amounts tied to residence or contribution years, not earnings: "[e]ight OECD countries have a residence-based basic pension for future retirees while Norway is replacing it with a targeted scheme that involves a means test. Nine OECD countries feature a contribution-based basic pension" [13]. **Targeted plans** means-test the floor: "[i]n these plans, the value of the benefit depends on income from other sources and possibly also assets. Hence, poorer pensioners receive higher benefits than better-off retirees" — with entitlement reaching full-career workers on very low earnings (30 per cent of average) in eight OECD countries [13]. **Minimum contributory pensions** guarantee a floor inside the earnings-related system itself: they "are currently found in 18 OECD countries, with Chile and Italy phasing it out for future retirees," and work either as "a minimum for total lifetime entitlements, which may increase in level once the length of the contribution period exceeds certain thresholds," or as "minimum pension credits that calculate year-by-year entitlements of low earners based on a higher earnings level" [13]. The broken-career evidence of the previous chapter is written in this taxonomy: residence-based basics ignore career shape entirely; targeted floors catch falls; minimum credits soften them; pure earnings-related tiers transmit them one-for-one.

The second tier has three designs. **Defined benefit (DB)** remains the modal form: "public pay-as-you-go schemes follow a general defined benefit format in 20 OECD countries, with pension's dependent on the number of years of contributions, accrual rates and individual pensionable earnings. In another eight countries, DB schemes apply to current retirees but have been or will be closed to new workers" [13] — a generational transition visible in the rules themselves. Private occupational DB is "[m]andatory or quasi-mandatory in two OECD countries – Switzerland and the Netherlands, respectively – however, in the Netherlands, they are being replaced by defined contribution (DC) pensions from 2028 at the latest" [13]. **Points schemes** operate "[i]n five OECD countries: French occupational plans managed by social partners under public supervision and the Estonian, German, Lithuanian and Slovak public schemes. Workers earn pension points based on their earnings. At retirement, the sum pension points is multiplied by the pension-point value to convert them into a regular pension payment" [13] — a design whose adjustment lever (the point value) is collective and visible.

Defined contribution splits by funding. "Funded defined contribution (FDC) plans are compulsory for future retirees in 14 OECD countries. In these schemes, contributions flow into an individual account. The accumulation of contributions and investment returns is usually converted into a monthly pension at retirement," with Denmark, Iceland, the Netherlands, Sweden and the United Kingdom also running "quasi-mandatory, occupational FDC schemes" alongside public provision [13]. "The notional defined contribution (NDC) schemes are at the core of the pension system in five OECD countries (Italy, Latvia, Norway, Poland and Sweden)... These are pay-as-you-go public schemes with individual accounts that apply a notional rate of return to contributions made, mimicking FDC plans. The accounts are 'notional' in that the balances exist only on the books of the managing institution. At retirement, the accumulated notional capital is converted into a monthly pension using a formula based on life expectancy or mortality rates" [13]. NDC is the architectural answer to ageing inside PAYG: the life-expectancy conversion makes longevity risk self-adjusting rather than a recurring fiscal crisis. Finally, the exceptions prove the taxonomy: "[o]nly Ireland and New Zealand in the OECD do not have second-tier pensions" [13] — pure first-tier systems, which is precisely why their broken-career penalties were zero. The sections that follow stress-test this architecture: first against inflation, then against the reform wave of the past decade.

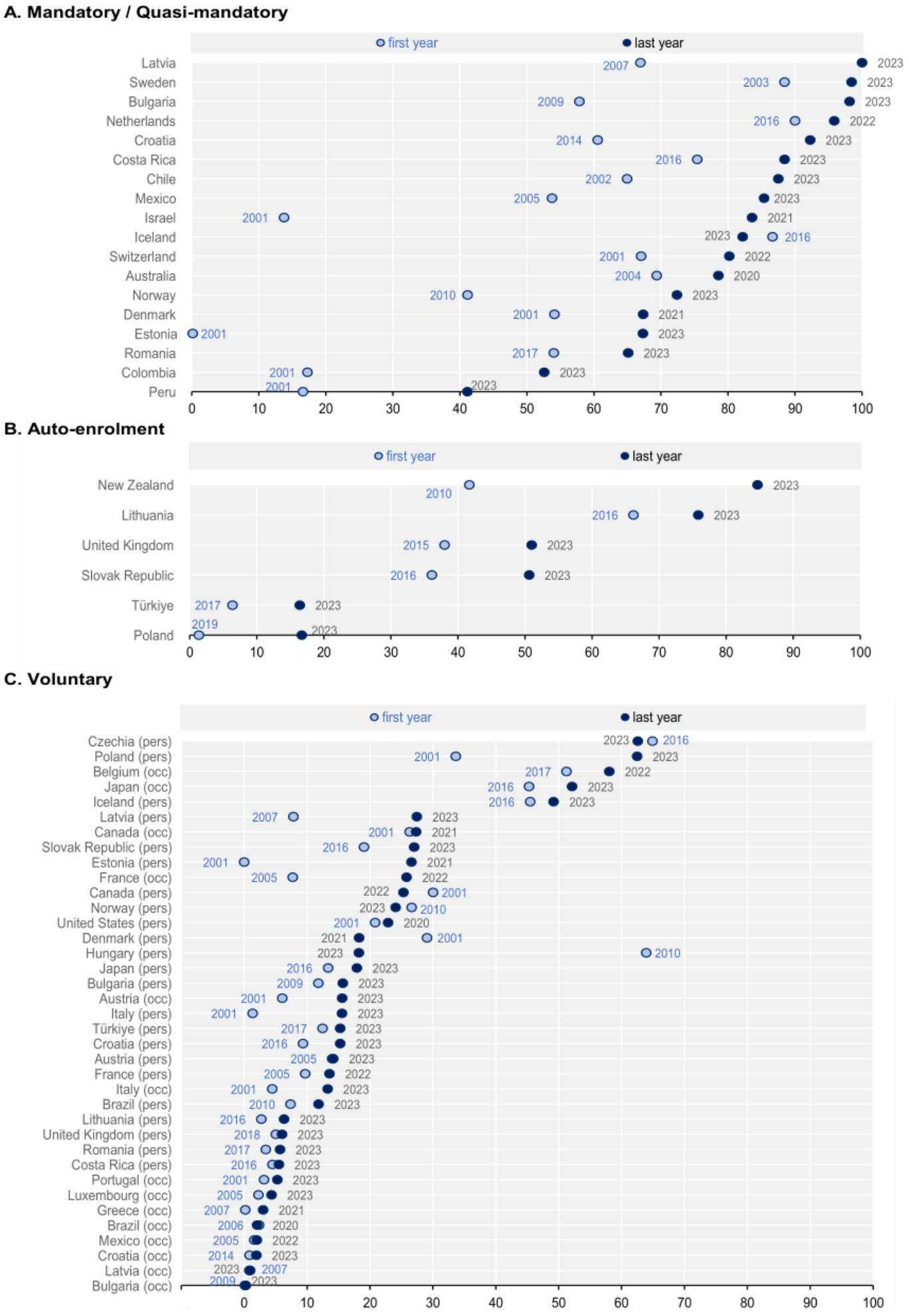


Exhibit. Proportion of the working-age population participating in asset-backed pension arrangements, as a percentage of the working-age population, across countries. Source: S0013.

Inflation and pensions

The inflation surge of 2021–23 was the first full stress test of pension indexation in a generation, and most systems failed it on timing rather than intent. “Bringing together both inflation trends and pension adjustments, the real value of pensions was lower in January 2023 compared to January 2022 in most OECD countries” [13]. The OECD adds a methodological warning that is itself a finding: “[t]hese results should be interpreted with care as the precise indexation timing plays a key role in the outcome shown in the chart for many countries… In a given country, the graph would show no change in the real value of pensions if indexation to inflation in the previous year were applied in January whereas it would show a significant loss if the same indexation rule is only applied in February” [13]. An annual, backward-looking adjustment guarantees a period of real loss during any surge; the only design questions are how long and how deep.

The losses sorted by tier and rule. “Earnings-related pensions in payment lost more than 5% of their real value in Austria, Costa Rica, Estonia, Poland and Sweden. Among these countries, Sweden does not index to prices at all and pensions are only partially adjusted to prices in Costa Rica and Estonia (20%). In Austria and Poland, there is some delay in pensions adjusting to price increases as they all apply a smoothed price indicator comparing inflation between two 12-month periods” [13] — smoothing, designed to damp volatility, systematically lags a surge. Targeted benefits fared worst where adjustment was rarest: they “were more than 5% lower in real terms in Costa Rica, Estonia, Latvia, Poland and the United Kingdom,” with “Latvia’s targeted pension lost around 18% and Poland’s 14% of their real value due to high inflation combined with an indexation rule that only requires adjustments once every three years so that no indexation took place during this period” [13]. Both governments broke their own cycles: “Latvia increased targeted benefits in nominal terms by 14.7% in July 2023 and by 9.6% in January 2024, and has moved to annual indexation from 2024… in Poland, targeted benefits are scheduled to increase by 39% in January 2024” [13]. Basic pensions indexed to wages lost where wages lagged prices: “[b]asic pensions were more than 5% lower in real terms in Czechia, Denmark, Estonia and the United Kingdom… Basic pensions in Czechia, Denmark and Estonia are adjusted to wages or the wage bill,” while “[t]he loss of value of the basic pension in the United Kingdom is related to the suspension of the triple lock indexing pensions to the highest of inflation, wage growth or 2.5%” [13].

The gainers are instructive. “In Türkiye, the real value of earnings-related and targeted pensions increased by around 17% over this period due to deviations from the indexation rule”; “[i]n Lithuania the steep increase is the consequence of the importance of food prices in indexation of targeted benefits”; Spain combined “the re-introduction of price indexation on contributory benefits at the end of 2021” with non-contributory benefits “increased by 6.5% above inflation in January 2023”; and “[i]n Mexico, the basic pension increased by about 16% in real terms… as part of a wider reform to bolster old-age income” [13]. Germany illustrates rule-bound symmetry: it “does not apply negative indexation, [so] a catch-up factor offsets non-implemented negative indexation by lowering the positive indexation in the following years. That factor was suspended in 2018… The catch-up factor was reinstated in July 2022, hence regular indexation was reduced by 1.17 percentage points to compensate for the lack of negative indexation in 2021” [13]. Around the edges, governments improvised: Australia “temporarily loosened (between July 2022 and June 2024) the means test to qualify for the targeted pension benefit for people with investment income”; “Hungary moved forward the introduction of its 13th-month pension payment, reaching the full amount in 2022 instead of 2024”; ad hoc payments went to pensioners in eight countries, “in Denmark and Greece ad hoc payments specifically targeted low-income pensioners” [13].

Underneath sits the structural dilemma. “The surge in inflation raises the competing objectives between sound pension finances and retirement income adequacy with some unusual acuity… Upholding pensioners’ purchasing power requires adjusting pensions to increasing prices. But, in the recent period, wages, and therefore most pension schemes’ revenues, have not increased at the same pace as the negative terms-of-trade shock… if benefits are adjusted in line with price inflation, this generates deficits in pension finances” [13]. Hence the design lessons the OECD draws: index quickly — “threshold indexation to prices with a low threshold such as Belgium’s 2% is very appealing to achieve this goal” — while recognising that “projection-based indexation rules… have not been effective at protecting pensioners’ purchasing power during the inflation surge,” and that “[r]educed indexation of higher pensions as is often done in Austria and Italy can be an effective strategy to control pension expenditures while fully protecting the purchasing power of low-income pensioners who spend a large share of their incomes on essential goods such as heating and food” [13]. Pensioner-specific price indices offer no escape: in France a retiree basket “would have had a very limited impact, estimated at a total of -0.3% cumulated over 1998-2015,” and the US CPI-E, after outpacing the working-age index by 0.4 points a year over 1983–2002, showed “virtually no difference between 2002-21” [13]. The verdict of the surge: price indexation works, but only if it is frequent, prompt and — for adequacy

— defended most fiercely at the bottom of the distribution. The next section turns from the inflation episode to the reform wave it interrupted: recent pension reforms across the OECD and the EU.

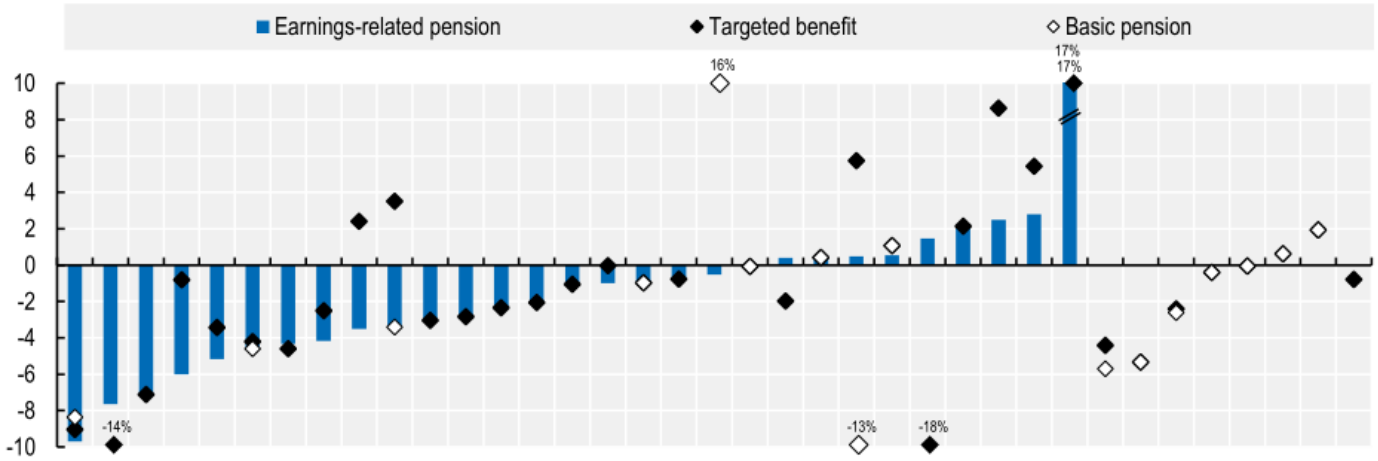


Exhibit. Percentage change in the real value of pensions in payment between January 2022 and January 2023, after CPI inflation and pension indexation, for earnings-related pensions, targeted benefits and basic pensions. Source: S0002.

Recent pension reforms and pension reforms in EU Member States

Pension reform is where the projections of this report become politics. Eurostat's survey of the field groups the EU's initiatives "under two main headings: on the one hand, safeguarding the sustainability of pension systems and more generally public finances; on the other, labour market reforms that are designed to keep older people at work, thereby reducing the number of early exits from the workforce" [14]. The 2024 Ageing Report's catalogue of legislated measures with direct effects on older workers' exit behaviour — the situation as of 30 September 2023 — shows both headings in motion, with "the countries with the most significant new reforms since the 2021 Ageing Report are France, Slovakia and Sweden" [4].

The sustainability measures work the parameters this report has tracked throughout: retirement ages, career-length requirements and early-exit penalties. One member state's 2015 reform "raised the minimum early retirement age to 63 years and the minimum required career length to 42 years... The reform also raised the statutory retirement age in the three main public old-age pension schemes (wage earners, self-employed and civil servants), from 65 years to 66 years in 2025 and to 67 years in 2030. A career of 45 years remains required for drawing a full pension" [4]. Another phases its statutory age "to 65 years following the 2015 reform: by 2029 for men and by 2037 for women (63 years in 2029). After 2037, the statutory retirement age is supposed to increase in line with gains in life expectancy, though no clear rule has been legislated" [4] — an intention without a mechanism, and therefore excluded from the projections. A third tightens through career length: "[t]he required career length for workers in normal work conditions to qualify for retirement is rising by two months annually, until it reaches 40 years for men and 37 years for women in 2027," with early retirement inside the final year priced at "0.4% for each month of anticipation" [4]. Reform also runs backward: in one member state, "[i]n 2017, a reform was adopted that caps the increase in the statutory retirement age at 65, reversing the 2011 reform that increased retirement ages approximately in line with gains in life expectancy... As of 2036, it will be 65 years for all" [4], in a system where women's statutory ages vary with the number of children raised. And eligibility can tighten through insurance periods: "[e]arly retirement is possible 3 year prior to the statutory retirement age, with a minimum insurance period of 40 years (2023 reform, 35 years before)" [4].

The constraint on all of this is consent, and the survey evidence shows a specific asymmetry. Europeans are anxious: asked in late 2016 "about their concern over not having sufficient income in old-age, adult respondents across the EU-28 had an average score of 5.8 — on a scale from 1 (not worried) to 10 (extremely worried). Greeks expressed the highest level of concern (8.0), followed by the inhabitants of Latvia and Portugal (both 7.0). At the other end of the range, the lowest levels of concern were recorded in the Netherlands, Sweden and Denmark" [14]. Notably, "[o]lder people (aged 65 years or more) were generally less concerned than their fellow adult citizens about having sufficient income in old-age... an average score of 4.8 compared with 5.8 for the total population" in every member state except Bulgaria [14] — current systems serve current pensioners better than future ones expect to be served. And the public endorses preparation in the abstract: "[a] survey conducted in 2018 reveals that more than three quarters (77 %) of all adults (aged 15 years or more) in the euro area (EA-19) agreed with the premise that

governments needed to save more today in order to prepare their public finances for population ageing," with a majority in every euro-area country, "the highest share recorded in Ireland (91 %) and the lowest in Greece (67 %)" [14].

But the specific instrument fails: "[w]hen asked in the same survey about their views on whether or not the retirement age should be increased to ensure the sustainability of pension systems, just over one quarter (26 %) of all adults in the euro area agreed... Almost half (45 %) of the population in Ireland agreed that it would be necessary to increase the retirement age, while there were eight euro area members where fewer than one quarter of all respondents agreed" [14]. The gap between 77 per cent (save more) and 26 per cent (work longer) is the political economy of pension reform in one pair of numbers: sustainability is popular, its most effective lever is not. That asymmetry explains the reform catalogue's shape — the drift toward career-length requirements, insurance periods, actuarial penalties and indexation adjustments, which raise effective retirement ages and trim liabilities without a headline vote on the statutory age; toward automatic mechanisms legislated once rather than repeated discretionary fights; and, in the Slovak-style case, toward reversals when the politics overwhelm the arithmetic. It also explains why the projections treat enacted reforms as adjustments to exit behaviour rather than certainties: a reform's half-life is a political variable. The next section quantifies what all this machinery delivers — pension entitlements under the base case and alternative scenarios.

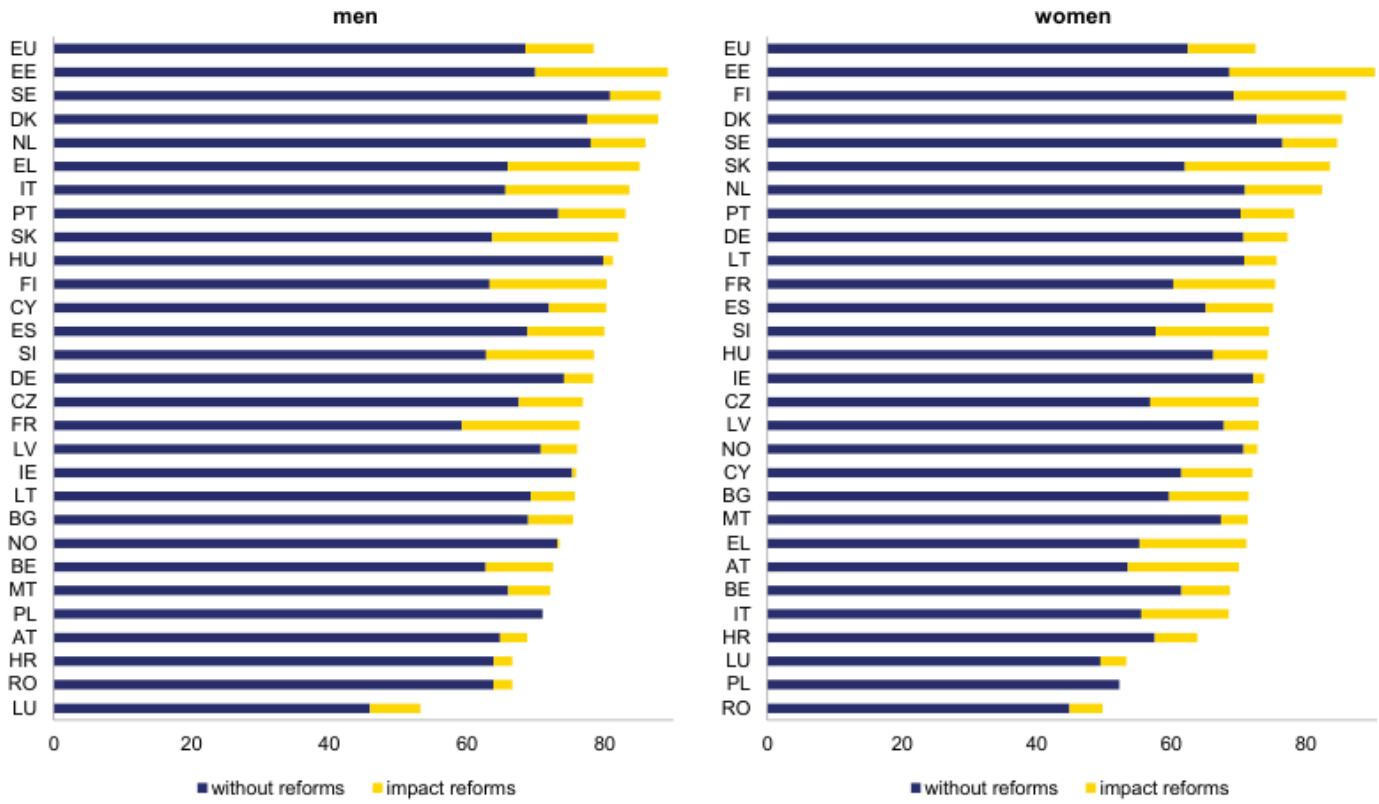


Exhibit. Impact of pension reforms on the projected participation rate of people aged 55–64 in 2070, in percent, for men and women across EU Member States. Source: S0015.

Pension entitlements: base case and alternative scenarios

What will the architecture actually pay? The OECD's entitlement modelling starts from a deliberately stylised worker: "[t]he base case in Chapter 4 concentrates on full-career replacement rates when individuals are at a constant level of earnings relative to the average during their whole career" [13], entering work at 22 and retiring at the normal retirement age. On those assumptions, "[f]ull-career male workers at the average wage throughout their career have, on average, a future gross replacement rate of 50.7%, when they start working at age 22"; the equivalent figure for women is 50.1% [13]. The base case is a measuring rod, not a forecast — its full, unbroken, constant-earnings career is precisely what the alternative scenarios relax, one assumption at a time (the unemployment and childcare breaks were quantified in the working-longer chapter; here the earnings path and employment status vary).

First, the earnings profile. Real careers slope upward, so the alternative scenario has the worker start lower and progress: "the relative wage increases throughout the career – from 60% of the average wage at age 22 to 123.33% at retirement age, ensuring the same lifetime earnings" [13]. With identical lifetime earnings, "[u]nder this scenario the benefit level for male workers is 52.0% of the average wage, slightly higher than for the base case at 50.7%. For

women, it is 51.3%, compared to the base case of 50.1%" [13]. But measured against final salary the picture inverts: "whilst in the base case final earnings and lifetime average earnings are the same this is not the case for the alternative profile case as the final earnings are higher, implying a benefit level of 42.2% of final earnings on average," and 41.6 per cent for women [13] — replacement of living standards at retirement is systematically lower than headline replacement rates suggest for anyone whose earnings grew.

Whether the profile matters at all is a formula question. "In some countries, the pension benefit level is identical in the earnings profile and the base cases, as pension systems that have flat-rate benefits, or points systems or constant accrual rates with wage valorisation of past earnings are not affected, as career average earnings are the same… These countries are Austria, Canada, Czechia, Germany, Hungary, Ireland, Japan, Lithuania, Luxembourg, Mexico, New Zealand and the Slovak Republic" [13]. "By contrast, countries that do not use the entire career earnings and price uprate past wages when calculating pensions have higher benefit values using the earnings profile scenario compared to the base case. The countries in question are Colombia, Costa Rica, France, Portugal, Slovenia, Spain and the United States as only 10, 25, 25, 40, 24, 27 and 35 years of earnings, respectively, are used" [13]. Best-years and final-salary formulas quietly privilege steep late-career earnings paths — a distributional choice embedded in arithmetic, favouring professionals whose pay peaks late over flat-profile manual workers with the same lifetime earnings.

Second, employment status. "Self-employed workers with a taxable income… equal to the net average wage before tax can, on average in the OECD, expect to receive an old-age pension equal to 79% of the pension of the average-wage dependent worker in the private sector" [13]. The gap is manufactured by contribution rules: the self-employed pay the combined employee-plus-employer rate in only a minority of countries; "[i]n 13 countries… pension coverage is limited because they are allowed to contribute less than employees, through reduced contribution rates (France, Iceland, Israel, Italy, Latvia, Norway, the Slovak Republic, Sweden and Switzerland), or flat-rate contribution (Colombia, Greece, Poland and Spain)"; and "[i]n Australia, Denmark, Germany, Japan, Mexico and the Netherlands, the self-employed are, in contrast to employees, not required to join earnings-related schemes" [13]. The outcomes follow the rules: where earnings-related participation is optional, "the relative pension of the self-employed is about half that of employees in Denmark, Germany, Greece and the Netherlands and is much lower in Mexico (17%) and Japan (34%)"; with flat-rate contributions, "between 40% and 60%… in Greece, Poland and Spain"; Switzerland reaches only 51 per cent because its self-employed skip the occupational pillar, while Estonia, France, Iceland, Korea, Lithuania and Norway all exceed 90 per cent [13]. Design can neutralise the gap: Austria's state "contributes 4.3% for the self-employed to fully offset the lower contribution rate they pay," Czechia's progressive formula yields 87 per cent "even though the contribution base is set at only 50% of taxable income," Australia's means-tested basic produces the outlier of 109 per cent, and in Austria and Luxembourg gross-income calculation bases make the self-employed pension "slightly higher than that of employees" [13].

The scenario method's lesson generalises: headline replacement rates describe a worker who barely exists — full-career, constantly paid, dependently employed. Every deviation the real world supplies (rising earnings, self-employment, breaks) moves the outcome, and the direction and size of the move are set by formula choices — valorisation, earnings windows, contribution bases, floors — that are rarely debated as the distributional decisions they are. The next section closes the loop with the labour market: how pension reforms themselves change older workers' participation.

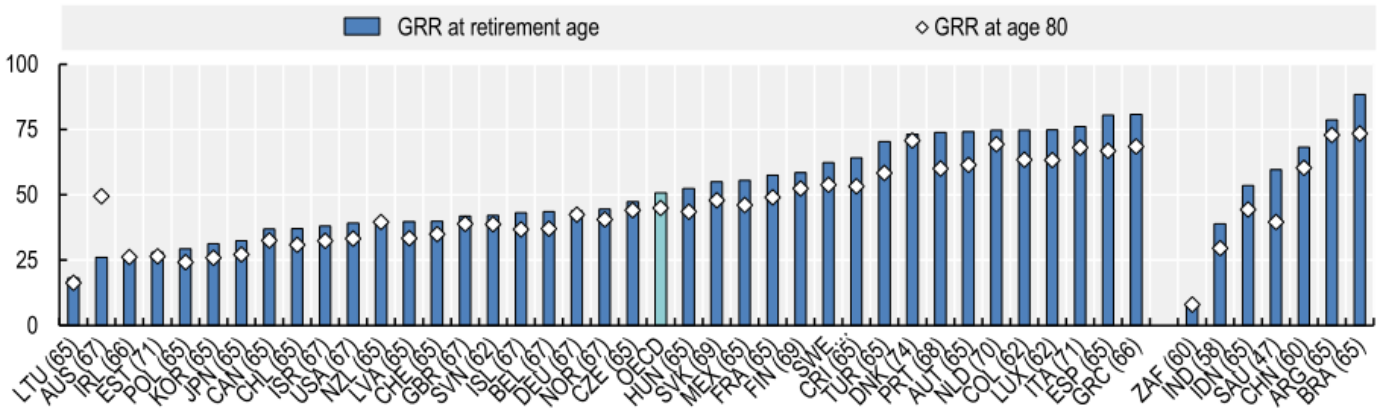


Exhibit. Gross pension replacement rates for average earners, in percent, shown both at retirement age and at age 80 to reveal how indexation erodes or preserves pension value across retirement. Source: S0002.

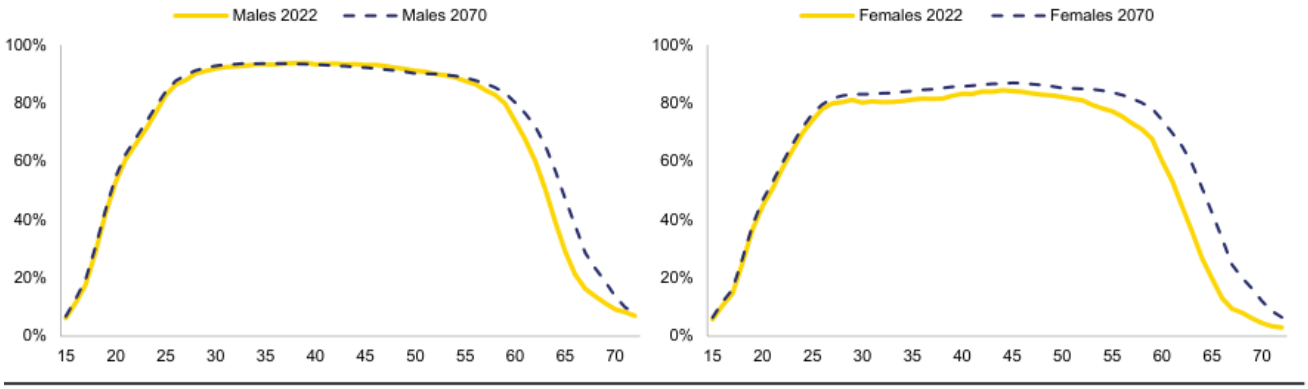
The impact of pension reforms on older workers' participation

Do pension reforms actually keep people working? The past decade offers the closest thing to a natural experiment, and the headline outcome is strong: "[t]he good news is that progress in the labour force participation of older workers across the OECD has been impressive. The employment rate of 55-64 year-olds in the OECD reached a record 64% in the second quarter of 2023, almost 8 percentage points higher than a decade ago" [13]. The reform wave behind it was broad and combined: "many OECD countries have combined action to increase statutory retirement ages, curb early retirement and offer incentives to work longer with efforts to foster employability, job mobility and labour demand for older workers. Normal retirement ages are set to increase in 23 out of 38 OECD countries, reaching an average of 66.3 years for men and 65.8 years for women starting their career today. In Denmark, Estonia, Italy, the Netherlands and Sweden, the normal retirement age will rise to 70 years or more if life-expectancy gains materialise as projected and legislated links with life expectancy are applied" [13]. The mechanism is the one the EU's projection machinery formalises: reforms shift the exit rates of workers aged 51–74 — later statutory and early-retirement ages, converging female ages, life-expectancy links and repriced early/delayed incentives all operate by making exit later or costlier at each age, and the participation profile follows.

The context made the reforms bite. Post-pandemic, "most OECD countries are facing labour shortages across many sectors and occupations. Unfilled job vacancies reached record-high levels in 2022; they remain high even in 2023 amidst a significant slow-down in economic activities. With large numbers of baby boomers retiring in the coming years, it is becoming even more important to promote the labour market participation of underrepresented groups in general, and older workers in particular" [13]. Demand pulled in the same direction as pension incentives pushed. Longevity underwrote feasibility: "[l]onger working lives are supported by growing life expectancy that, at age 65, increased by 4.8 years between 1970 and 2021, on average across the OECD, and by 1.6 years between 2000 and 2021. Despite the slowdown in life expectancy gains over the last decade, projections suggest further growth in the future" [13]. And the policy debate itself has rotated: "[s]ince the last edition of Pensions at a Glance in 2021, the pension policy debate in OECD countries has switched from pandemic responses back to a focus on more long-term structural issues," with the 65+ share rising "at 18% in 2022… to 27% by 2050, on average in the OECD" [13].

Attribution, however, needs the caution of the projections chapter: a large share of the participation rise is cohort replacement — more-educated, more-attached cohorts ageing into the 55–64 bracket — which would have lifted employment even without reform. What reform demonstrably adds is the shifting of exit ages at the margin and the closing of early-exit routes; the correlation evidence of the three-pillars analysis (stronger incentives associated with both higher employment and later exit) supports a real, if partial, reform effect. The editorial is equally clear about what reform has not fixed. "Many older workers still struggle to keep their skills up to date, have limited access to good-quality jobs, and risk having an inadequate pension in old age because of short and unstable working careers… participation in formal and informal training among older workers is less than half of that of prime-aged workers, on average. In addition, when looking at the reasons for quitting jobs, about one in five workers aged 50-64 across the OECD did so because of ill health" [13]. And "[p]erceptions of age discrimination remain common despite the fact that it is banned by legislation in virtually all OECD countries" [13].

The deepest brake is distributional. "Estimates show that low earners have three years fewer of life-expectancy at age 65 relative to high earners, which reduces their pension wealth by about 13%. Increasing retirement ages should thus be seen in the context of persistent socio-economic inequalities, potentially fuelling resistance against pension reforms" [13]. A uniform rise in the retirement age is a proportionally larger benefit cut for shorter-lived groups — the same gradient that drove the hazardous-work debate — which is why the participation gains, however impressive, arrive with a legitimacy constraint attached: reforms sustain themselves politically only when the employability and job-quality pillars advance with the incentive pillar. The next sections turn to the system's resilience agenda: building public retirement systems that can absorb these pressures, and the defined-contribution payout phase where individual longevity risk now concentrates.



ource: European Commission, EPC.

Exhibit. Age profiles of labour force participation rates in the EU by sex, comparing 2022 with 2070 and showing how participation at older ages rises across the projection. Source: S0015.

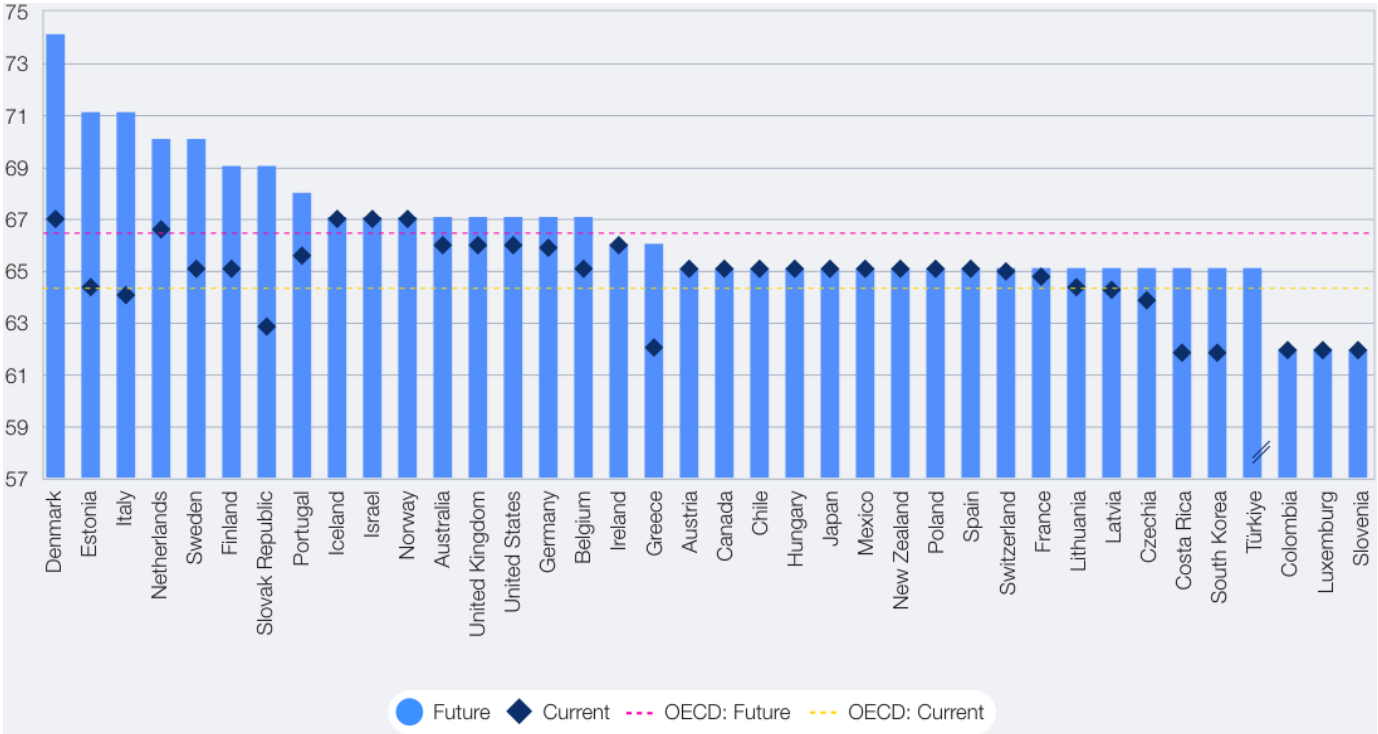
Building resilient public retirement systems

Resilience, in the World Economic Forum's framing, is a design property: systems that absorb demographic and financial shocks through built-in adjustment rather than recurring political crisis. The exemplar is the Netherlands. "The Netherlands has developed a globally recognized pension system that combines collective risk-sharing with adaptive mechanisms to ensure sustainability and equity" [7]. Its first pillar, the AOW, is "a universal, pay-as-you-go public pension funded through social security contributions and taxes. By linking the retirement age to life expectancy, the AOW addresses demographic pressures while ensuring basic income security for all residents" [7] — the automatic stabiliser this report has encountered repeatedly, here applied to the floor itself. The second pillar is mid-transition "under the Future Pensions Act of 2023," moving toward collective defined contribution with features that read as a checklist of the old system's diagnosed flaws: "[i]ntroducing tools for the pension fund to design a life-cycle strategy that better matches the members' (collective) risk profile"; "[m]ore transparency of pension pots and how they evolve during the accumulation phase"; "[r]emoving ex-ante redistribution effects that no longer match the modern labour market"; and a choice for social partners "between two contract forms with varying degrees of risk sharing and freedom of choice: the solidary defined contribution scheme and the flexible defined contribution scheme. Both contract forms are collective defined contribution schemes and share the characteristics of personal pension pots, lifelong annuities, diversified portfolios (including alternative asset classes) and the ability to absorb demographic and financial shocks" [7]. With "90% of workers enrolled in occupational pensions and a focus on intergenerational equity, the Dutch system consistently ranks highly in global pension indices" [7]. The architecture's lesson: individual accounts for transparency and mobility, collective buffers for shock absorption, annuitisation for longevity — risk shared where pooling is efficient, personalised where it is not.

The transition itself is the second lesson. "Transitions from pay-as-you-go systems to funded schemes, or from defined benefit to (collective) defined contribution plans can take years, if not decades, and necessitate compromises on intergenerational redistribution, clear communication and long-term political commitments. Gradual changes rather than abrupt shifts are necessary to help mitigate disruptions and allow individuals and organizations to adjust to new rules over time" [7]. Gradualism is not timidity; it is the price of consent in systems where, as the reform chapter showed, the public backs preparation but not its sharpest instruments — and where trust is itself a resilience input: "[i]n some countries, trust in public pensions is eroding due to fears of insolvency, causing workers to hesitate to participate in systems they perceive to be unreliable" [7].

The third element is the funded complement. "Over the past two decades, asset-backed pensions have become a key pillar of retirement systems, with assets more than tripling to over USD 56 trillion in 2023. These pensions diversify resources for financing retirement and complement pay-as-you-go systems, enhancing resilience to economic shocks and demographic challenges" [15]. Diversification is the operative word: PAYG income depends on the wage bill, funded income on asset returns, and the two fail in different states of the world. The OECD Pensions Outlook 2024 agenda for the funded pillar targets its weak points — "how employers, particularly smaller ones, and the self-employed, can be better integrated in these systems," "how investment strategies can balance sustainability and risk," "the importance of longevity protection and innovative approaches to providing retirement income, such as options for pooling risks and leveraging home equity," and the saver's toolkit, since "digital tools, such as pension dashboards, are important to empower savers and foster awareness" [15].

The fourth element is behavioural. “Making sound long-term financial decisions can be challenging, especially given individuals’ behavioural hurdles that encourage inertia and short-term thinking. Some tools that retirement systems can use to combat this include auto-enrolment and mandatory participation, which help people overcome inertia, as well as balancing levers such as premiums and tax incentives” [7]. And the fifth is the coverage frontier, where resilience is currently absent: “[g]lobally, an estimated 60% of workers operate in the informal economy, and the gig workforce is growing rapidly. These workers frequently lack access to contributory pension schemes, which depend on stable employer–employee relationships... pension systems often lack portability, making it challenging for workers who change jobs frequently or move across borders to build adequate retirement savings. As gig and informal work reshapes labour markets, public pension systems must evolve to include those workers or risk leaving a growing segment of the population unprotected in retirement” [7]. Meanwhile the parameter drift continues — retirement ages “will increase by about two years for those now entering the workforce” across more than half of OECD countries [7]. A resilient system, assembled: a universal floor with automatic longevity adjustment; a funded, collectively buffered earnings-related tier; behavioural defaults doing the enrolment work; portability and coverage designed for the labour market that exists, not the one the system’s founders knew. The next section examines where risk has migrated in the funded world — defined-contribution pensions and the design of the payout phase, where accumulated pots must somehow become lifetime income.



SOURCE: OECD. (2023, December 13). *Pensions at a glance 2023*. https://www.oecd.org/en/publications/2023/12/pensions-at-a-glance-2023_4757bf20.html

Exhibit. Normal retirement ages across the OECD for men entering the labour market at age 22 with a full career, showing legislated increases of about two years for new entrants. Source: S0010.

Defined contribution pensions and designing the payout phase

Defined contribution systems solved the accumulation problem and created a harder one: at retirement, an individual holds a pot of assets and must personally convert it into income for an unknown lifespan. The OECD Pensions Outlook 2024 approaches the payout phase as a needs-matching problem — retirees need lifetime income for essential spending, liquid reserves for unexpected spending, and flexible access for discretionary spending — and its central finding is that left to themselves, people systematically fail to assemble that portfolio. “Even though a large proportion of individuals state a preference for a lifetime or guaranteed income, the proportion of individuals who voluntarily choose such a product is much lower, even when products are readily available. One randomly controlled trial in the United States indicated that over 50% of those having at least USD 100 000 in retirement savings would be willing to purchase a life annuity at prevailing rates, in contrast to only around 12% who actually do (Arapakis and Wettstein, 2023). Similarly, in the United Kingdom, 92% of survey respondents felt that a guaranteed regular income in retirement was important, yet fewer than 10% of individuals accessing their retirement savings for the first time

use their savings to purchase an annuity" [15]. The annuity puzzle is not a preference problem; it is an access and comprehension problem.

The barriers stack. Awareness: "[i]n the United Kingdom, 32% of people aged 50-69 saving for retirement had not heard of a life annuity, and an additional 21% thought the product was a risky product with volatile income… In the United States, one third of savers were not familiar with lifetime income products" [15]. Comprehension: "[o]nly 26% of pension savers in the United Kingdom feel that they understand their pension statement" [15]. Advice: "[i]n the United States, 10% of advisors do not recommend annuities to anyone, and two-thirds of advisors recommend them to fewer than half of their clients" [15]. And friction: savers "are less likely to make the effort to shop around for a decumulation product compared to an investment product, and have a tendency to stay with their existing providers" [15]. For the spending side, the product mapping is clear enough — "[f]lexible drawdown products may be best suited for occasional discretionary spending. These allow individuals to decide when and how much they withdraw," while "[l]ifetime income products can be valuable for regular discretionary expenses as they allow people to easily budget their expenditure and avoid underspending" [15] — but "ensuring that suitable options exist for retirees to utilise their savings in retirement does not ensure that they will do so" [15].

The Outlook's answer is a three-level assistance architecture: "[t]he first is a default solution, where individuals will automatically have a payout solution chosen for them. The second is the provision of guidance to give individuals the information and tools needed to make an informed decision on their own and then execute it. The final is financial advice, where individuals can have a personalised recommendation for a suitable financial solution" [15]. Defaults carry the strongest warning label in the report: "[d]efault options can be very effective in nudging individuals into a specific financial option, but they do not engage people in the decisions about how to manage their retirement savings… Any default option for the decumulation of retirement savings should be implemented in a way that is not likely to cause undue harm to any group of the population. Ideally, it should allow for the possibility for the retiree to adapt if the default turns out to not be suitable for their needs. This implies that life annuities are not appropriate as a default option in most cases because they are irreversible. Features such as cooling off periods or products that provide regular incomes before converting savings into a lifetime income can provide individuals with the opportunity to experience a regular income while still allowing them a window during which to change their mind. Alternatively, managed drawdown products can provide a regular income while allowing retirees to retain the flexibility to adapt the solution at a later point in time" [15].

Defaults must also be market-aware: "[i]ndividuals should not be defaulted into a product that locks in investment losses from which they would have been likely to recover if delaying the transfer of their savings. Similarly, if individuals do not have a significant level of savings, defaulting them into a lifetime income product is not likely to be appropriate" [15]. Around the default, engagement infrastructure matters — communication campaigns, digital tools and dashboards, and choice architecture — with a governance caveat: "[s]upervisors should ensure that the use of behavioural insights results in good outcomes for consumers and is not used primarily for commercial gain" [15]. The payout phase is thus where the DC model either completes itself or fails quietly: accumulation built the assets ($56 trillion of them), but retirement security is decided by conversion design — defaults, guidance and advice standing in for the risk-pooling that DB provided automatically. The next section applies the needs framework in detail: payout options for essential, unexpected and other spending.

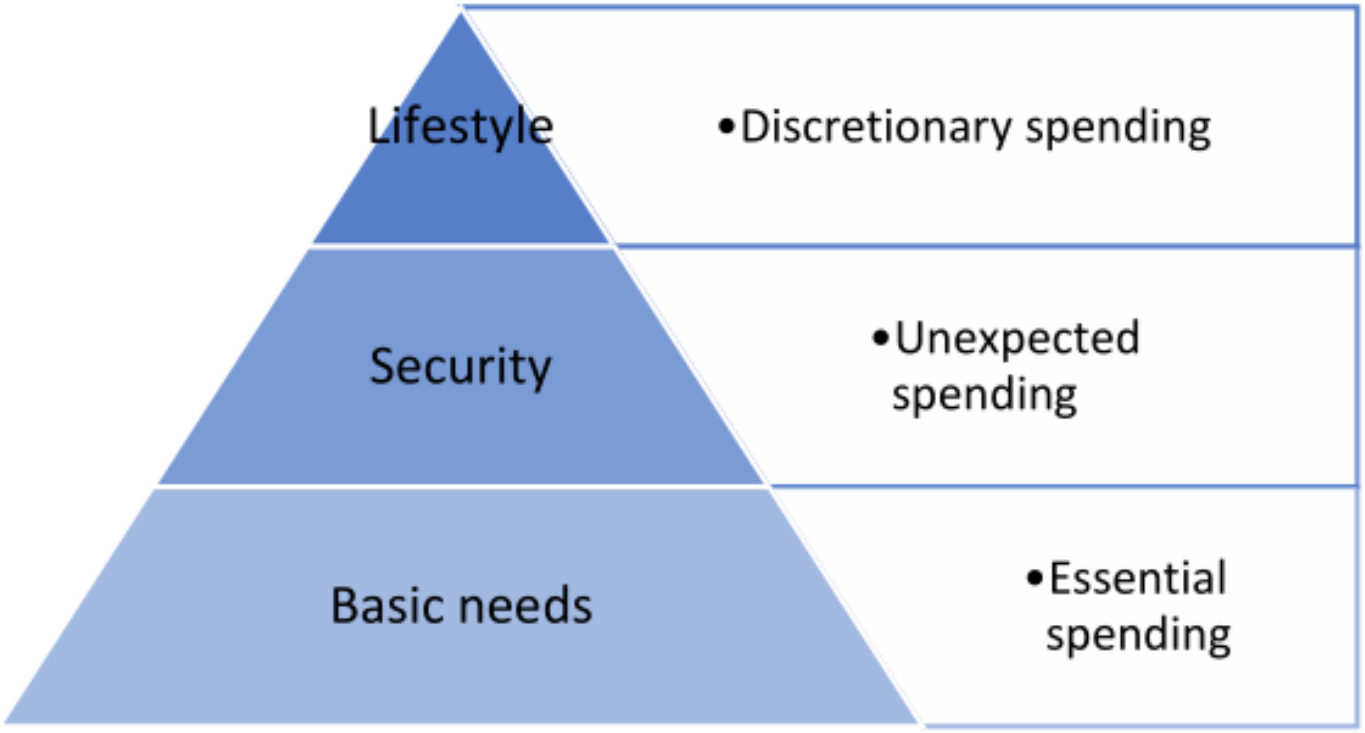


Exhibit. Components of financial needs in retirement, organised as basic needs requiring essential spending, security against unexpected spending, and lifestyle funded by discretionary spending. Source: S0013.

Payout options for essential, unexpected and other spending needs in retirement

The OECD's payout framework begins by refusing to treat a pension pot as one undifferentiated sum. Retirement spending comes in three kinds — essential spending on basic needs, unexpected spending for security, and discretionary spending for lifestyle (the Outlook's Figure 1.5 hierarchy) — and each kind carries a different risk that the payout product must mitigate. The first design step is diagnostic: "a well-designed framework for the payout of retirement savings in defined contribution plans should consider the role of those plans in financing different needs in retirement in the context of the pension system, as well as how individual and economic circumstances may influence the optimal solution for payout" [15]. Concretely, "[c]onsumption benchmarks are useful to illustrate how much savings is needed to finance different living standards in retirement. Comparing expected retirement income from other sources with these benchmarks can provide an indication of whether savings in defined contribution plans will likely be used to finance essential, unexpected or discretionary spending needs" [15]. Where a generous public pension already covers essentials, the DC pot is lifestyle money and can stay flexible; where it does not, the pot is survival income and must be annuitised. The same balance is a different product in different systems — which is why payout design cannot be separated from the architecture of the previous sections. Overarching all of it: "[t]he design of the retirement phase of asset-backed pensions should ensure that people are protected against longevity risk" [15].

For **essential spending**, the match is longevity-plus-inflation protection: "Essential spending needs are well served with a guaranteed lifetime income that also mitigates inflation risk" [15]. Rent, food, heating and care costs do not retire when the retiree's horizon extends, and they cannot be exposed to market or longevity risk. Yet even here the guideline is not compulsion: "[w]hile guaranteed lifetime incomes are well suited to meeting essential spending needs in most cases, flexibility may be needed in certain circumstances. Individual circumstances may imply that alternative options would be better suited for their retirement, and unfavourable market conditions may be detrimental to people if they have to lock in losses to obtain a guaranteed lifetime income" [15] — a retiree with a short life expectancy, a working spouse, or a pot crystallised at a market trough may rationally defer or decline annuitisation. The policy task is to make the lifetime-income route available, understandable and fairly priced — not to force conversion at the worst moment.

For **unexpected spending**, the requirement reverses: liquidity. "Unexpected spending needs will require liquidity while allowing for returns to at least keep up with inflation" [15]. Health shocks, home repairs and family emergencies arrive as lump sums, and an annuitised income stream cannot answer them. Hence a hard design rule: "[a]llow retirees to set aside a portion of their retirement savings to cover unexpected expenses. Requiring that retirees convert their entire balance into an income may restrict their ability to adapt to unexpected events, particularly if that income

is relatively low" [15]. Full annuitisation mandates — historically common — fail precisely the retirees with the least slack: their income is guaranteed and inadequate, and their reserve is zero. A protected liquid tranche is not a leakage from the system; it is the insurance layer that makes the guaranteed layer sustainable, since a retiree with no buffer will raid, borrow or distress-sell at the first shock.

For **discretionary spending**, heterogeneity rules: "[d]iscretionary spending will require more flexibility, but lifetime incomes can also be useful for regular discretionary spending needs to allow individuals to optimise their spending budget over their retirement," and "[d]iscretionary spending needs are likely to be more heterogenous across the population, and thereby require more flexibility as to the options available at retirement. Nevertheless, regular discretionary spending needs are well met by options providing a regular income, and these options should be available. Individuals having higher balances or savings in voluntary pension plans are more likely to use their savings for discretionary spending needs" [15]. Flexible drawdown serves the occasional large purchase; a regular income serves the standing lifestyle — and, counterintuitively, protects against underspending, the quiet failure mode in which anxious retirees hoard against a longevity risk that pooling would have removed.

The framework's supporting guidelines tie the three layers together: defaults established "with care and only when the solution is not likely to cause undue harm," because "[m]ost individuals defaulted into a particular payout solution are likely to simply accept the default"; awareness and education; digital tools with personalised information; behavioural nudges accompanied by engagement tools; and personalised guidance, with the honest admission that "regulatory requirements do not always distinguish between guidance and advice in a sufficiently clear manner," and that digital advice, though promising for cost, is "not yet developed for the payout of pensions, and individuals remain reluctant to fully rely on technology-based solutions" [15]. The composite picture is a layered decumulation portfolio — guaranteed floor, liquid reserve, flexible remainder — assembled at retirement with help proportionate to the stakes. The next section turns from spending needs to the asset side that funds them: equity investments in pension funds.

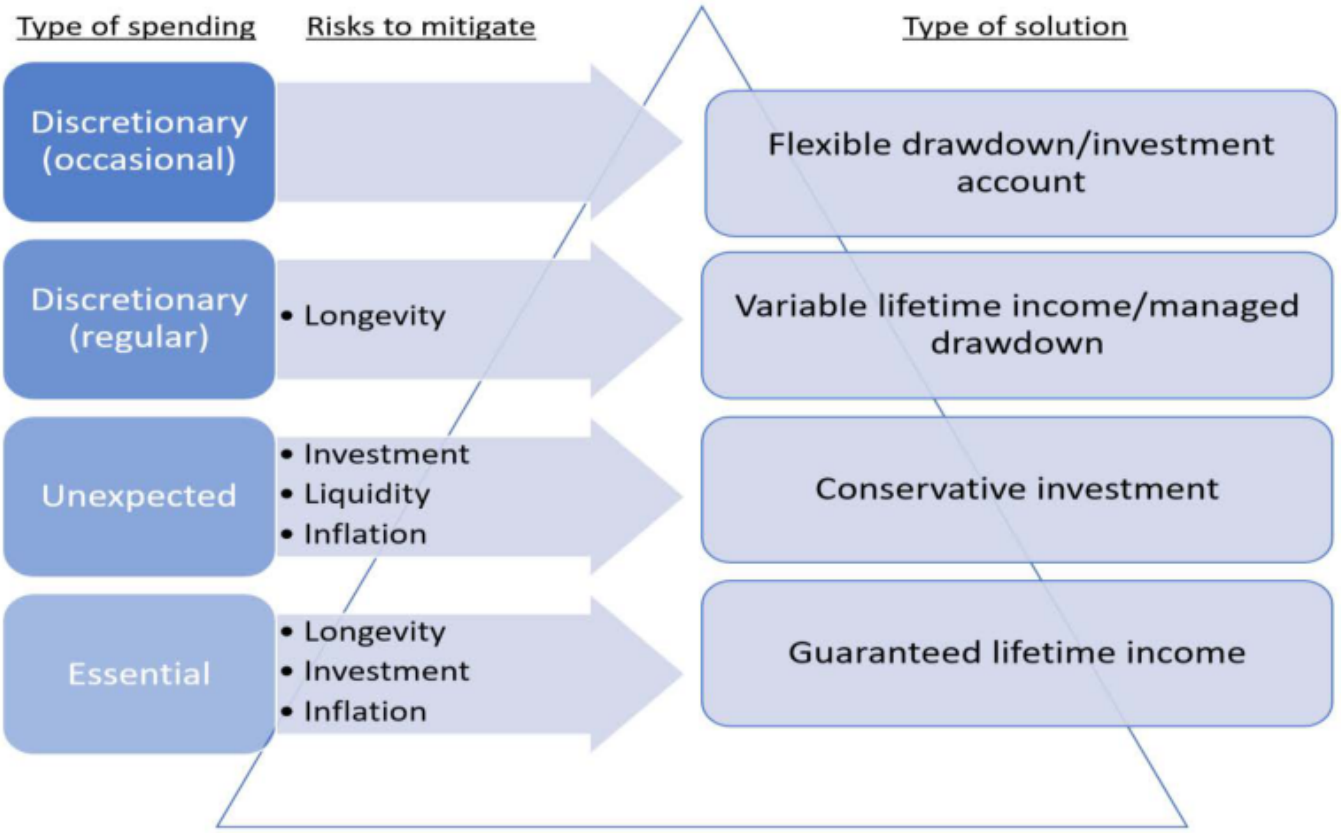


Exhibit. The risks that should be mitigated for different retirement spending needs — essential, unexpected and discretionary — and the corresponding types of payout solutions suited to each. Source: S0013.

Equity investments in pension funds: current practices, trends and performance case studies

The asset-backed pillar runs on equities. “Equity investment represents a significant share of the portfolio of DC pension schemes and has been rising steadily over the past 20 years in many jurisdictions. At the end of 2022, the total equity exposure of DC schemes, including public and private equities, represented more than 40% of total investment in 13 out of 38 jurisdictions analysed. By contrast, total equity exposure represented less than 20% of total investment in 7 jurisdictions. Average equity exposures tend to be lower in jurisdictions capping equity investment. Moreover, there has been a general upward trend in the equity exposure of DC pension schemes in many jurisdictions, with an increase of more than 20 percentage points between 2001 and 2022 in 5 OECD countries” [15]. The OECD Pensions Outlook 2024 asks the obvious question directly — does this exposure actually pay the member?

The answer is yes, on average and over time. “Investing in equity markets would lead to better retirement income outcomes for members of DC pension plans. The analysis shows that higher equity investment usually brings higher average performance when comparing DC pension funds that have different levels of equity exposure within countries or within pension entities. Still, diversification in terms of asset classes and geographical coverage plays a role in investment performance for a given equity exposure” [15]. The cohort evidence is broad: “[b]ased on historical returns on selected asset classes, the analysis shows that a vast majority of cohorts (at least 80%) across 19 different OECD countries would have accumulated more assets at retirement had they invested at least part of their retirement savings for 40 years in a mix of domestic and foreign equities instead of only investing in domestic fixed income securities (government bonds plus cash and deposits). The analysis from the stochastic model considering uncertain risk variables provides consistent results, with a probability of 87% to 94% of getting a higher replacement rate when investing in equities compared to only investing in fixed income, depending on the investment strategy and payout option” [15]. Regulation should therefore “avoid default investment strategies that are too conservative, like fixed income only strategies, because of low returns” — the safe-looking default is, in expectation, the expensive one.

Three caveats discipline the finding. First, horizon: equities pay “when people save for retirement for long periods, either by starting to save early or by delaying retirement… cohorts that would have saved for 20 years instead of 40 would have accumulated less at retirement with all investment strategies, but the difference is much larger for portfolios fully invested in diversified equities” — compounding is the mechanism, and it needs years; still, “even with only 20 years of contributions, at least 69% of cohorts across countries would have been better-off if they had invested in equities rather than in domestic fixed income” [15]. Second, volatility and cohort fairness: higher-exposure schemes “have usually achieved more volatile annual investment returns,” payout-phase benefits stay volatile if savings remain in equities, and “[v]olatile equity returns imply that two individuals born just one year apart could end up with large differences in accumulated assets at retirement despite having the exact same earnings history, contribution rate and investment strategy” — mitigated only by the fact that “both individuals would in general be better-off with a portfolio invested in equities than with any other investment strategy” [15]. Third, sequence risk at the retirement gate: “the median replacement rate of the equity portfolio declines from 39% for all simulations to 31% for simulations with a fall in equity markets of at least 10% in the year just before retirement, when assuming that individuals buy a life annuity at retirement” [15]. A crash in the final year taxes the whole career’s saving.

Life-cycle design is the market’s answer to the third caveat, and the case studies show its dialects. Chile’s multifunds hard-code it by age: five funds “from Fund A that can invest 40% to 80% in equities, to Fund E that can invest no more than 5%,” with defaults stepping down — “Fund B (25% to 60% in equities) is the default option for men and women up to age 35, Fund C is the default option for men aged 36 to 55 and women aged 36 to 50, and Fund D is the default option for men aged 56 and older and women aged 51 and older,” and the riskiest fund closed to older members [15]. Sweden’s AP7 default is more permissive for longer: “100% of the assets are allocated to the equity fund up to the age of 55. From the age of 56, the allocation is annually and linearly rebalanced towards the fixed income fund until reaching an allocation of two-thirds in the fixed income fund and one-third in the equity fund at the age of 75” — de-risking continuing a decade into retirement — while Alecta’s default steps 60 per cent equities to age 63, 50 per cent to 65, 40 per cent thereafter [15]. Denmark’s PFA offers four life-cycle profiles of rising risk; Australia’s QSuper shifts members at ages 40, 50 and 58 and at balance thresholds; and Czechia illustrates the conservative pole — transformed funds with “an annual nonnegative return guarantee,” closed to new entrants in 2013 yet still holding 63 per cent of members at end-2022 [15], inertia preserving precisely the low-return design the evidence warns against. The equity chapter’s verdict: equities are the returns engine of funded pensions — four-fifths-plus of cohorts gain — provided the glidepath, not the member’s nerve, manages the final decade. The next section turns to the levies on that engine: the tax treatment of retirement savings.

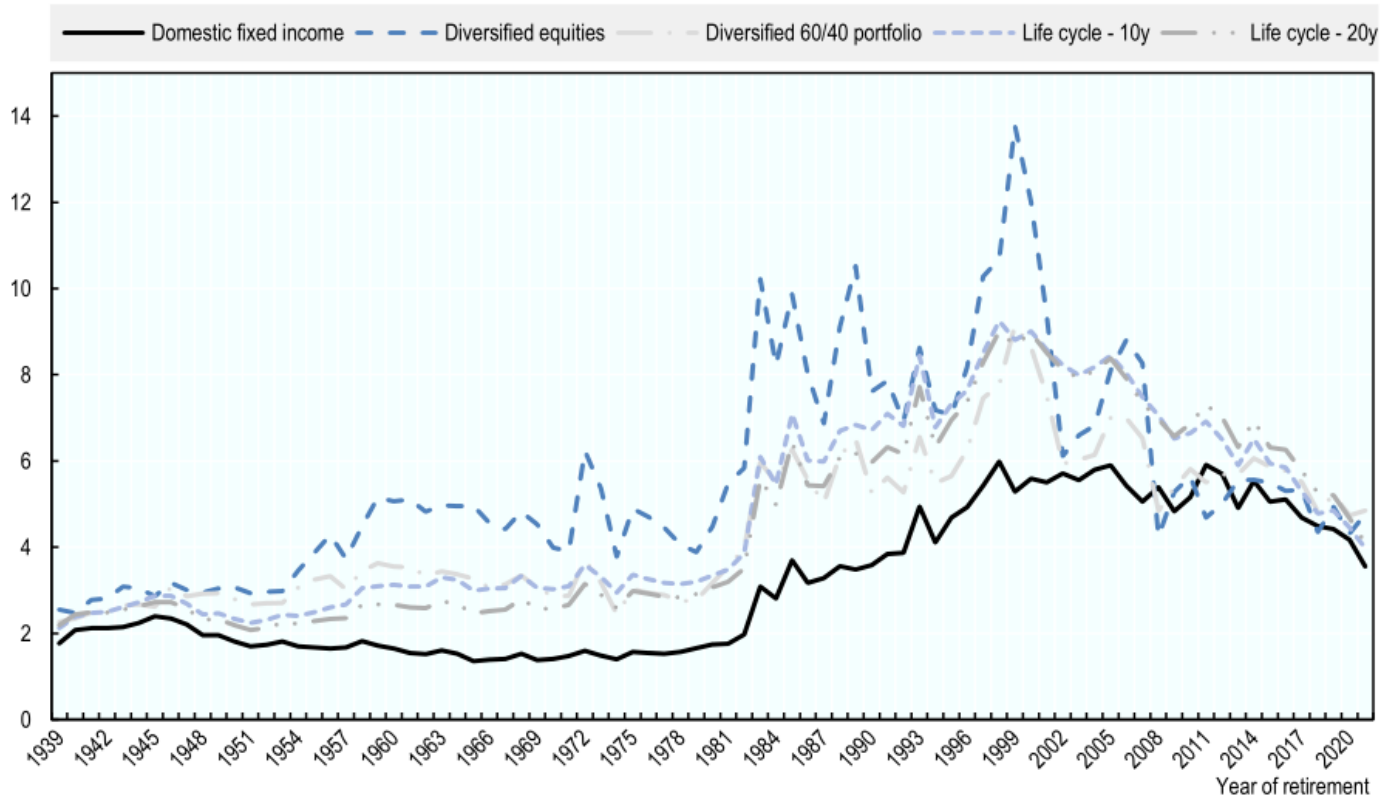


Exhibit. Accumulated retirement savings as a multiple of total contributions paid, by year of retirement, comparing domestic fixed income, diversified equities, a diversified 60/40 portfolio and life-cycle strategies. Source: S0013.

Tax treatment of retirement savings and incentives for retirement saving

Every asset-backed pension is co-financed by the tax system, and the terms of that co-financing are summarised in a three-letter code: whether contributions, investment income and benefits are Exempt or Taxed. The modal OECD regime is EET — relief on the way in and during accumulation, taxation on the way out — applying to the main plan in countries from Belgium, Canada and Germany to the United Kingdom and the United States; the alternatives span TEE (contributions taxed, benefits exempt, as in Czechia, Hungary and Mexico), ETT (Australia, Denmark, Italy, Sweden), TET (Austria, France, Korea) and, at the poles, EEE in Colombia, Estonia and the Slovak Republic and TTE in New Zealand and Türkiye [15]. Beneath the codes, "[t]ax rules for retirement savings remain complex in many countries. Tax rules tend to vary depending on the type of pension plan, the voluntary or mandatory nature of the plan, and who pays the contributions. Even in countries where only one tax treatment generally applies to all retirement savings, differences may exist in tax-deductibility limits, tax rates or tax credit levels. Moreover, frequent changes to tax rules have occurred in some countries over the past decade, potentially deterring individuals from saving in asset-backed pension plans, as changes make people uncertain about the tax treatment that may apply to them in the future given the long-term nature of saving for retirement" [15]. A forty-year savings contract denominated in rules that change every parliament is a contract with a political-risk clause.

The deeper design flaw is distributional. "The tax treatment of retirement savings tends to favour middle and high-income earners in most OECD countries… As most countries have progressive personal income tax systems, all these forms of tax relief, except tax credits, favour higher earners as they have higher marginal tax rates" [15] — a deduction is worth the saver's marginal rate, so the state pays most to those who need the inducement least. The corrective instruments exist but are blunted: "[b]y contrast, tax credits provide the same tax relief on contributions to all individuals and can even be set at a higher level for lower earners. In all OECD countries that use tax credits, these credits are non-refundable. This means individuals with a tax liability lower than the value of the tax credit can only partially benefit from it" [15]. The better-targeted tools sit outside the tax code proper: "[m]atching contributions provide the same incentive to all individuals as a proportion of earnings. However, state matching contributions tend to be capped, thereby reducing the value of the incentive for higher earners. Additionally, some countries use matching contributions to target specific groups of workers, in particular low earners. Finally, fixed nominal subsidies favour low earners as the fixed amount represents a higher share of their income" [15]. On the employer side, contributions are "always deductible from corporate income tax like any other business expenses," and "[i]n many countries, employer contributions are exempt from social security contributions or subject to reduced rates. This

preferential treatment of employer contributions makes them a more cost-effective option for employers than paying the same amount as salary" [15] — a wedge that quietly routes pay into pensions.

Two maintenance failures compound the design issues. First, parameter drift: "[m]any countries fail to update income thresholds and contribution limits for tax relief purposes in line with inflation or wages on an annual basis. Moreover, countries offering government matching contributions or subsidies tend to update the maximum entitlements, at best, only discretionally. Failing to update limits and thresholds regularly reduces the attractiveness of financial incentives over time" [15] — fiscal drag applied to the incentive itself. Second, behavioural steering is rare: "[c]ountries generally tax pension benefits and apply the same tax treatment across all types of retirement income payments (e.g. annuities, drawdowns, lump sums). A minority of countries use the tax system to either encourage people to take their pension benefits in a certain form or to discourage early withdrawals" [15] — despite the payout-phase evidence that form is exactly where retirees err.

The reform direction of the past decade has been to raise incentive value while nudging form and targeting. Estonia made "lifetime annuities and pensions paid over the average remaining life expectancy… no longer taxable since 2021," and "Korea reduced the taxation of pension benefits taken over periods exceeding 10 years in 2020" [15] — tax as annuitisation policy. Caps have been cleaned up: "Australia eliminated the excess concessional contributions charge on 1 July 2021," "[t]he United Kingdom eliminated the tax charge that applied when an individual had built up pension savings worth more than the lifetime allowance as of the tax year 2023/24" and "increased the annual contribution limit (annual allowance) from GBP 40 000 to GBP 60 000 in 2023" [15]. Levelling and targeting appear too: "[i]n 2023, the Netherlands increased the tax-deductibility limit for contributions to voluntary personal plans from 13.3% to 30% of annual income… to create a level playing field… between occupational and personal plans"; Germany's employer incentive gives "a tax allowance of 30% of the contribution" for low-income earners; and Denmark's extra exemption — raised in 2020 to 32 per cent within 15 years of retirement — means "132% or 112% of contributions are tax exempted instead of 100%" [15]. Spain moved the opposite way on individual plans — the personal contribution limit fell to EUR 1,500 by 2022 while occupational limits rose — steering saving toward employer schemes [15]. The overall lesson matches the chapter's pattern: incentives work best when they are stable, indexed, refundable or matched for low earners, and pointed at the behaviours (participation, annuitisation, employer provision) the system actually needs. The next section turns to the saver's information problem: individual pension dashboards.

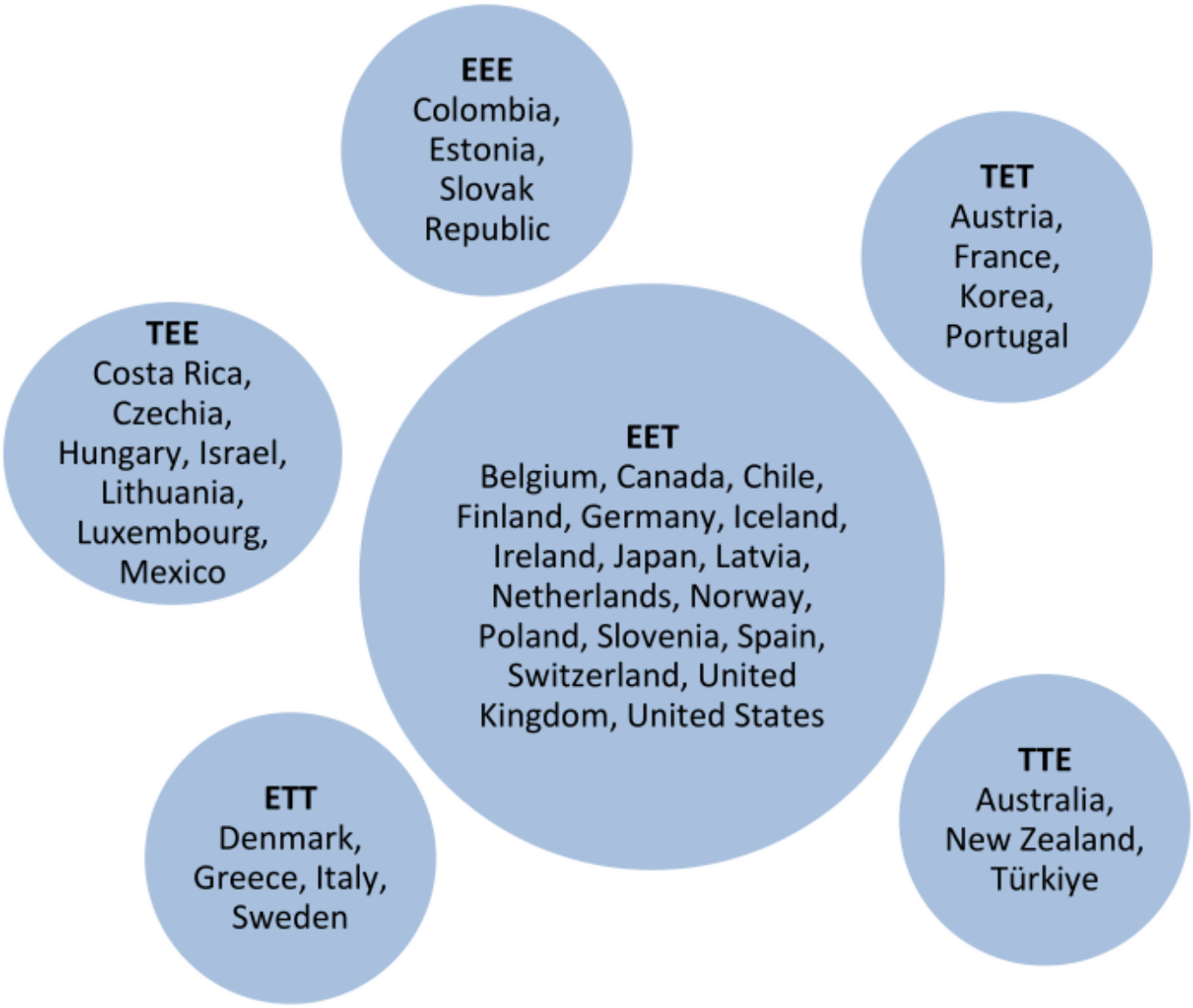


Exhibit. Overview of the tax treatment of retirement savings in OECD countries in 2023, grouping countries by regime — including EET, TEE, EEE and TET — according to how contributions, investment returns and benefits are taxed. Source: S0013.

Individual pension dashboards: design, development and operation

A multi-pillar pension system fragments a person's retirement wealth across employers, providers and state schemes accumulated over a forty-year career. The individual pension dashboard is the consolidation technology: a single login showing everything. "Individual pension dashboards often automatically link to an individual's pensions so they can see their consolidated information simply by logging in. Some dashboards also allow users to add additional accounts or financial resources that may not be automatically linked, as in Chile and Sweden. Sweden even allows users to input information regarding their housing assets. Denmark and Norway invite users to contact their pension providers if one of their accounts does not show up in the dashboard. Denmark and the Netherlands allow users to include their spouse's pension accounts in the dashboard to facilitate household retirement planning" [15]. The counter-model is manual: "[o]ther dashboards – typically those that are privately developed without government involvement – require that users manually link their pension accounts to the dashboard, as in Estonia, Ireland, and the Slovak Republic" [15] — automatic, state-backed linking is the difference between a tool everyone has and a tool only the diligent assemble.

Content builds in layers. "The minimum information provided by dashboards is the amount of pensions accrued at the time of the request. This amount is stated in terms of accumulated assets for asset-backed individual retirement savings, or as pension income for public benefits or asset-backed defined benefit plans" [15]. Above the minimum, "[m]ost dashboards also include an estimate of expected pension income at the statutory retirement age" — with instructive exceptions: "the myGov platform in Australia, whose primary objective is to help individuals find all their superannuation accounts and to consolidate smaller accounts, rather than helping them to have a view on how

much they will have at retirement," the income calculator being a separate Moneysmart service [15]. Scenario breadth varies: "[t]he dashboards in Chile, the Netherlands and the Slovak Republic show expected retirement income under good and bad investment scenarios, as will Belgium's in 2026," while "[d]ashboards in the United Kingdom will show both currently accrued values and projected values that assume that the current level of contributions will continue until retirement. They will only show the information submitted by the providers and will not allow the user to manipulate the information to model different scenarios or projections" [15] — a read-only design that trades engagement for standardisation.

Interactivity is where dashboards earn their keep. "The most common parameter that platforms allow users to change is retirement age" — the single most powerful lever, made tangible — and "[b]elgium, Chile, the Netherlands and Sweden also allow users to modify their employment characteristics, such as hours worked or salary. In addition to a simple calculator with these basic parameters, Norway provides an advanced calculator where individuals can see the impact of their marital status… of taking partial retirement while continuing to work, and of any employment outside of Norway. Chile also allows users to include spouses and/or dependent children in the calculation… The Netherlands, where pension assets must be split with ex-spouses, allows people to see the impact that a divorce has on their pension. Belgium is considering including the impact of taking parental leave on retirement income calculations" [15]. Two honest limitations recur: "[m]ost dashboards show a single estimated income at retirement in real terms, so that individuals can compare the estimate with their current salary. A main shortfall in this approach is that individuals do not see how their purchasing power may erode over retirement. Sweden is considering how to provide this information to users" [15]. Richer dashboards add the decision-relevant detail: contributions history (the Slovak Republic's Orange Envelope tracks "the amount and date of each contribution"), benefits "both gross and net of tax," Denmark's per-source tax treatment, survivor and disability entitlements, and — where choice is the point — fees and performance, with Australia letting users "select up to four funds to compare their performance and fees" and classifying funds as performing or underperforming over nine years [15]. Behavioural design closes the loop: Sweden's dashboard lets users compare their expected income with similar-profile peers and respond with a happy, neutral or sad face — "[i]f they click the sad face, they will be directed to information on how they can improve their retirement outcomes. Sweden found that rather than make people worried, providing information on how they could improve their situation kept them engaged" [15].

None of this matters if nobody logs in, so promotion is a design layer of its own. Mexico's regulator "runs regular communication campaigns to get people to use the AFORE Movil app"; Hong Kong's MPFA ran awareness campaigns before launch; Sweden, finding its early users were the already-engaged, targeted "those with low knowledge and interest in pensions… via telephone or employer meetings rather than via internet channels" [15]. Where the operator lacks a marketing budget, the ecosystem promotes: in Denmark the dashboard "is promoted by both banks and providers, in part because banks request information on pensions for mortgage applications. Belgium and the Netherlands have made it mandatory for providers to signpost the dashboard in the statements they provide to members. Providers in Chile who have not developed their own pension simulator are required to provide a link directly to the regulator's simulator" [15], and Swedish unions promote it to members. Dashboards are, in short, the visible tip of the funded system's promise — personal ownership requires personal information — and their development is iterative, "informed through the monitoring of metrics that indicate usage, engagement, and ultimate improvement in retirement outcomes" [15]. The pensions chapter closes by returning to the floor beneath all this machinery: basic, targeted and minimum contributory pensions, and who actually participates and is covered.

Basic, targeted and minimum contributory pensions; participation and coverage

The architecture chapter introduced the first tier's three instruments; this section examines their gates — how many years of residence or contribution a person needs before the floor actually pays, and at what age — because a floor with a high gate is coverage in name only. For contribution-based basic pensions, "[o]n average 34 years of contributions are needed for a full contribution-based basic pension, with 13 years contributions to be entitled to any benefit. Six countries currently require 10 years to be eligible for such a benefit, while Czechia requires 35 years for eligibility, which is also the same level for the full benefit" [13]. The pattern beneath the average is proportional entitlement: "whilst full entitlement to the contribution-based basic pension is achieved after 40 years in Canada, Japan and Luxembourg, only 10 years of contribution are required for eligibility for a reduced benefit… In Czechia 35 years are required for eligibility, with Argentina at 30 years and no other OECD or G20 country requiring more than 15 years" [13]. Residence-based basics soften the gate differently — entitlement accrues with years of residence rather than contributions, with "proportionally reduced benefits in many countries" for incomplete residence [13]. Czechia is

the outlier that proves the design point: a basic pension demanding 35 contribution years excludes precisely the interrupted careers the first tier exists to protect.

Minimum contributory pensions — the most widespread floor, found in 18 OECD countries — gate more gently at entry and more steeply at the top. “On average 19 years of contribution are required for eligibility to a minimum contributory pension, with 29 years required on average for the full pension. In France and Switzerland, only one period of contribution is required for a minimum contributory pension, whilst over 40 years are required for the full benefit. In the Slovak Republic, the minimum contributory pension is achieved after 30 years, with no explicit maximum duration. Full minimum pensions are achieved with 25 years of contributions or fewer in Chile, Colombia, Costa Rica, Hungary, Italy, Mexico, Poland, Slovenia, Spain and Türkiye” [13]. Two structural choices matter. First, most minimum pensions pay a single monetary value regardless of career length — “fewer than half of countries applying higher rates for longer careers of contribution” [13] — making them pure floors rather than graduated entitlements. Second, the interaction with targeted benefits is rationed: “[p]eople in Greece, the Netherlands and New Zealand cannot receive a targeted benefit on top of a full residence-based basic pension” (Table note, Figure 3.2) [13] — where the universal floor is judged sufficient, the means-tested top-up is switched off, a fiscal-economy choice that binds exactly when the basic level stagnates.

Claiming ages complete the coverage picture. “It is generally not possible to retire before the standard statutory age within residence-based basic pensions or for safety-net benefits” [13] — the floor pays late by design, which is one reason early exit runs through earnings-related schemes and disability routes instead. Those schemes allow early claiming “[c]ommonly between two and five years below the normal statutory retirement age,” with “a further reduction for each year that the pension is taken early. Belgium and Luxembourg do not apply a penalty,” and no early option at all in the DB schemes of “Austria (for women), Colombia, Costa Rica, Hungary (men), Türkiye and the United Kingdom” [13]. A distinct tradition ties early access to career length rather than age: “[e]arly retirement age is based on having made a given number of years of contributions in Austria (40 years), Germany (35), Greece (40) and Luxembourg (40)… in Estonia, early retirement is possible one year early with 20 years of contribution, increasing to five years with 40 years of contribution” [13]. For the longest careers the gates open widest: “[r]etirement is possible without penalty at age 60 with 44 years of contributions in Belgium or at age 57 with 40 years of contributions in Luxembourg,” and in Portugal “there is no sustainability factor… if there are 46 years of contribution by age 60” [13]. Funded systems gate by adequacy instead: “[i]n the FDC schemes of Chile, Colombia and Mexico and the DB scheme in the Slovak Republic, early retirement requires that the pension entitlements exceed a floor” [13].

Read together, the floor’s coverage logic is a set of trades: universality against contribution records, entry thresholds against full-benefit thresholds, and claiming flexibility against adequacy floors. The systems that score best on this report’s broken-career test combine low entry gates (France and Switzerland’s single contribution period), proportional accrual, and a targeted top-up that remains switched on. Those that demand near-full careers for their “basic” pension have, in effect, two earnings-related tiers and no floor at all. With the pension system now fully mapped — architecture, inflation behaviour, reforms, entitlements, participation effects, resilience, payout, investments, tax and information — the report turns to outcomes: the current income of pensioners and the sources it comes from.

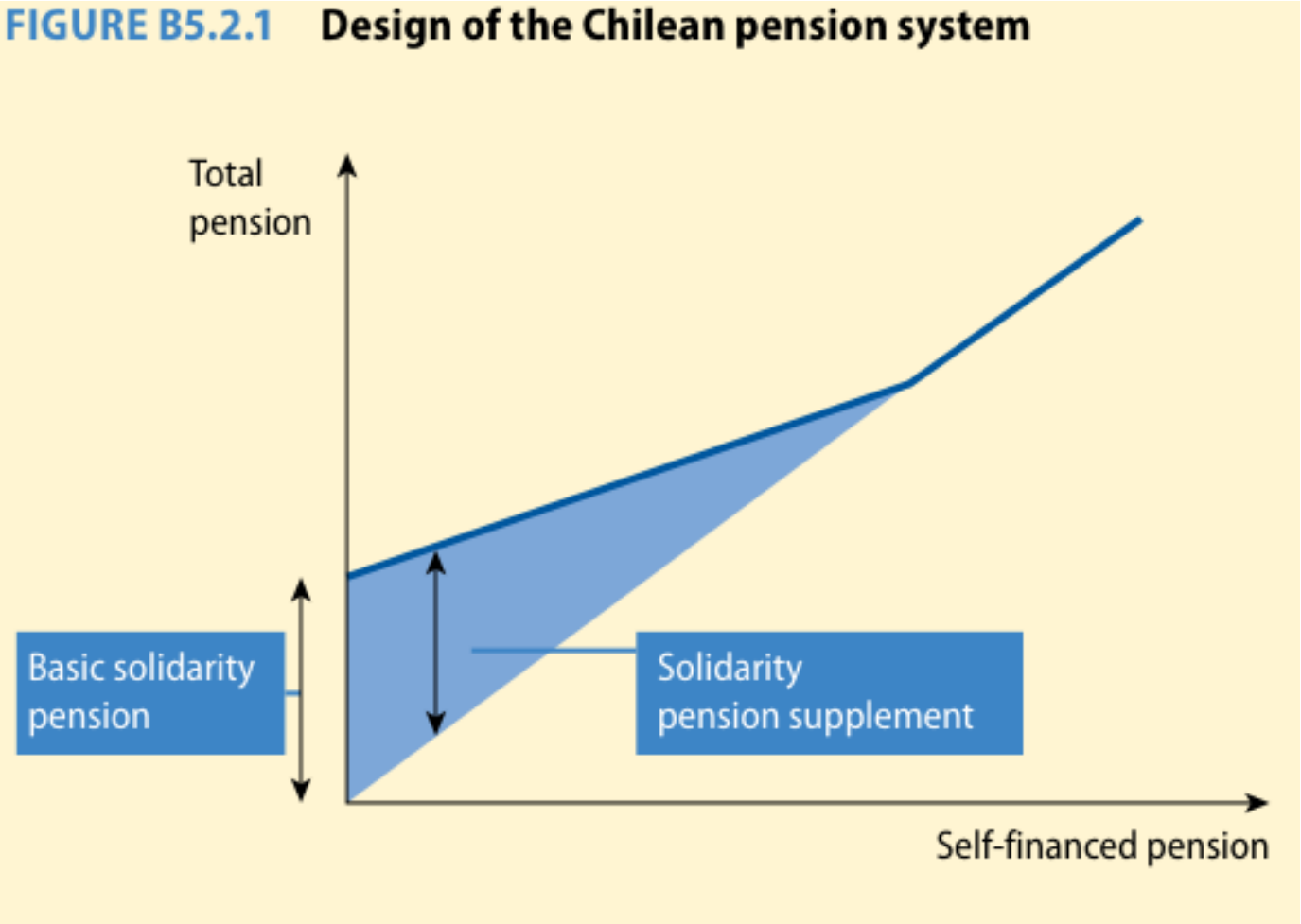


Exhibit. Design of the Chilean pension system, showing how the total pension combines a basic solidarity pension and a solidarity pension supplement with the self-financed pension from individual accounts. Source: S0029.

Income, wealth and poverty in old age

Current income of pensioners and sources of income in retirement

What do today's pensioners actually live on? The OECD's income accounts divide the disposable income of people over 65 into four sources, and the composition is the pension system's architecture expressed as household cash flow. "Of the four main sources of income on which older people draw, public transfers (earnings-related pensions, resource-tested benefits, etc.) and private occupational transfers (pensions, severance payments, death grants, etc.) account for around two-thirds of the total income. Public transfers account for 57% and private occupational transfers represent 7% of older people's incomes on average" [13]. The remaining third is not pension income at all: "[w]ork accounts for 26% and capital for about 10% of older people's incomes on average" [13]. The averages conceal three distinct national models, each recognisable from the systems chapters.

The transfer state dominates in continental Europe: "[t]he countries where over-65s are most reliant on public transfers are Austria, Belgium, Finland and Luxembourg: more than 80% of their incomes come from that source" [13]. In these systems retirement is what the public pension says it is — adequacy and fiscal exposure are the same number viewed from opposite sides, and the indexation battles of the inflation chapter are household-budget battles. The occupational model peaks in the Netherlands, where "[p]rivate occupational transfers are relevant in 13 OECD countries, with the Netherlands being highest at 40%" [13] — the quasi-mandatory second pillar of the architecture chapter, visible in retirees' bank accounts.

The third model is continued work. "Public transfers represent only 10% and 18% of all income in Mexico and Chile, respectively," and "[w]ork is especially important in Chile, Korea and Mexico, where it accounts for around half of old-age income; it also represents a large share of income in Costa Rica, Estonia, Iceland, Japan, Latvia, Lithuania, New Zealand, Poland, the Slovak Republic and the United States" [13]. Where pensions are thin or still maturing — Korea's system above all — "retirement" is a partial state financed substantially by earnings into late life, which reframes the working-longer chapter: in these countries older people do not work longer by policy encouragement

but by arithmetic necessity. A measurement caution applies: "[a]s incomes are measured at the household level, work is likely to be a more important income source for older people where many of them live in multi-generational households" [13] — part of the recorded work share is younger relatives' earnings, equivalised across the household ("[i]ncomes are measured on a household basis and equivalised with the square-root equivalence scale... for disposable incomes (i.e. net of personal income tax and social security contributions)") [13].

The fourth source marks the funded frontier: "[c]apital, mostly private pensions, represents over 40% of all income sources of older people in Canada. In Denmark, Korea and the United States, capital represents over 20% of all income" [13]. These are the countries where asset-backed provision has had decades to mature into actual retirement cash flow — the $56 trillion of pension assets from the resilience chapter, decumulating. The composition data thus read as a lagging indicator of system design: today's income mix reflects the architecture of thirty years ago, and the funded shares now accumulating will rewrite it for the cohorts retiring in the 2040s and 2050s. Two implications follow for the analysis ahead. First, the public-transfer share is the fiscal transmission channel — every point of indexation, every reform parameter, lands on household income almost one-for-one in the transfer-dominant countries. Second, where work and family households carry old age instead, income statistics flatter pension adequacy: the pension system can look sufficient because grandmothers are still working or living with earners. The next section looks beneath the averages at distribution — incomes and poverty of older people.

Incomes and poverty of older people

Retirement, measured in income, is a pay cut that deepens with age. "People aged 65+, and in particular those aged 75+, have a lower disposable income than the total population in most OECD countries. On average across the OECD, people aged 66-75 have a disposable income of 93% of the disposable income of the total population, falling further to 81% among people aged over 75" [13]. The dispersion is wide: relative income is "below 80% of the disposable income of the total population in Estonia, Korea and Lithuania in the age group 66-75, and below 65% in the same three countries and Latvia in the age group 76+," while "[t]he relative income of people aged 66-75 is higher than that of the total population in 11 OECD countries and exceeds 110% in Israel, Italy and Luxembourg. Luxembourg is the only country where the relative income of people over 75 exceeds that of the total population" [13]. The slide from 93 to 81 per cent across the retirement span is mechanical as much as biographical: pensions indexed below earnings growth lose relative value every year a retiree lives, so the oldest old drift down the distribution even in systems that served them adequately at 66.

Poverty concentrates accordingly. "On average in the OECD, 14.2% of individuals aged over 65 live in relative income poverty, defined as having an income below half the national median equivalised household disposable income. Their income gap to the relative poverty line is 23.1% on average... The average for the total population is 11.4%, some 2.8 percentage points below the old-age level" [13], with older people more likely to be poor in 23 countries — although "[i]n recent decades, relative poverty has tended to shift from people aged over 65 to people aged 18 to 25" [13]. The extremes are stark: "[a]ccording to the latest available figures, relative poverty rates of people aged over 65 exceeded 40% in Korea, were above 30% in Estonia and Latvia, and 20% or more in Australia, Costa Rica, Japan, Lithuania and the United States. By contrast, Czechia, Denmark, France, Iceland, Luxembourg and Norway have the lowest relative poverty rates, close to 5% or below. First-tier pension levels are important factors influencing old-age poverty rates" [13]. Korea's gap between old-age and total-population poverty — 25 percentage points — is the OECD's largest [13]; at the other end, "[o]lder people are less likely to be poor than the total population in several countries, especially France, Greece, Luxembourg, Norway and Spain where the old-age poverty rate is at least 4 percentage points lower" [13]. Two caveats frame the league table: the measure is relative (half of each country's own median) and income-only — "the considerable country differences in wealth (housing or otherwise) held by older people may not be reflected in income poverty rates" [13], a gap the wealth section takes up.

Within old age, poverty is gendered and aged. "Poverty among the 'younger old' (aged 66-75) is less frequent than among the 'older old' (aged 76 and over); the OECD average poverty rates are 12.5% and 16.6%, respectively. The difference between the two is particularly high in Korea (+20.6 percentage points), Latvia (+17.5 percentage points) and Estonia (+15.4 percentage points)... In Korea, the pension system is still maturing, and current generations of very old people still have very low pensions. Moreover, in all three countries, individual pensions are indexed to less than earnings growth" [13]. The exceptions instruct: "[i]n five OECD countries – Chile, Germany, Hungary, Iceland and Poland – the over 75s fare slightly better than their younger counterparts do," while "[p]ension reforms that have reduced the generosity of pension systems typically lower the relative income of new generations of retirees" [13] — where the age gradient inverts, it is usually because today's new retirees are poorer than yesterday's, not because the old grew richer. For gender: "[t]he average old-age poverty rates for women and men in the OECD equal 16.6%

and 11.1%, respectively. Lower earnings-related pension income and longer life expectancy are among the main drivers of higher poverty incidence among women than among men. Older women are at greater risk of poverty than older men in all countries except Chile, Costa Rica and Iceland" [13], with the largest gaps — 20 points or more — in the Baltic countries and about 11 points in Korea [13]. Inequality among the old mirrors inequality among the working: the Gini of disposable income over 65 averages 0.306, from Costa Rica's 0.500 and the United States' 0.409 down to Czechia's 0.198 [13].

Governments have begun answering at the floor. Chile "replaced its targeted public pension scheme with a quasi-universal scheme in January 2022. While the previous scheme (basic solidarity pension) was targeted at those aged 65+ belonging to the poorest 60% of the population, the new scheme (universal guaranteed pension) was initially accessible to the 90% poorest people aged 65+," expanded in February 2023 with "another 70 000 beneficiaries," its benefit "4% higher in real terms than its predecessor, or about 19% of gross average earnings," and its withdrawal redesigned so that "no withdrawal takes place in the new scheme until the FDC pension renders 3.4 times the amount of the universal guaranteed pension" — where the old scheme clawed back a third from the first peso [13]. Elsewhere: "Canada increased its residence-based basic pension by 10% for people aged 75 and over in July 2022. Estonia increased its basic and targeted pension by EUR 20, or about 7% above indexation in January 2023. Sweden increased its targeted pension benefits by about 12% in August 2022" [13], with France, Italy, Lithuania, Spain and Türkiye also raising first-tier benefits. The poverty data deliver the chapter's verdict in advance: the floor's level and indexation — not the replacement rate of the average earner — decide whether old age means poverty. The next section examines the working poor of the older population: older people in work who remain at risk of poverty.

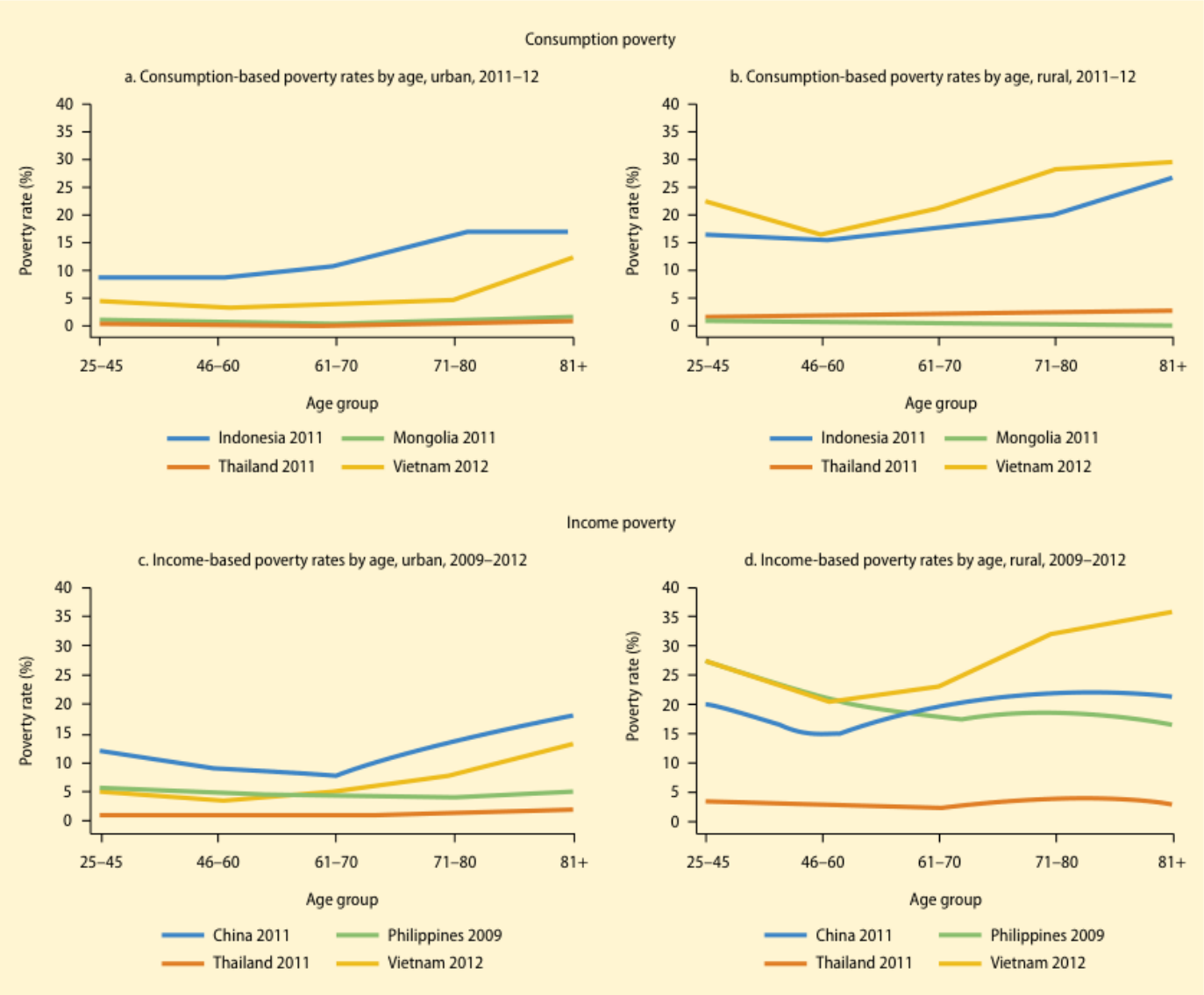


Exhibit. Percentage of people below the poverty line by age and rural or urban location in selected countries, showing that poverty among the rural elderly remains significantly higher than among the urban elderly. Source: S0029.

Older people in work at risk of poverty

Working longer is the report's master remedy for ageing — but for a substantial minority of older workers, work itself does not pay enough to escape poverty. In the EU-28, "[a]lmost 10% of older people in work were at risk of poverty," and "[a] relatively high share of the EU-28 population at the end of their working lives (55-64 years) faces various forms of in-work poverty, deprivation and exclusion" [14]. The deprivation measure makes the squeeze tangible: in 2015, "17.0% of people aged 55-64 years in the EU-28 could not afford to spend a small amount of money on themselves, while just 9.6% of very old people aged 85 years or more were unable to afford to spend a small amount of money on themselves" [14]. The gradient runs the wrong way for the working-longer agenda: the years immediately before retirement are the most deprived of later life, and pensions — where adequate — relieve rather than deepen the squeeze. The burden is gendered: this deprivation "was systematically more prevalent among older women (than men) for each of the age groups presented" [14].

Geography decides how harsh the end of working life is. Among 65–74-year-olds, "more than half of the older people (both sexes) living in Romania faced this problem (52.6%), while upwards of 40.0% of the older people living in Greece and Bulgaria could not afford to spend a small amount of money on themselves each week. By contrast, there were six EU Member States where the share… was less than 5.0%; the lowest share being recorded in Finland (0.2%)" [14]. Gender gaps in this deprivation were largest in Bulgaria and Portugal, with Lithuania, Greece, Denmark, Finland and Ireland the only exceptions to women's disadvantage [14]. These are the same member states whose first-tier floors and pension levels anchor the bottom of the income tables — in-work poverty among older people and pensioner poverty are one continuous distribution, not two problems.

The OECD's life-course analysis explains why late-career poverty is so persistent. "'Ageing unequally' refers to inequality that develops throughout the life course and materialises in old age. It is often the result of specific episodes during people's lives that tend to cumulate their detrimental effects on health and income at old age" [3]. Health is the compounding mechanism: "[a]t all ages, men and women in bad health work less and earn less when they work. Over the whole career, bad health reduces lifetime earnings by 33% and 17% for men with low and high levels of education, respectively" [3]. And the transmission into retirement is nearly mechanical: "[o]n average, about two-thirds of lifetime earnings inequality passes on to pension inequality – from less than 25% for many Anglo-Saxon countries to more than 85% in about one-third of OECD countries" [3]. The forward warning completes the section's logic: while today's older people enjoy "higher incomes and lower poverty risks than previous ones in most countries, the younger generations are likely to face again higher inequality in old age," having known "more unstable labour market conditions and widening inequalities" [3]. In-work poverty at 60 is therefore a leading indicator: it forecasts both inadequate pension accrual and the political limits of asking low-paid older workers to simply work longer.

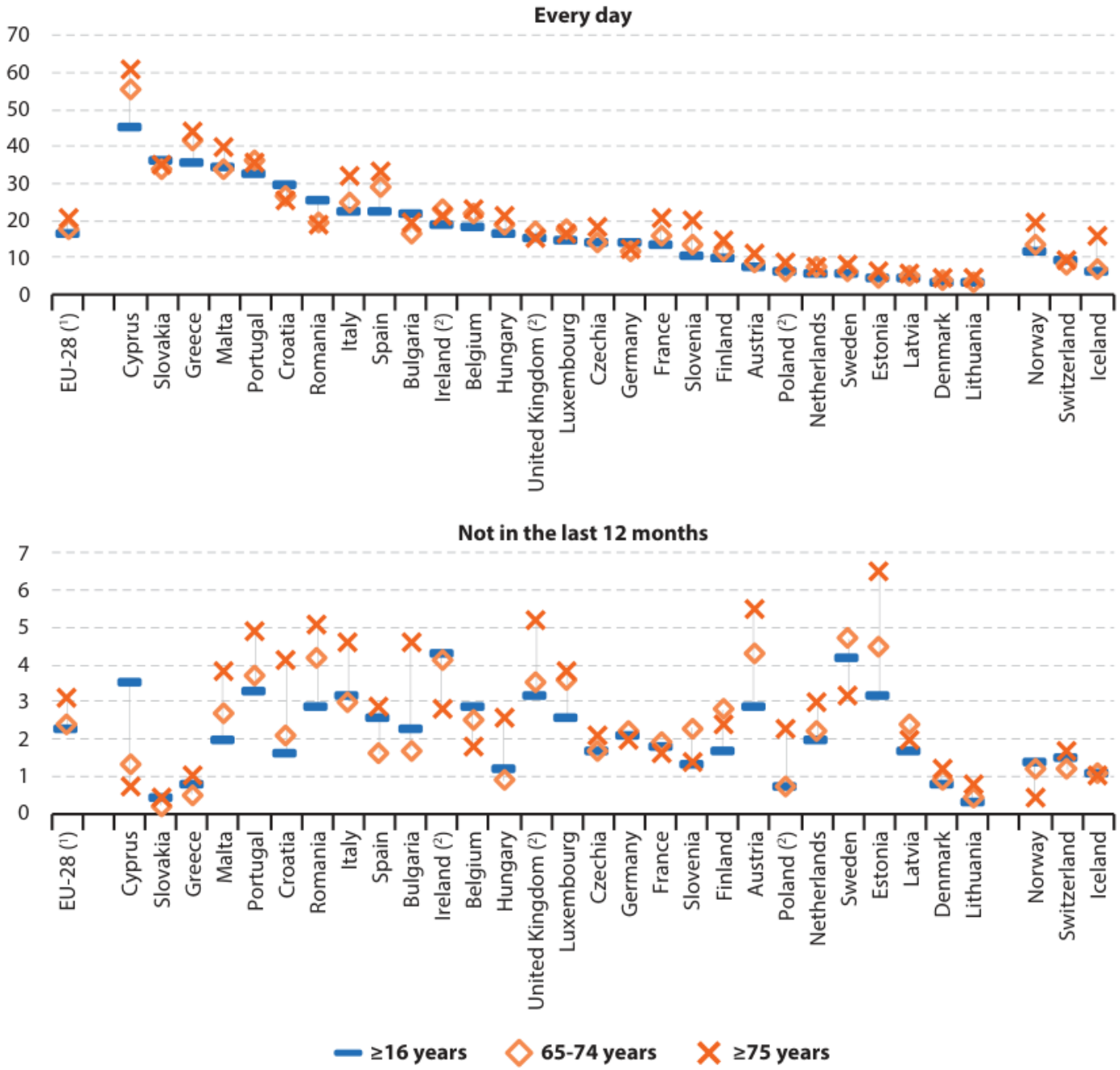


Exhibit. Frequency of getting together with family or relatives by age class in 2015, from daily contact to no contact in the previous 12 months, for people aged 65–74 and 75 and over compared with those aged 16 and over. Source: S0014.

Wealth and debt of older people

Income poverty statistics understate older people's resources in one direction and overstate them in another, because old age is financed from a balance sheet as well as a benefit statement. "Wealth assumes an increasingly important function for economic security throughout old age. Individuals and households use wealth accrued throughout their lives, including physical assets such as homes or land and financial assets, to finance consumption and withstand shocks once they stop working. The distribution of household wealth is typically more unequal than the distribution of income within and across countries" [1]. The life-cycle model predicts the shape the data show: "[a]verage household wealth typically rises with the age of the household head up to the official retirement age, when older persons start to spend their savings or else stop accruing assets and the burden of liabilities declines" [1].

The euro-area evidence (ECB Household Finance and Consumption Survey, 2014) quantifies the peak: "[i]n 2014, households in the euro area with a reference person aged 55-64 years had a net wealth that was 57.1% higher than the median for all households; a slightly higher share (59.8%) was recorded for households where the reference person was aged 65-74 years. Although the net wealth of households where the reference person was aged 75 years or more declined (compared with other older people), it remained above the median for all households" [14]. "In most of the EU Member States shown... the highest levels of net wealth were recorded for households where the

reference person was aged either 55-64 years or 65-74 years," and in western and Nordic states all three 55+ groups sit above average — "this pattern may be linked, at least in part, to a dramatic increase in property prices that benefitted (some) older people while making it difficult for younger generations to get on the housing ladder" [14]. Net wealth here is comprehensive — assets "include real assets (property, vehicles, valuables) and financial assets (deposits, savings, funds, bonds, shares, pension plans, life insurance); liabilities include mortgages, loans, overdrafts and credit card debt" [14] — which is why the housing windfall dominates: for most older households the home is the balance sheet.

Debt, symmetrically, is a young household's condition. "In 2014, close to half (42.4%) of all households in the euro area were holding debt... older people generally had lower levels of debt than younger generations: while 9.3% of households in the euro area with a reference person aged 75 years or more were holding debt, this share was considerably higher (60.9%) for households where the reference person was aged 35-44 years" [14]. "The likelihood of holding debt fell as a function of age: in each of the EU Member States shown, the lowest proportion of households holding debt was recorded for households where the reference person was aged 75 years or more" [14]. Retired households also spend less: "[i]n 2015, mean consumption expenditure among households where the reference person was retired averaged 21 106 PPS across the whole of the EU-28. Expenditure peaked at 52 946 PPS in Luxembourg... At the other end of the range, the lowest levels of mean consumption expenditure among households where the reference person was retired — less than 10 000 PPS — were recorded in Estonia, Bulgaria, Romania and Latvia (where the lowest level of expenditure was recorded, an average of 8 132 PPS per household)" [14] — and "households where the reference person was retired usually had a lower level of consumption expenditure than the average for all households" [14].

The balance sheet, however, rescues only those who have one. "[T]here are large gaps in wealth between people living in relative income poverty and those with incomes above the relative poverty line... These differences increase up to ages 70 to 74. Gaps in wealth widen among survivors despite the fact that the poorest people tend to die earlier than people with higher incomes and more wealth" [1] — wealth compounds the income divide rather than cushioning it, and survival itself selects for the wealthy. And accumulation is far from universal: "[a]bout 46 per cent of adults save for old age in high-income countries, compared to 16 per cent in middle- and low-income countries," disparities reflecting "people's varying capacities to accrue wealth based on their incomes and their unequal access to asset markets. In 75 countries, women's property rights are still restricted" [1], with financial literacy, digital capability, discrimination and missing infrastructure compounding the exclusion. The section's balance-sheet verdict: old age in the data is asset-rich and debt-light on average — a median position built substantially on housing appreciation — but the average conceals the split that runs through the whole report, between households whose wealth can absorb shocks and the income-poor whose wealth gap to the rest widens with every year of survival. The chapter closes by asking what "enough" means: a framework to assess financial needs in retirement.

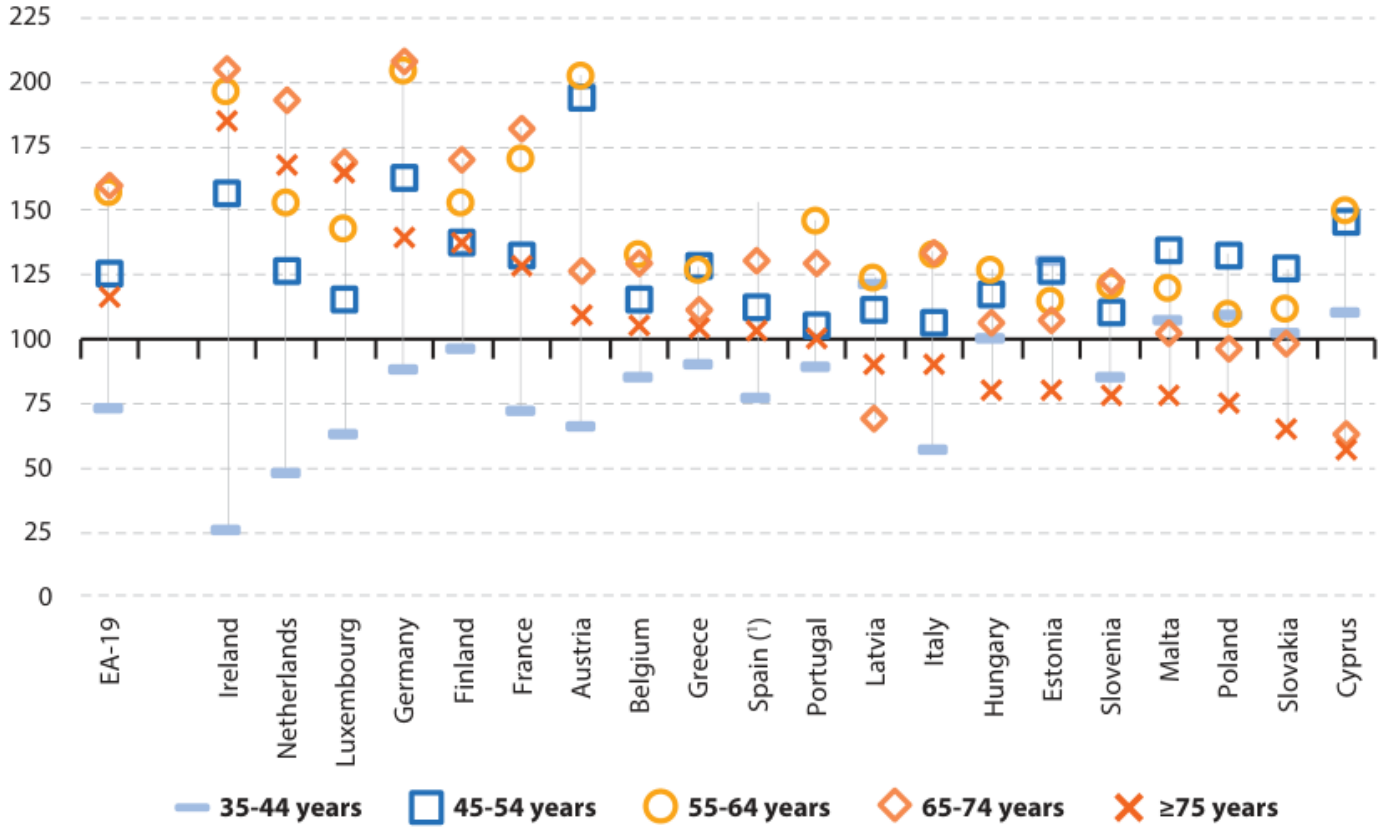


Exhibit. Median net wealth of households in 2014 by age of the reference person, expressed relative to the median for all households (100), across age classes from 35–44 to 75 and over. Source: S0014.

A framework to assess financial needs in retirement

Every number in this chapter — replacement rates, poverty lines, income shares — answers "how much?" The framework that closes the outcomes analysis asks the prior question: enough for what, in what form, against which risks? The OECD's assessment framework for retirement finance (Pensions Outlook 2024) sorts spending needs into three classes with different product logics, and tests any pension arrangement — public, occupational or personal — against all three. "Individuals should ideally have guaranteed lifetime income to be able to cover essential spending needs throughout retirement. Essential spending needs include housing, food, transportation, and healthcare costs. Guaranteed lifetime income products that provide a constant stream of income for life are well suited to cover these needs" [15]. The first test is therefore coverage: does the retiree's guaranteed lifetime income — starting with the public pension, for "[t]he guaranteed lifetime income that pay-as-you-go public pension benefits provide in many jurisdictions can be sufficient to cover basic income needs," though "[t]he extent to which pay-as-you-go public pension benefits are sufficient to cover basic income needs varies across population groups" [15] — meet the essential floor for life, inflation included? Preferences ratify the priority: "[a] recent survey across 15 European countries showed that security was the most important priority when investing for retirement. Around 80% of respondents chose safety over performance, with women having a larger preference for security compared to men. Additionally, 62% of respondents stated that they preferred an annuity solution in retirement for at least part of their savings, though the majority of individuals still preferred a lump sum over an annuity when shown projections" [15]; in the UK "92% of respondents felt that a guaranteed regular income in retirement was important" [15].

The second test is shock absorption. "Retirees will ideally be able to keep some of their financial resources aside to use in case they experience an unexpected expense or emergency… Common financial shocks that retirees face relate to healthcare, home repairs, or the loss of a spouse. Funds used for unexpected expenses need to be available and easy to access when individuals need them, and therefore need to be held in relatively safe and liquid assets" [15]. The exposure is common and gendered: in Australia and the Netherlands "retirees saved primarily to have some precautionary savings for health reasons, and this was a more important motivation for women than for men," while "Northern Europe tends to also have higher public healthcare coverage, indicating less of a need for precautionary savings" [15] — the framework's second lesson: the liquidity test is country-specific, scaled by what the health and care systems already absorb. "One survey of the United States population indicated that over 40% of retirees below age 80 have had an unexpected financial shock, and that nearly a quarter of these shocks were due to major home repairs" [15]; "[i]n a survey of children in the United States with recently deceased parents, one-third felt that the

death of one parent had reduced the financial security of the surviving parent, and a quarter reported that the financial situation became much worse" [15]; and the two shocks people are least prepared for are this report's recurring ones — "long-term care costs and divorce… Around 40% of lower-income people in the United States and Canadians report being financially unprepared for long-term care expenses, and 35% unprepared for divorce" [15]. Insurance covers the insurable tail; the set-aside covers the rest; neither works if the pension was fully annuitised on day one.

The third test is the discretionary life — and here the framework's behavioural evidence bites. "People value flexibility about as much as they value security… nearly 90% also thought that being able to have flexible access to money was important" alongside the UK's 90-plus per cent for guaranteed income, and "[a] survey of individuals in the United States revealed that 41% of retired individuals preferred investment solutions to finance their retirement compared to 20% preferring an annuity" [15]. Regular income nonetheless anchors wellbeing: "[r]etirees financing more of their consumption through lifetime annuities seem to have higher levels of satisfaction and may remain more satisfied throughout retirement," and "90% of those who took an annuity reported that their budget was more predictable, that they felt more financially secure and that it was easier to pay for their needs" [15]. The failure mode the framework guards against is underspending: "[i]ndividuals often refer to some reference withdrawal amount to budget their regular spending, even if this level of spending is not optimal. More than half of retirees over the age of 65 in Australia draw down their superannuation account at the minimum required rate," and in the US "90% considered that the Required Minimum Distribution rates were important in deciding how much to withdraw, and over half viewed these rates as a good guide" — so "[u]sing such references to budget can lead retirees to not spend down their retirement savings and have a lower standard of living than what they should be able to afford" [15]. Inflation protection, conversely, matters less here: "individuals' consumption levels in retirement tend to decline over time across all income levels," with reported enjoyment of spending itself falling with age [15]. And because needs are this heterogeneous, the framework's final test is decision support: generic information demonstrably fails — in the UK "27% of individuals felt that general communication did not help at all with their decision making, with an additional 54% reporting that it only helped a little," and in an experiment "only 14% of individuals made the best decision regarding the amount to withdraw from their pension pot" given general tax information alone — while interactive tools work: Sweden's Minpension planning tool for over-54s "has contributed to an increase in the number of people choosing a lifetime income for their retirement" [15]. Assess any retirement system, then, on four questions: Is the essential floor guaranteed for life? Is liquid shock capacity preserved? Does the discretionary layer flex without inviting underspending? And are people guided, not merely informed? The next chapter turns to the largest asset most retirees hold against all four tests: their home.

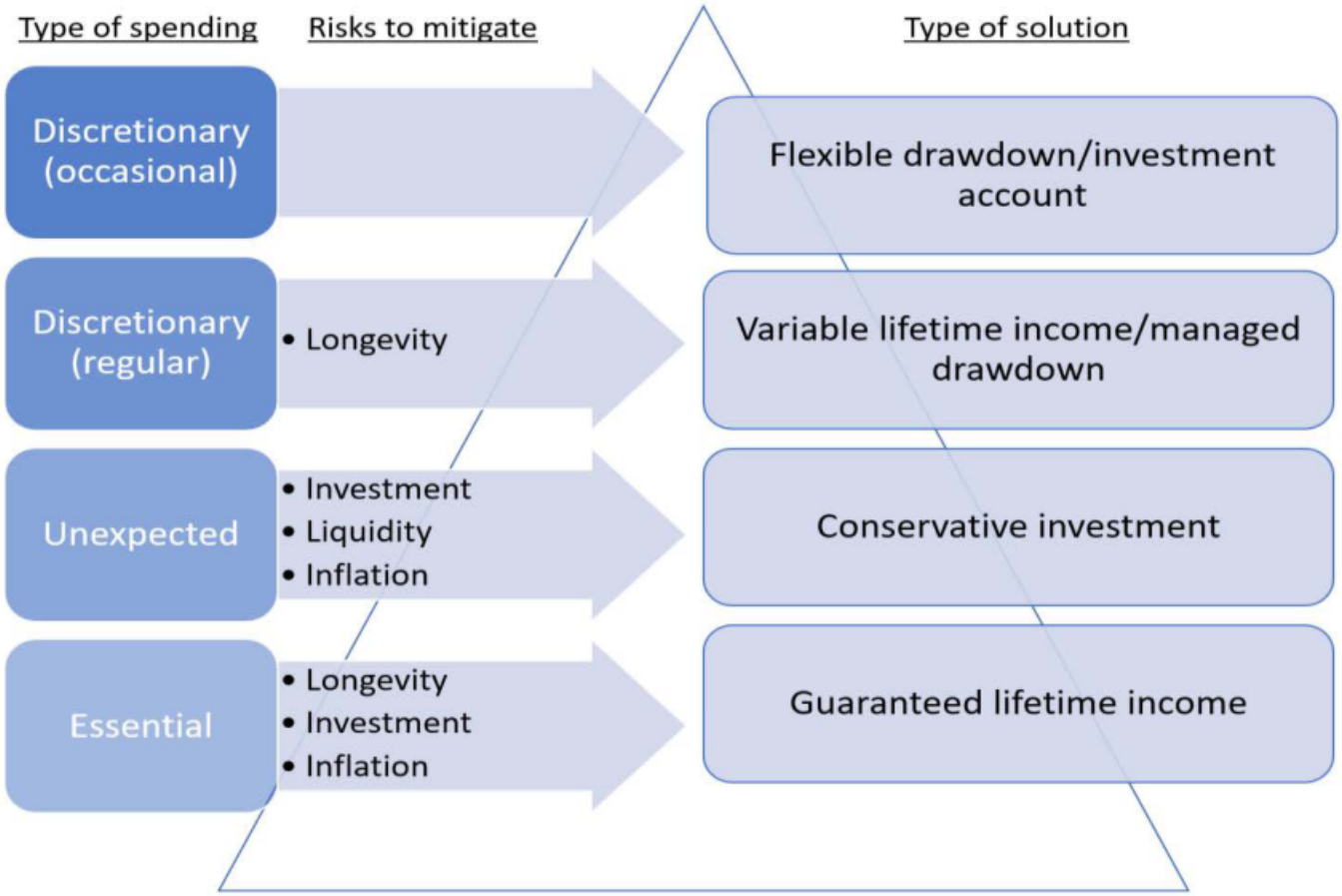


Exhibit. The risks to be mitigated for each class of retirement spending need and the corresponding solutions, with essential spending on basic needs matched to a regular, stable source of finance. Source: S0013.

Housing wealth and home equity release

Overview of home equity release products

The wealth chapter ended on a paradox: the typical older household is asset-rich and income-poor, its balance sheet dominated by a home it cannot spend. Equity release is the product family built to resolve that paradox — converting housing wealth into cash while the owner stays put. Three designs dominate. A **reverse mortgage** is a loan secured on the home with no required repayments until the borrower dies, moves into care or sells; interest rolls up against the property. A **home reversion** sells a share of the home to a provider at a discount to market value in exchange for a lump sum or income and a lifetime right to remain. A **sell and rent back** scheme sells the home outright, with the former owner staying on as a tenant. All three trade the same asset — future housing equity — for present liquidity, and the OECD's survey of the market (Pensions Outlook 2024) shows how tightly the terms are engineered around the central risk: that a long life plus compounding interest outruns the value of the house.

Access is deliberately easy at the front door. "One advantage of home equity release products for retirees is that the eligibility requirements are generally easier to meet, because these products do not generally require income checks or credit assessments. However, there are certain criteria that individuals must meet. These include minimum age requirements, the property value or type and the individual's relationship to it, and the outstanding mortgage remaining to be repaid" [15]. Age gates do the underwriting that income checks do elsewhere: "[m]inimum age limits are the most common feature of reverse mortgage products... The youngest age allowed where limits are formally imposed is 55. This is the case in Canada, Ireland, Korea, and the United Kingdom. A minimum age of 60 is the most common, as in Australia, France, Italy, Japan, Norway and Sweden. The United States requires borrowers in its Home Equity Conversion Mortgage (HECM) programme to be at least 62... Spain requires that borrowers be 65, though Spain is unique in allowing an exception to the minimum age requirement where borrowers demonstrate 33% disability or dependency" [15] — the logic being that "[f]or reverse mortgages, debt can accumulate rapidly over time, and younger borrowers would end up paying a large proportion of their property value in interest" [15]. Collateral rules follow: products "often require that the mortgage is largely paid off or that the home is owned outright... The Housing Pension Programme in Korea (JooTaekYeonKeum, JTYK) requires that the mortgage be fully paid off to

qualify," while "[r]everse mortgage products in Canada generally require that the loan proceeds are used to pay off any outstanding debt and to close any open Home Equity Line of Credit" [15]. Property rules bite unevenly — the US HECM requires principal-residence occupancy, France permits renting the property out with lender agreement, Japanese lenders often refuse condominiums and "the collateral often only counts the land on which the property sits" — and minimum property values "can exclude lower wealth households from being eligible" [15]. The United States added a safeguard after experience: "[t]he HECM programme in the United States has required a financial assessment of borrowers since 2015 to ensure that they will be able to cover insurance and taxes owed on the property and for upkeep of the home… in response to many borrowers maxing out the value of the loan then having to foreclose because they did not set aside funds to cover these costs" [15].

How much can be released is an actuarial dial. "The maximum amount that homeowners can borrow with a reverse mortgage is commonly expressed as a loan-to-value (LTV) ratio… The percentage allowed generally increases with age. This is to mitigate the risk that the outstanding balance of the loan will eventually exceed the property value" [15]. "Maximum LTVs vary across countries, but are generally between around 15% for younger borrowers and 50% for older ones… in Australia the maximum LTV for someone aged 65 is around 20-25%… Canada imposes a maximum LTV of 55%," with one Swedish product reaching 45% at age 90, Norway capping at 49% at 80, and Ireland's Spry product at 40% for 85+ [15]. "The United Kingdom is unique in offering underwritten reverse mortgages, which offer higher LTVs for those in worse health and/or with lower life expectancies" [15] — impaired-life pricing imported from the annuity market. Australia's public scheme ties the dial to the pension itself: "maximum loan amounts in the Home Equity Access Scheme are related to the public means-tested pension benefit to which an individual is entitled… The sum of the public pension and loan payments must be below 1.5 times the maximum rate for the public pension" [15].

The public model shows what the product can do when the state absorbs the risks: Australia's HEAS "allows retirees to convert part of their home equity into a steady income stream at a low interest rate… Retirees can choose regular fortnightly payments, lumpsum withdrawals or a combination of the two… The HEAS offers a government-backed interest rate of 3.95% per annum, significantly lower than commercial reverse mortgage rates," with "[a] vital safeguard ensures that borrowers or their estates will never owe more than the home's value at the time of repayment" — and "[t]he rapid increase in participation – a surge of more than 300% since 2020 – highlights the growing appeal of home equity access" [7]. Private markets, by contrast, barely exist in most countries: "[t]here are many barriers for providers to enter and stay in the equity release market. As a result, markets tend to be very small and concentrated in most jurisdictions… In Australia, two competitors wrote 80% of the market during 2013-2017… there is only one product available in the Netherlands… In Germany, there were only 200 reverse mortgages outstanding in 2015" [15]. The barriers are structural: "[s]tarting in the business requires a very large amount of upfront capital, which the provider will not normally receive back until decades later. Lenders can have difficulty to access wholesale funding. Indeed, the availability of funding has been the largest challenge for providers in the equity release market" [15], compounded by property-price risk — "a drop in home values increases the risk of crossover, where the accumulated loan value could exceed the proceeds obtained from selling the home," the loss crystallised by no-negative-equity guarantees, which is why "[t]here were 20 lenders in Australia before the crisis, and now there are only 3" [15]. The overview's verdict: the mechanism works, the demand is proven where a public scheme de-risks it, and the private market's thinness is a supply failure, not a demand failure. The next three sections take each product in turn, beginning with the reverse mortgage.

Reverse mortgages

The reverse mortgage inverts the ordinary loan: the borrower draws against the home, makes no scheduled repayments, and the debt — principal plus rolled-up interest — settles when the home is sold, the borrower dies or moves into care. Its central control is the loan-to-value cap. "The maximum amount that homeowners can borrow with a reverse mortgage is commonly expressed as a loan-to-value (LTV) ratio… The percentage allowed generally increases with age. This is to mitigate the risk that the outstanding balance of the loan will eventually exceed the property value," a risk the United States priced in after the crisis: it "introduced borrowing limits in 2014 following the financial crisis when property values fell, often below outstanding loan values" [15]. The caps are conservative: "[m]aximum LTVs vary across countries, but are generally between around 15% for younger borrowers and 50% for older ones" — "[o]ne product in Sweden and the Spry product in Ireland offer 15% at age 60, and in Australia the maximum LTV for someone aged 65 is around 20-25%," rising to 45 percent at age 90 in one Swedish product, capped at 49 percent at 80 in Norway, while "Canada imposes a maximum LTV of 55%, which is also the maximum

LTV for the only product offered in the Netherlands" [15]. A 65-year-old can thus typically release only a fifth to a quarter of the home's value — the product is a supplement to pension income, not a substitute for it.

Local calibration fills in the cap. In Norway the LTV also reflects "the housing market and expectations regarding its future appreciation" and is reduced for jointly held cooperative debt; US HECM limits "also vary by area"; "Australia['s] maximum loan amounts in the Home Equity Access Scheme are related to the public means-tested pension benefit," with pension plus loan payments capped below "1.5 times the maximum rate for the public pension"; German providers "can deduct expected maintenance costs" [15]. The United Kingdom is "unique in offering underwritten reverse mortgages, which offer higher LTVs for those in worse health and/or with lower life expectancies" — impaired-life pricing imported from the annuity market [15]. Some products let borrowers step up later: HEAS borrowers "can request an increase in the LTV that corresponds to their age," and one Swedish product raises limits when home equity grows, for a fee [15]. Absolute bounds also apply — Canada's minimum loan is CAD 20,000 and maximum CAD 750,000; the Dutch product runs from EUR 30,000 to EUR 250,000 [15].

Payout form is the second design axis, and it prices the borrower's discipline. A lump sum means "borrowers receive the maximum loan amount all at once" and "will owe interest on the entire amount from inception" — costly if the money is really for gradual spending. A fixed annuity spreads the maximum "over a fixed number of years... but it will not provide an income for life so they will see their finances decrease after the fixed term ends." A line of credit "minimises the interest paid, because the borrower will not owe interest on funds they have not used," though flexibility "may come at an increased cost (i.e. higher interest rates)." A life annuity pays "a regular fixed amount until the borrower dies. This is most useful to provide a regular income in retirement and protect the borrower from longevity risk, but again may come at a cost" [15]. The menu is the needs framework in product form: the same house can fund a lump-sum repair, a term income bridge, a standby shock reserve, or lifetime income — at four different interest bills and four different risk allocations between borrower and lender.

Home reversions

The home reversion replaces debt with a sale. The owner sells part — or all — of the home to a provider for a lump sum or instalments, and in exchange keeps the right to live there for life. There is no loan, no interest clock and no crossover risk for the household: the provider simply owns its share from day one and realises it when the owner dies, moves into care or sells. The price of that simplicity is paid up front and permanently: the share is bought at a discount to its market value — the provider's compensation for an unknown wait, an illiquid asset and house-price risk — and the sold share, with its future appreciation, is gone for good. The reversion is therefore the mirror image of the reverse mortgage: certainty about what is given up, in exchange for certainty about what is kept.

The product's documented forms are few and recent. In Australia, "[i]ndividuals can sell their home partially or in full in exchange for a lump sum. There is usually an upfront fee of around AUD 2 000, plus transaction costs. Homeowners may have the option to buy back the sold share, and products can provide a rebate to heirs if the owner dies earlier than expected" [15]. New Zealand got its first reversion only in 2023: "[t]he provider Lifetime introduced a home reversion product to the market in 2023, the first product of this type to be available in New Zealand. In exchange for a partial sale of the home equity, individuals receive payments in instalments over ten years, with the option to extend by ten more years if they are under age 90. Lifetime occupancy is granted" [15]. Germany is the outlier where the sale model beats the loan model: "[h]ome reversions remain more popular than reverse mortgages, but both markets are small. In 2015, there were less than 1 000 outstanding home reversions, and only 200 reverse mortgages" [15]. The public sector experiments too: "[s]ome municipalities have home reversion programmes where they pay residents monthly payments for life in exchange for their home" [15] — a municipal annuity secured on the housing stock. In one further market covered by the OECD annex, "[h]ome reversions are more prevalent [than reverse mortgages], and pay life annuities" [15].

Because the transaction is irreversible in a way a loan is not, the fairness mechanisms cluster around early exit. "Home reversions and sell-and-rent-back schemes can provide rebates in the case that the seller dies or sells their property early, though they are not normally required. Lifetime's home reversion product in New Zealand will reduce the acquired equity accordingly if the home is sold before the ten-year fixed annuity is fully paid out. In both Australia and the United Kingdom, products can provide a rebate to heirs if the seller dies earlier than expected. In Germany, providers may pay a lump sum or a fixed term annuity to compensate for the unused right of residence" [15]. Without such rebates, a seller who dies at 70 having sold a share priced for occupancy to 95 transfers a windfall to the provider — the reversion's version of the annuity's mortality cross-subsidy, except that here it is visible, contractual and therefore contestable.

The reversion also sits closest to the chapter's consumer-protection frontier, because its customers sign away property, not just borrowing capacity. "Equity release products are very long-term and complex products which can provide a sizable payment to homeowners, with comparatively little direct cost for them or ongoing payment obligations. It can therefore be difficult for homeowners to fully understand the potential consequences that the product could have for them twenty or thirty years in the future" [15]. "Disclosure is one approach to promote understanding, though alone it is generally not sufficient" [15]: Australia's responsible-lending regime, introduced in 2012, requires "an assessment of the needs of the borrower... the disclosure of home equity projections from MoneySmart's calculator, an information statement, a tenancy protection warning, and annual account statements" [15], and its suitability rules require providers "to assess the future needs of the borrower, not only their current situation" [15]. The UK requires providers to take "'reasonable steps' to ensure that the product is suitable. However, the Financial Conduct Authority found that this assessment is often treated as a box-ticking exercise" [15]. Vulnerability is the deeper problem: "[a] particular concern with these products that target elderly individuals is that they are at higher risk of having cognitive decline and may not be capable of fully understanding the product and its implications... the United Kingdom requires that an independent solicitor certify that the person making the decision to take the product is competent to do so. Providers in France face marketing restrictions, and are prohibited from using certain tactics such as door-to-door sales" [15]. "The higher potential for cognitive decline also raises the risk of elder abuse, where relatives or other people close to the individual may convince them to enter an equity release agreement in order to take advantage of the financial gains... ASIC found that providers in Australia were not doing enough to watch for signs of elder abuse" — red flags include "repayments being made by an adult child, money transfers to a non-borrower, money transferred to a child, the involvement of children in the application process" — while working safeguards include "limiting the power of attorney where non-borrowers may benefit, and only disbursing money into the borrower's account" [15]. Advice mandates are the standard backstop — "[i]ndependent financial advice is required in Canada (in some provinces), New Zealand, Spain and the United Kingdom," with counselling-style information sessions in Japan and the US — yet "[a] requirement to obtain financial advice may not always be sufficient... The FCA found that advisors in the UK... did not sufficiently challenge the individual regarding whether the product was suitable for them," and "[i]n Australia, advisers have been reluctant to provide guidance about equity release products for fear of liability, given the prevailing negative perceptions around the products" [15]. The reversion's lesson generalises across the chapter: the products are sound, the protections are written, and the weakest link is the human checkpoint in between. The next section examines the third design — sell and rent back schemes, where the former owner stays on not as owner-occupier but as tenant.

Sell and rent back schemes

The third design completes the spectrum of ownership surrender. A reverse mortgage keeps the owner an owner, borrowing against the home; a reversion sells a share while guaranteeing occupancy for life; a sell and rent back scheme sells the home — in whole or in part — and the former owner stays on by paying for the privilege, as a tenant in what was their own house. It is the most complete conversion of housing wealth into cash, and the one where the right to remain is weakest: not a property right but a contract term.

The payment mechanics show how the equity drains. "The payments from sell and rent back schemes come in a variety of designs. In Australia, proceeds from the partial sale of the home can be paid out as a lump sum or in instalments. The seller then pays 'rental' fees to the buyer as a percentage of the buyer's equity in the home, which serve to transfer that additional equity to the buyer. As such, the transfer of equity accelerates over time as the payments by the seller increase" [15]. The Australian annex states the endgame plainly: sellers "pay fees as a proportion of the sold equity, which are paid through an additional transfer of equity... The individual may have the right to stay in the home when they retain 0% equity only if the contract allows it" [15]. Rent, in other words, is paid in slices of the house, on an accelerating schedule, and whether the seller can remain once nothing is left is a matter of drafting. The Dutch design shares risk more symmetrically: "[i]n the Netherlands, the Verzilvermijvast product pays 50% of the home value as a lump sum to the seller. The seller then pays 6% of the house value annually. When the seller moves out or dies, Verzilvermijvast makes a final payment of 50% of the difference in the home value. As such, Verzilvermijvast shares the profits and losses from the variation in the home's value with the seller" [15] — half the upside and half the downside stay with the household, a genuine partnership rather than a clean sale. Early-exit fairness exists here too, as with reversions: such schemes "can provide rebates in the case that the seller dies or sells their property early, though they are not normally required" [15].

Across all three designs, the contract is where protection lives. "The contractual terms of the product must lay out the various rights and obligations of the homeowner and the conditions to terminate the product. These include the

costs and fees that the homeowner must cover, their right to exit the contract, the obligations to maintain the property, rights for lifetime residency and tenancy protections, the ability to transfer the contract to a new property, the delay allowed to transfer owed funds to the provider, and insolvency protection for the homeowner" [15]. Costs are real but easily hidden: "the transaction itself can incorporate some of these costs by adding them to the loan value or reducing the benefit transferred, so homeowners do not necessarily feel the full initial costs of the product. Homeowners are normally obliged to cover all costs relating to the establishment of the contract. This often includes an initiation fee, an ongoing servicing fee, and any third-party charges such as the valuation of the property, inspections, and the like" [15], with fee caps where regulators impose them — for US HECM loans, "[o]rigination fees are capped at USD 6 000" [15]. Regret is provided for: "[s]everal countries require that providers give homeowners time to change their mind about signing the contract. Such cooling-off periods are normally around 30 days, and are required in Canada, France, and Poland" [15]. And occupancy is conditional everywhere: "[c]onditions of the contract normally require homeowners to meet all tax obligations, insurance premiums (e.g. fire, flood), and ensure the normal maintenance of the home… Not fulfilling these obligations can be grounds for early termination of the contract and the forced sale of the home" [15] — enforced with varying intrusiveness, from the HECM's upfront demonstration "that they have sufficient funds to cover the expected costs," to German set-aside funds, to Sweden, where "homeowners must allow for regular inspections to ensure the upkeep and maintenance of the home" [15], and in Canada extending to termination for "using the money for something illegal or lying on the application" [15]. Australia's disclosure regime even names the tenure risk directly, requiring "a tenancy protection warning" among its mandated disclosures [15].

The chapter's synthesis can now be stated. Housing is the largest single asset of the older population — median net wealth peaks at 65–74, dominated by property — and three workable mechanisms exist to spend it without moving: borrow against it, sell a share of it, or sell it and rent it back. They differ in who keeps the appreciation, who bears the price risk, and how secure the roof is at the end; they agree in being small, complex and guard-railed, with public schemes — Australia's HEAS, Korea's JTYK, America's insured HECM, municipal reversions — repeatedly supplying the trust and risk absorption that private markets alone have not. Against the needs framework, equity release scores well on shocks and discretionary top-ups, cautiously on essentials (only the life-annuity payout forms guarantee lifetime coverage), and poorly wherever minimum values and loan floors exclude exactly the low-wealth households whose incomes are thinnest. The asset is there; the plumbing is half-built. The report turns next to the distribution that runs underneath every chapter so far: ageing unequally.

Ageing unequally: inequality across the life course

Preventing ageing unequally

Every preceding chapter has reported averages with a distribution hiding inside them: life expectancy with an education gradient, poverty concentrated among women and the oldest old, training that flows to the already skilled, housing wealth that rescues some balance sheets and not others. This chapter makes the distribution itself the subject. Its source text, the OECD's Preventing Ageing Unequally (2017), opens with a warning that reverses the report's founding optimism: "[t]he health status and income levels of older people have substantially improved over the last decades in OECD and emerging countries. This Preventing Ageing Unequally report shows, however, that risks of increasing inequality among future retirees have been building up" [3]. The drivers are the report's own themes turned adverse: "[w]ith longer life expectancy, declining family size, higher inequality over the working lives and reforms that have reduced future retirement income and tightened the links between pension contributions and benefits, the experience of old age is going to change dramatically for today's younger generations. Financing of pensions and caring for elderly people in particular might become more difficult, aggravating the burden for the most disadvantaged. The sustained and fairly generalised improvement in the living standards of older people is unlikely to be prolonged in the forthcoming decades" [3]. Today's pensioners, in other words, may be the high-water mark.

The mechanism is cumulative advantage and disadvantage across the life course. "Preventing Ageing Unequally clearly shows that socio-economic outcomes of individuals build up from childhood to adulthood, and inequality of a given cohort later in life largely results from differences in individual experiences accumulated over the life course. Better health early in life tends to be correlated with better educational attainment and they both contribute to better labour market outcomes. In turn, people with stable and well-paid jobs also tend to enjoy good access to health and other services, and are likely to retire with adequate income. These people are also more likely to have insurance or financial assets to cover them in case they need long-term care. By contrast, people who had low-quality jobs, low incomes and unstable careers are likely to be in worse health and face higher risks of poverty and if they need long-

term care, what they get is likely to be rather dismal" [3]. The compounding begins before school: "[u]nequal ageing begins in childhood. Children from more privileged socio-economic backgrounds are healthier, perform better in school, obtain higher degrees and have better chances of succeeding later in the labour market… Finally, despite the continuing growth in life expectancy, people do not necessarily spend the end of their working lives and retirement years in good health – there are, for instance, wide disparities in disability rates along gender, educational and geographical lines. The wide variety of health and socio-economic outcomes over people's working lives can add up to significant inequality in old age" [3]. Each domain hands its inequality to the next: health to education, education to earnings, earnings to pension entitlements, entitlements to care access — which is why single-point interventions late in life arrive after the multiplication has already happened.

The context is a generation-long widening that ageing policy did not cause but must now absorb: "[e]arnings and household income inequalities have increased over the past three decades in most countries, fuelled by the rising share of non-standard work, difficulties encountered by many young people in securing a foothold in the labour market and high unemployment in some countries" [3]. Tighter contribution-benefit links — the actuarial fairness reforms of the pensions chapter — mechanically convert that wider working-life dispersion into wider pension dispersion, removing part of the redistribution that used to compress it. The OECD's response is organised by timing: "Preventing ageing unequally requires a comprehensive policy approach to help individuals overcome disadvantages that could cumulate over their life course and result in bad health, low income and poverty at old age… education, health and employment experiences interact and compound inequality. This also means that inequality-reducing policies in one area will spill over into other areas generating additional welfare effects. The report identifies policies that can contain if not fully address the risks of unequal ageing. It suggests a policy agenda to prevent, mitigate and cope with these inequalities, drawing on good practices in OECD countries and emerging economies, which led to the OECD Action Plan on Preventing Ageing Unequally endorsed by the OECD Ministerial Council Meeting in June 2017" [3]. Prevent early (childhood health and education, school-to-work transitions), mitigate mid-course (health at work, training, career repair), cope late (floors, care access, poverty relief) — with prevention carrying the highest return precisely because compounding works in policy's favour too when it starts early. The analysis that follows tests the framework against the generational evidence: how life course inequality is changing across generations.

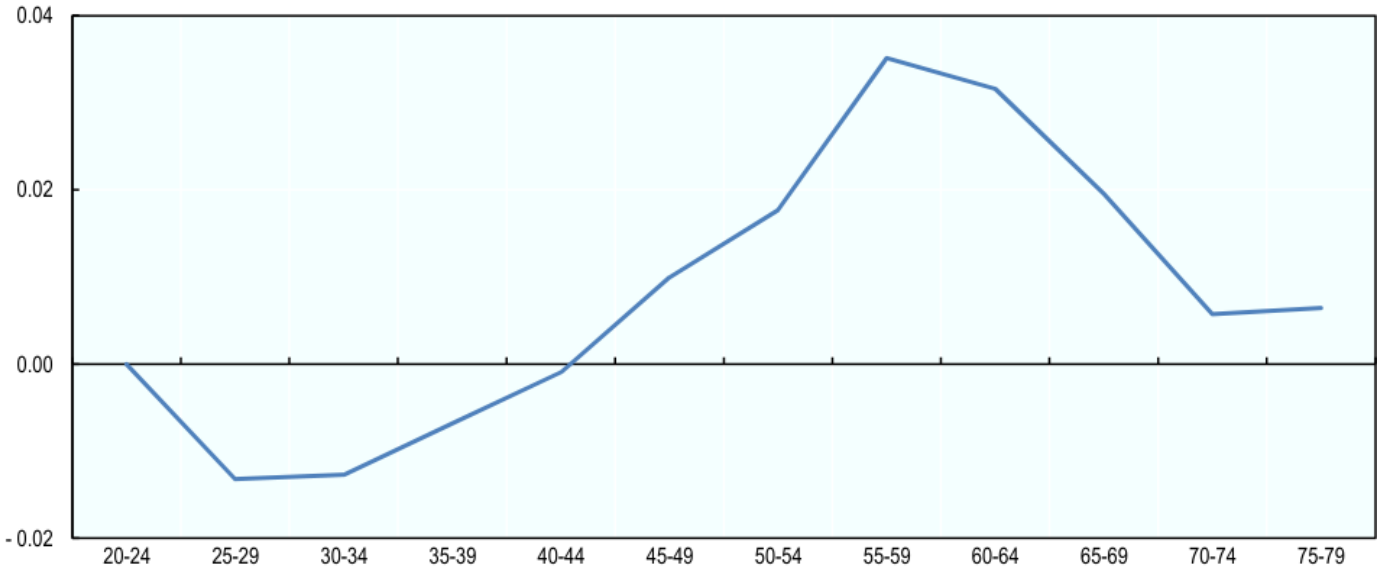


Exhibit. Estimated age pattern of income inequality within cohorts, measured by the Gini index relative to the 20–24 reference age group and averaged across cohorts and countries, from early adulthood into old age. Source: S0016.

Life course inequality across generations

How much inequality a generation carries into old age is decided less by its pension formula than by the shape of its working lives — and those shapes are changing in opposite directions for women and men. The OECD's generational analysis (Preventing Ageing Unequally, ch. 4) measures the input directly: expected lifetime earnings for five-year birth cohorts from 1940–44 to 1970–74 across 13 OECD countries, built from hourly wages, hours worked, employment probability and survival probability. The first finding is that education divides lifetimes far more than it divides pay packets: "[d]ifferences in lifetime earnings by level of education are greater than in average hourly wages given the education gradient in the other components of lifetime earnings – employment, hours worked and survival probabilities – particularly among women. For example, the ratio of lifetime earnings of highly educated women born in

1970-74 to those of their low-educated peers is estimated at 3.10" [3]. The decomposition reverses the usual intuition: "[t]his ratio would be reduced to 1.95 if both groups had the same employment rates" [3] — the employment gap, not the wage gap, does most of the work — while "differences in survival probabilities at working ages between education groups are rather small and contribute little to the education premium in lifetime earnings" [3] (equalising survival would move the ratio only from 1.95 to 1.90).

Across generations, the two genders' trajectories diverge. "On average among the 13 OECD countries, women's lifetime earnings have increased from one generation to the next at all levels of education. For men, they have risen slightly among the highly educated and fallen a little among the low-educated" [3]. Convergence followed: "[f]irst, the lifetime earnings of younger generations of women have caught up with those of men educated to the same level. The gender gap in lifetime earnings has shrunk by about one-third – from 45% among highly educated cohorts born in 1940-44 to about 30% among those born in 1970-74, and from 80% to about 50% among the low-educated in the same birth cohorts. The pattern is driven chiefly by employment trends. While women's lifetime employment rates have risen robustly from older to younger cohorts, men's have tended to decline. Nevertheless, remaining gaps are wide" [3]. Since pension entitlements are capitalised lifetime earnings, this is tomorrow's gender pension gap narrowing — from 80 per cent to 50 per cent at the bottom of the education distribution — even as it remains, in the OECD's own words, wide.

The second movement is the education premium splitting by sex: "[s]econd, the education premium of lifetime earnings has declined over generations among women, but has risen among men. The prime reason is that the education gap in employment rates has moved in opposite directions for women and men" [3]. As women's employment generalised, low-educated women's employment rose toward their educated peers', compressing the ratio; as low-educated men's employment eroded, their lifetime earnings fell away from educated men's. The levels remain strikingly national: "[i]n the 1970-74 female birth cohort, it ranges from 1.7 in Denmark to 4.3 in Greece, and from 1.6 in the Netherlands to 2.9 in the United States among men" [3]. "As for patterns in the high-to-low education ratio, it has declined among women from one generation to the next, except in Finland, France and Switzerland, and has increased among men, except in France and Italy, where it has remained broadly constant, and in Australia where it has decreased slightly. Among women, the cross-generational fall has been very steep in Greece, Spain and Ireland. In Ireland, for example, the ratio of lifetime earnings has fallen from 24 for the oldest cohort to about 4 among the youngest" [3] — the 24 being what a generation of low-educated Irish women, largely outside paid work, earned against graduates; its collapse to 4 is female employment itself arriving.

The mortality side of the ledger moves the other way. New measurement finds the education gap in remaining life at 65 "substantially larger in most countries than those found in previous studies, due to more precise information on mortality after the age of 75"; in Norway "the life expectancy gap between high and low educational groups was four years among both men and women in the 1970s, widening to five among women and seven for men in the late 2000s" [3], and across 17 European countries "absolute inequalities in self-assessed health were mostly constant… between the 1990s and the 2000s, whereas relative inequalities grew" [3]. Longer lives, then, are being distributed more unequally even as women's earnings converge: the generations now retiring carry a compressing earnings distribution for women, a polarising one for low-educated men, and a widening longevity divide for everyone — the raw material from which the pension system, pass-through included, will manufacture the old-age inequality of the 2040s. The next section looks at where that inequality lands: living conditions in later life.

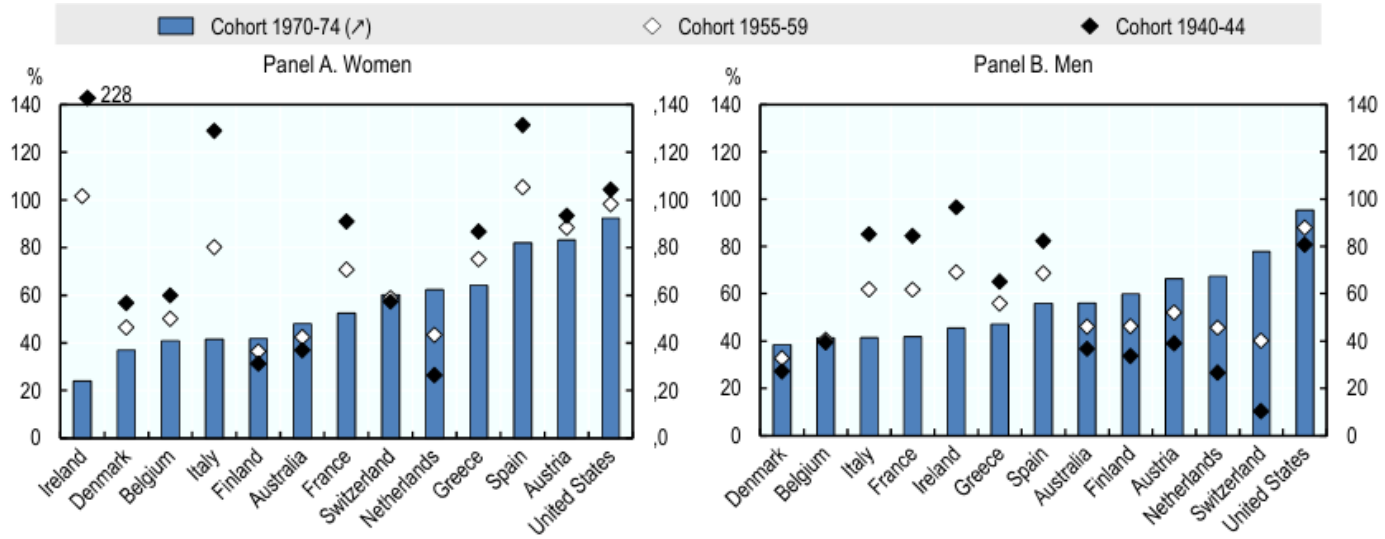


Exhibit. Career average of the education premium in real hourly wages for three birth cohorts — 1940–44, 1955–59 and 1970–74 — showing how the earnings advantage of higher education has evolved across generations. Source: S0016.

Living conditions in later life

Accumulated inequality is finally lived somewhere: in a region, a household, a body. The UN's World Social Report states the chapter's balance sheet plainly: "[o]ld-age pensions and access to health care are critical in reducing poverty and inequality among older persons. But they are not sufficient on their own. Ensuring economic security and health for all in old age calls for promoting equal opportunity from birth, including through universal access to health care and education as well as opportunities for decent work" [1]. The reason is structural dependence: older people "begin to work less or stop working altogether due to health reasons or family responsibilities, or because they are obligated or wish to retire at a statutory retirement age. Discrimination may undermine further employment. While many older persons remain productive, their economic well-being depends on the availability of public income support, affordable health care, family support and personal savings to a greater extent than for the working-age population" [1]. "Not everyone has benefitted equally from the advances in health and living conditions that drive population ageing. Many of today's older persons are in good health. Others live with multiple ailments or severe disability. Some are economically active and enjoy income security but many live in poverty. Some have strong social ties. Others lead isolated lives. Population averages conceal these inequalities and the diverse needs and abilities of older people" [1].

Geography is the first lived dimension. "Regional differences in the share of the older population are very large within many OECD countries. Combined with a wide disparity in availability of health services across regions, this reveals that access to health services for the elderly is problematic in some of them. Recently, rural areas have aged faster than urban areas, which is likely to stretch health services and public transport to their limit and amplify the risks of isolation" [3]. Access is not a soft variable: "[p]hysical obstacles to accessing health care services not only translate into higher user costs, they may even contribute to higher rates of morbidity. Because very old people are most affected by poor accessibility, they pay the price of higher costs, possibly with their lives" [3], and "[t]he availability of physicians is positively related to a region's life expectancy at birth – an effect that remains significant even when controlling for GDP per capita" [3]. Place, in short, is a longevity policy whether governments treat it as one or not; "preventive health care or the use of nurse practitioners can help alleviate strained health care services in difficult to service regions. However, they cannot fully replace doctors and hospitals" [3].

The household is the second dimension, and it is feminised and very old. "[T]he over-80s are in better health than previous generations. However, a large fraction of the elderly report depressive symptoms, while caring for a frail partner at older ages is likely to become more frequent, thereby putting a strain on mental and physical well-being. Moreover, a high proportion of women aged 80 and over live alone, which combined with a persisting gender pension gap, increases the risk of falling below the poverty line" [3]. Care itself is the third dimension, where the life course presents its final bill: "[i]nequalities that have cumulated over the life course are magnified if the costs of LTC borne by poorer individuals are not well covered by social protection systems... The proportion of home care costs covered by social protection systems differs widely across countries. This implies that in some countries people who have LTC needs might in addition fall into poverty, choose not to have those needs met or opt for institutional care. Going into institutional care is typically not people's first choice, but in many OECD countries the costs of insti-

tutional care are better covered than costs of home care" [3] — a coverage inversion that pushes people toward the setting they least want — alongside "substantial inequalities in the provision of informal care across genders" [3].

The UN's verdict on trajectory closes the section and echoes the chapter's opener: "[w]ithout swift policy action to prevent it, future cohorts of older persons are likely to be economically more unequal than those today," and "[w]hile worries over fiscal sustainability have dominated pension reforms, pension benefits are still too low in many countries and beyond the reach of some groups. Inadequate investment in economic security at older ages risks jeopardizing progress in reducing poverty among older persons. It may diminish the well-being of future cohorts of older persons and further undermine already shaky trust in Governments" [1]. The labour-market root is named too: "[i]nvestments in decent work are paramount. Decades of wage stagnation are jeopardizing people's ability to save enough to support themselves in older ages. Informal employment and various non-standard forms of employment limit workers' social protection coverage and their contributions to public revenues" [1]. Living conditions in later life are thus not a welfare afterthought but the observable sum of every earlier chapter — and the strongest lever sits at the very beginning of the chain, as the closing section shows: education, and what it does to later-life outcomes, with early investment returning a cumulative 13 per cent a year by age 65 in the US evidence [3].

Education and later-life outcomes

If one variable threads every gradient in this report — earnings, employment, disability, longevity, pension wealth — it is education. The closing section of the inequality chapter examines it as the master determinant and asks what follows for policy timing. The stakes are visible in a single comparison: "[i]n 2006, 25-year-old men without a high school diploma had on average a life expectancy 9.3 years less than those with a bachelor's degree or higher; women without a high school diploma had a life expectancy 8.6 years less than those with a bachelor's degree or higher" [16] (US National Center for Health Statistics). The association is not merely correlational bookkeeping: "[n]umerous studies document that education is strongly associated with health over the lifecourse... Education is strongly associated with psychological, social and financial resources that are helpful to deal with health shocks" [9], and quasi-experimental evidence finds educational attainment itself reduces cardiovascular risk factors — "reduced smoking, depression, triglycerides, heart disease, and better HDL" [9]. "Formal education has preventive effects as it lowers health risk, improves cognition and prolongs life expectancy, regardless of being in the labour force or not... Not only formal education, but also lifelong learning through volunteering and training seem to support mental health and cognitive development among adults and older adults" [9] — though "[p]rotective health effects of education may weaken with age" [9], which is an argument for starting early, not for despairing late.

The OECD's causal chain runs through childhood: "[c]hildhood circumstances affect education and later life health, as well as future labour market experiences and career progression. For example, effects from adverse health events early in life on health at older ages can be indirect, e.g. channelled through lower educational achievement and restricted life opportunities, and/or remain latent for a long period" [3]. Latency is what makes late correction so weak: "[i]n contrast, corrective measures at later stages have to overcome what may be more deeply entrenched problems... and can produce benefits only over a shorter remaining period of life. Likewise, pension arrangements can correct some but not all of the inequality among the retirees that results from employment and earnings inequalities. Thus, substantial savings of public expenditure could be made if income, wealth, education and health inequalities were picked up earlier and addressed at younger ages" [3]. The returns are quantified: "[p]olicies to support children's educational, health and material well-being must start in early childhood and be sustained throughout childhood... US research suggests that each dollar invested in high quality programs before the age of five years yields a cumulative annual return of 13% at age 65 (Garcia et al., 2016), and that these benefits are especially large for children from a disadvantaged background. OECD Programme for International Student Assessment (PISA) data show that 15-year old students who attended preprimary education for at least one year are likely to have higher maths scores than those who did not, the gap being equivalent to almost one year of formal schooling after accounting for students' socio-economic status" [3]. "[P]reventing child income poverty and material deprivation is a priority. Beyond their obvious negative impact on children's material well-being, they generate malnutrition and bad health, which in turn affect later life outcomes" [3].

Education's compositional power even bends population aggregates. Microsimulation for Belgium, Italy and the United States (OECD Global FEM) shows that "[s]hifts in the composition of populations in terms of educational attainment over time have a significant impact on these trends. Even though disability-free years are expected to decrease by more than half a year in each educational group in the United States, the aggregate is expected to remain almost constant. Similarly in Belgium, the aggregate increase in disability-free years across cohorts is over three years, whereas the increases in each educational group are around two years" [3] — rising attainment quietly subsid-

ising the health statistics. The same projections show post-50 working lives converging internationally ("Italians and Belgians born in the late 1960s are expected to work only four months less after age 50 than their US peers; this difference exceeds four years for Belgians and five for Italians born in the early 1940s… strongest among women, from 4.4 to 10.7 years in Belgium and from 3.4 to 9.4 years in Italy, against only 1.0 year from 9.7 to 10.7 years in the United States") while dividing by education at home: "[i]n all three countries, there are substantial and persistent differences in the length of working lives by level of educational attainment. Highly educated born in the late 1960s are expected to work longer after age 50 by 5.2 years in Belgium, 6.3 years in the United States and 6.9 years in Italy. These duration gaps are widening from 4.8 years for the early 1940s cohort in Belgium and 6.1 years in Italy. By contrast, the duration gap is narrowing slightly from 6.6 years in the United States" [3]. Pension-receipt duration swings with policy: "stable on average across cohorts at about 17.5 years" in the US, rising in Belgium "from about 14 to 16.5 years… due to life expectancy gains," and in Italy falling "sharply from slightly more than 18 years to about 14 years due to higher effective retirement ages" [3] — the same country holding both the longest and shortest claiming spans, a generation apart.

The institutional conclusion belongs to the New Map of Life: the money follows the young. Philanthropic and public investment in education remains "overwhelmingly targeted toward Americans under the age of 25," even though learning's returns now accrue across a hundred-year life; what is needed is to "[m]ake targeted investments to encourage education and learning across the entire life course… to change the cultural script so that learning becomes recognized as a lifelong endeavor," mobilising early-childhood and lifelong learning together "as a civic project" [16]. Education closes the inequality chapter where the chapter began: old-age inequality is manufactured early, priced by employment, and delivered by pension formulas — so the cheapest, fairest longevity policy is a school. The report turns next to the opportunity side of the ledger: the global longevity economy.

The longevity economy and age-friendly societies

The global longevity economy: outlook

The longevity economy is already the world's third-largest market hiding in plain sight, and its centre of gravity is moving south. "As of 2020, 1.9 billion people—or nearly one quarter of the global population—were age 50 or older. This group is expected to grow by roughly 70% during 2020-2050, so that one in three people (representing 3.2 billion individuals) will be age 50 or older by 2050" [17]. The distribution inverts over the same period: "[n]early two-thirds of the global 50-plus population currently resides in high-income and upper-middle-income economies," but "[b]y 2050 they are projected to be home to 44% of the world's 50-plus population, up from 34% in 2020" for low- and lower-middle-income economies, "driving expanded economic impacts from the 50-plus population" [17]. The outlook's core claim follows: "[h]ow rapid aging leads to a larger 50-plus cohort with strong consumption power, which in turn fuels the broader economy" is the dynamic policymakers and businesses must plan around [17].

The surprise is how international this economy already is. "Among the 76 economies in this report, 23 derive more value from the incoming spending by foreign 50-plus populations than domestically. This is common even in economies with the oldest populations: of these 23 economies, 14 have relatively large 50-plus population shares (above 30%), yet they still rely more heavily on older people abroad as a dominant source of growth. For instance, in Malta, where 40% of the population is above age 50, foreign impacts far outpace domestic impacts (22% vs. 9%) —driven in large part by Malta's massive tourism sector" [17]. Youth is no exemption: "[c]ross-border impacts are similarly crucial for five of the ten youngest economies—including Botswana, Ghana, Rwanda, Senegal, and the Democratic Republic of Congo (DRC)—all of which saw overseas contributions drive around half or more of the impacts to their GDP in 2020. This highlights how all economies—even those with small 50-plus populations—benefit from older consumers around the world" [17]. The exceptions prove the trade mechanism: "[i]n Zimbabwe, Nigeria, Ethiopia, and Kenya, just 9-11% of the population are age 50-plus, yet their domestic impact easily outpaces the relatively minimal impacts from older consumers abroad," a pattern the report ties to lesser reliance on global markets [17].

Read against the chapters behind it, the outlook reframes the fiscal story. The same cohort that pension projections treat as a dependency cost is, in market terms, a consumption engine whose spending crosses borders through tourism, remittances of demand and traded services — large enough that a fifth to a half of its GDP impact lands in other countries, including countries too young to have a longevity economy of their own. The strategic implication cuts both ways: ageing economies that suppress older people's incomes and participation are shrinking their export

markets as well as their tax bases, while young economies are stakeholders in rich-world pension adequacy whether their own demographics show it or not.



Exhibit. The economic contribution of people aged 50 and older to the global economy, quantifying the spending and production generated by the longevity economy. Source: S0030.

Future-proofing the longevity economy and longevity economy principles

Measurement invites design. The World Economic Forum's Future-Proofing the Longevity Economy (2025), building on its 2024 Longevity Economy Principles, starts from an explicit reframing: "[t]he idea that an ageing society is a burden, and will deplete resources and tax social services, needs to be reframed: in a multistage life, everyone can contribute and benefit at different times depending on their circumstances. The challenge, then, is creating systems that allow people to contribute to society at various stages of life, whether that means transitioning into entrepreneurial ventures, shifting careers or engaging in informal care provision, while also increasing the number of years in which people are healthy enough to make a positive contribution to the economy" [7]. The principles that follow are less a programme than a stance toward every system this report has audited: "[a]n inclusive and financially resilient future is not just about making adjustments to existing systems, but about rethinking how individuals can be supported in their economic roles over time. Whether what is needed is providing access to lifelong learning, supporting carers or ensuring adequate retirement savings, the future of work (and life) – and the financial systems that support it – needs to be flexible and inclusive" [7].

The urgency is demographic arithmetic, not advocacy: "[m]ore than one in four people worldwide now reside in a country where the population has already peaked in size, signalling a profound shift in labour markets and economic systems. Meanwhile, global life expectancy has increased significantly over the past century; however, this progress masks stark disparities – life expectancy can exceed 80 years in high-income countries yet remain below 60 in some low-income nations. As birth rates decline, some of the world's most advanced economies would need to at least double their productivity growth rates to sustain historical improvements in living standards" [7]. Two further transitions intersect the demographic one — "[t]he rapid acceleration of technological advances and artificial intelligence, as well as the escalating impact of climate change and extreme heat. Taken together, these forces create both an urgent challenge and an unprecedented opportunity to rethink financial resilience and economic participation across all life stages" [7]. The climate link runs both ways: by 2050 "the interplay between climate change and longevity will shape both human well-being and economic development. Climate change directly impacts health and longevity through rising temperatures, extreme weather events and the spread of infectious diseases, which disproportionately affect vulnerable populations, including older people. Conversely, the longevity economy presents an opportunity to drive sustainable development… pension funds and their $28 trillion in assets can finance green growth, ensuring a healthier planet and population" [7].

The work programme is organised in five focus areas, each mapping to a chapter of this report: "[e]conomics of caregiving and long-term care: Providing care, for both children and older relatives, disproportionately affects women and has long-term implications for financial stability… Pathways for economic growth: Examining how financial systems must adapt to support people through different life stages, career changes and caring responsibilities…

Building resilient public retirement systems… ensuring they are financially sustainable and inclusive… From accumulation to decumulation: With people living longer, traditional retirement systems must be revamped to ensure that retirement savings can sustain individuals throughout their lifetimes… Role of employers in financial well-being: Employers can play a crucial role in enhancing financial well-being through programmes that provide support for employees at all stages of their lives" [7]. The accompanying trend judgements are blunt: "[a]s populations age, public retirement systems face growing challenges with regard to financial sustainability… The global shift from defined benefit to defined contribution plans places greater responsibility on individuals to manage their own financial security… The growing demand for care services requires urgent policy and workplace solutions to support both carers and care recipients… Flexible social systems are essential for enabling individuals to navigate economic transitions, which in turn allows them to continue contributing meaningfully to long-term economic growth" [7].

The closing claim is that the future is not pending: "[t]he 'future' of the longevity economy is not in some distant moment – it is already here. Around the world, innovative solutions are emerging to meet the challenges of this demographic transition, and these can be scaled, adapted and integrated into broader financial and social frameworks… Without bold action, the world risks financial instability, overwhelmed care systems and a workforce unprepared for the future" [7]. "The question is not whether change will come – but whether stakeholders will shape it or be forced to react to it" [7]. And the dialogue deficit is named as part of the problem: "[a] profound public debate about the future of the longevity economy – in which a sound, resilient pension system is just one aspect – has been largely absent from global discourse" [7]. Future-proofing, on this account, is co-design across government, business and civil society — "creating new opportunities for engagement, economic growth and collective well-being" by designing "financial pathways… to support individuals at every stage of their life" [7]. Principles stated, the chapter returns to the ground: how age-friendly the built and social environment actually is, beginning with global age-friendly cities.



Exhibit. Growth in participation in home equity access since 2020 — an increase of more than 300 per cent — as retirees facing decumulation uncertainty turn to their housing wealth as an alternative income source. Source: S0010.

Global age-friendly cities

The longevity economy has an address, and it is increasingly urban. The WHO's Global Age-friendly Cities: A Guide (2007) — the founding document of the age-friendly movement, built from consultations with older people in cities worldwide — opens on the twin curves: "[t]he world is rapidly ageing: the number of people aged 60 and over as a proportion of the global population will double from 11% in 2006 to 22% by 2050. By then, there will be more older people than children (aged 0–14 years) in the population for the first time in human history. Developing countries are ageing at a much faster rate than developed countries: within five decades, just over 80% of the world's older people will be living in developing countries compared with 60% in 2005" [18]. Simultaneously, "[a]s of 2007, over half of the global population now lives in cities... By 2030, about three out of every five people in the world will live in cities and the number of urban dwellers in the less developed regions will be almost four times as large as that in the more developed regions" [18]. The intersection is where this section lives: "[i]n developing countries... the share of older people in urban communities will multiply 16 times from about 56 million in 1998 to over 908 million in 2050. By that time, older people will comprise one fourth of the total urban population in less developed countries" [18].

The Guide's premise reframes both trends as achievement: "[p]opulation ageing and urbanization are the culmination of successful human development during last century... As stated in the WHO Brasilia Declaration on Ageing in 1996, 'healthy older people are a resource for their families, their communities and the economy'... Yet to be sustainable, cities must provide the structures and services to support their residents' wellbeing and productivity. Older people in particular require supportive and enabling living environments to compensate for physical and social changes associated with ageing. This necessity was recognized as one of the three priority directions of the Madrid International Plan of Action on Ageing endorsed by the United Nations in 2002. Making cities more age-friendly is a necessary and logical response to promote the wellbeing and contributions of older urban residents and keep cities thriving" [18]. The conceptual anchor is WHO's active ageing framework: "[a]ctive ageing is the process of optimizing opportunities for health, participation and security in order to enhance quality of life as people age. In an age-friendly city, policies, services, settings and structures support and enable people to age actively by: recognizing the wide range of capacities and resources among older people; anticipating and responding flexibly to ageing-related needs and preferences; respecting their decisions and lifestyle choices; protecting those who are most vulnerable; and promoting their inclusion in and contribution to all areas of community life" [18].

What older people themselves reported, city by city, is less abstract — and the health-services findings read as a field test of this report's access chapters. "[H]aving well-located, easily accessible health services is fundamentally important for older people in every collaborating city. Older people in cities such as Amman, Rio de Janeiro, Sherbrooke and Tokyo value having health services near by, and in Geneva and Shanghai, older people appreciate good transportation to health facilities. Services that are far away or difficult to reach are often seen as barriers. Public transportation is considered inadequate in some places and in others, such as Delhi and Mexico City, transportation for people with disabilities is reported to be a particular problem. Access to health care in emergency situations is a frequent concern... ambulance services are either insufficient (Ponce) or too slow because of heavy traffic (Moscow). Ways to minimize geographical barriers include the idea of co-locating or decentralizing services so they are available in all neighbourhoods. Other ideas are to offer transportation by volunteers and to provide health emergency telephone services, as in Himeji, for older people living alone" [18]. The built fabric fails in small, fixable ways — "poorly maintained elevators and ramps in Udaipur, poor building access for people with disabilities and a lack of wheelchairs or walking frames for patients in Cancún, and overcrowded facilities in many other cities" — and information fails too: "[a]nother frequently mentioned barrier to accessing care is insufficient knowledge about the health services available in the city. As observed in Melbourne, if services are not known about they are not used" [18] — the guidance problem of the pensions chapters, at street level.

The deepest complaint is not logistical but human. "Finally, the attitudes of health service providers towards older people are frequently mentioned. Tuymazy is one of the few places reporting the polite and friendly behaviour of clinic receptionists and nurses. Negative attitudes and poor communication by health providers are common complaints about care. Problems voiced include indifference, disrespect, uncaring attitudes, and treating older people as a burden or as a drain on resources" [18] — crystallised in a London testimony the Guide prints verbatim: "When they came to wash and change her they treated her like a piece of furniture – no dignity, no respect" [18]. The service wishlist is equally concrete: "geriatric clinic services and hospital beds, adult day centres, care for the demented, mental health services, respite care and training for caregivers, rehabilitation and palliative care... However, the health services that receive the most attention worldwide are disease prevention and health promotion, home care, and nursing (long-term care) homes" [18], with local good practice already visible — "self-help groups or organizations offering sports activities for fitness and rehabilitation, and regular health checks at home" in the Ruhr metro-

politan area, and in Mexico City a "targeted vaccination campaign and free eyeglasses" that older residents praise [18]. The Guide closes by framing itself as a beginning: "[t]his project is a starting point for many more community development and research activities, as well as for the establishment of a larger global network of age-friendly communities" [18]. Nearly two decades on, that network is the operating system of urban ageing policy — and its most basic determinant, as the next section shows, is whether an older person can physically get anywhere at all: transportation.

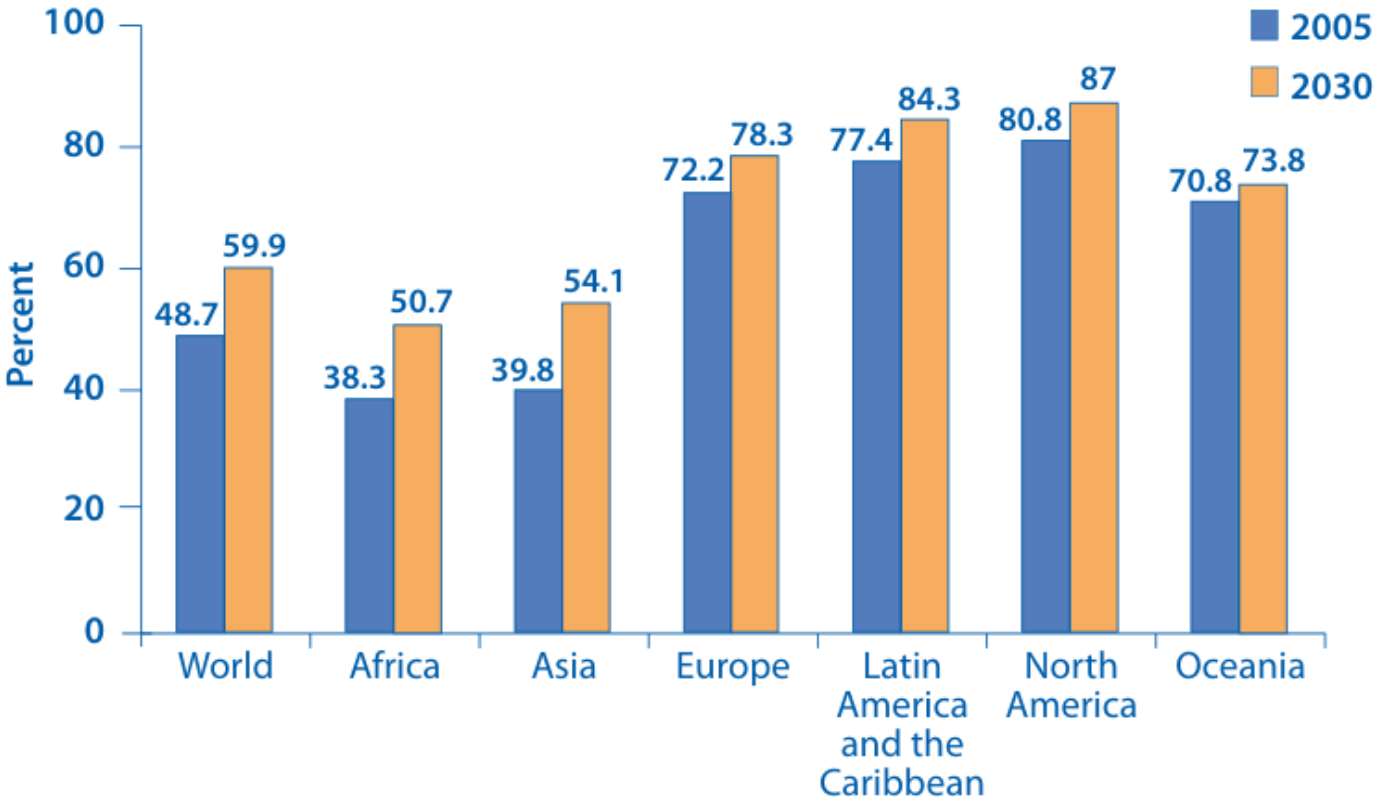


Exhibit. Percentage of the population living in urban areas in the world's major regions, showing the extent of urbanisation confronting cities as their populations age. Source: S0018.

Transportation for older people

Every service in this report — the clinic, the pension office, the grandchild — sits at the end of a journey, and for older people the journey is the fragile part. The built inheritance works against them: "[t]he communities we live in today were designed at the turn of the 20th century to accommodate young families, at a time when living to old age was a rarity… In 1900, there may have been one surviving grandparent in the household, but only four percent of the entire population was over 65" [9]. The farm gave way to the suburb — "[c]ommunities built for nuclear families, with houses and lots that offered space for many children… Those first suburban dwellers are now elderly grandparents, whose own children moved away… they find themselves living in spaces that have become ill-suited to their needs and ultimately difficult to manage. Especially in rural areas, isolation can become unbearable when people live at great distances from services and social life. Without public transportation, driving is the only option, and when driving becomes impossible, physical and social isolation follow. Even basic food and health services may be hundreds of miles away. In a world increasingly embedded in virtual connectivity, broadband access remains unavailable in large parts of the country, meaning no online grocery services or telehealth visits" [9]. The driving-cessation cliff — one day mobile, the next housebound — is a design failure, not a personal one.

The WHO consultations map the same dependence worldwide: "[d]riving as an essential transportation option for older people is discussed in a few locations. For example, it is observed that Melville is a city designed for cars, and in Himeji, cars are considered a necessity in the suburbs. In Ponce, cars are viewed as necessary because of the limited transport options available" [18]. Where driving works, it is because the environment forgives: light traffic in Portage la Prairie, "[i]n Saanich, the older people appreciate the advance warnings of crossroads. In Tokyo, older people mention that the road signals and signs are easy to see" [18]. "People in many more cities in countries at all stages of development report barriers to city driving. These include heavy traffic, poor condition of roads, ineffective traffic calming devices, inadequate street lighting, inadequate signage that is obscured or poorly positioned, and

the disregard for traffic rules and regulations" [18] — heavy traffic in Rio de Janeiro, potholed roads in Cancún, unlit streets in Mayaguez, signage "too small, too high and often obscured" in Halifax, aggressive drivers in Sherbrooke. And the social environment polices slower drivers out: "[i]n addition to the barriers identified above, the disrespect shown to older drivers discourages a number of them" — "I dislike driving. People curse you, make signs at you if you go slow. They are impolite" (older person, Tripoli); "[i]n La Plata, it is stated that older drivers are abused because they drive too slowly" [18]. Against that, supported driving has known tools: "[t]o ensure older people are confident drivers, refresher courses are recommended in some cities such as including Geneva and Portland. The special lessons given to older people in Himeji when they need to renew their licence is viewed as an age-friendly advantage" [18]. Parking is transport too: "[p]riority parking bays for older and handicapped people in close proximity to buildings, together with drop-off and pick-up bays, were seen as age-friendly features" [18], while failures are specific — bays "not wide enough for loading wheelchairs" in Portage la Prairie, too few drop-off points in Mayaguez, and in London, "[t]hey make disabled bays that are totally and utterly ignored" [18].

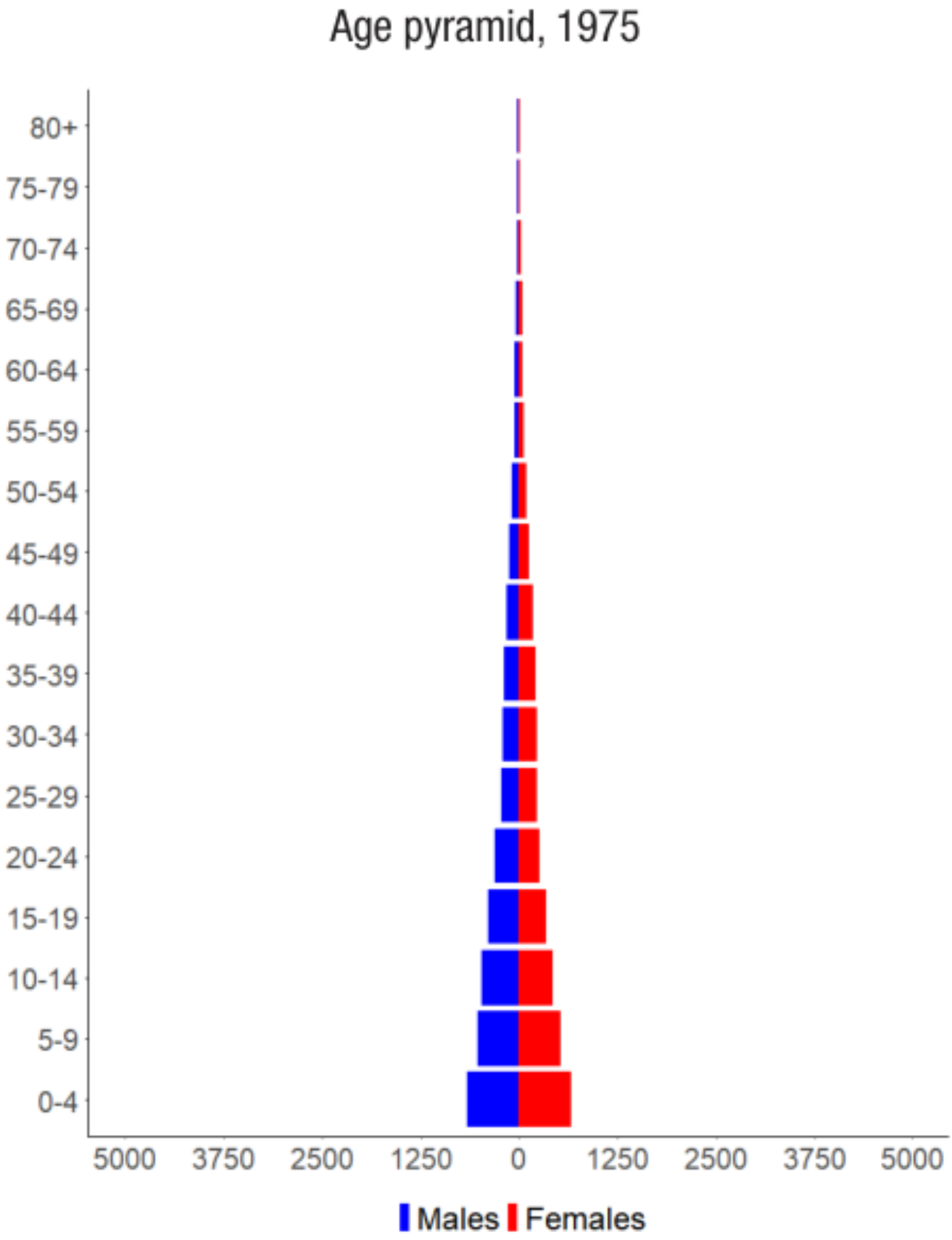
Public transport is the alternative the previous section already graded: "considered inadequate in some places," with "transportation for people with disabilities… a particular problem" in Delhi and Mexico City, volunteer transportation among the remedies older people themselves propose, and good transport to health facilities singled out for praise in Geneva and Shanghai [18]. The design answer integrates transport with everything else. The New Map of Life's exemplar, Fruitvale Village in East Oakland, sits "[a]djacent to the Bay Area Rapid Transit (BART) Fruitvale station and city bus lines," mixing "retail and restaurants" at ground level, "social service agencies and offices on the second floor, and subsidized apartments on the third," with a senior center, a charter high school and a Head Start centre in one complex creating "opportunities for spontaneous, daily interaction among generations," a public library with free internet, and a community clinic — "walkable, connected, and affordable, with easy access to public transit" [9]. Its lockdown stress test doubled as a mobility proof: "[w]hen Covid struck, local restaurants shifted to packaging thousands of hot meals each day and distributing them for free… 'People in this community are dealing with a lot of adversity, but they are doing it together instead of in isolation'" [9]. Demand confirms scarcity: "[m]ore than 4,000 applicants entered the lottery for the 94 apartments at Casa Arabella when they came on the rental market in 2019" [9]. What longevity-ready communities require, in the report's summary spec, is settlements "longevity-ready, resilient to climate change, connected to the internet, with housing and neighborhoods designed to bring generations together for mutual benefit, rather than continuing to segregate us by age" [9]. Transportation, in short, is not a sector but the precondition: health access, social participation, work and care all presume the trip. The chapter closes with three country case studies — Angola, Rwanda and Bangladesh — where these systems are being built, or not, at the frontier of ageing before wealth.

Country case studies: Angola, Rwanda and Bangladesh

The UN's three least-developed-country case studies test the report's central proposition — that ageing's outcomes are chosen, not fated — at the hardest end of the income distribution. The demographic positions differ sharply: "[w]hile some Asia-Pacific LDCs are approaching or have entered the late stage of the demographic transition, the majority of LDCs in sub-Saharan Africa are still in the early stages of the transition. The major demographic challenges facing these countries include persistently high levels of fertility and rapid population growth," yet all face the same second act: LDCs "are expected to experience a significant rise in both the proportion and absolute number of older persons between now and 2050 and an accelerated pace of population ageing beyond 2050," and "some Asia-Pacific LDCs already have a large number of older persons, but they lack well-established social protection programmes and health care services" [2]. The strategic window is the demographic dividend: countries in early and mid-transition "have an opportunity to achieve accelerated economic growth and social development, which may help them to avoid 'getting old before getting rich'" [2].

Each case isolates a different lever. "The experience of Angola highlights that rapid economic growth must be accompanied by creating ample opportunities of decent employment to help people escape poverty and reduce inequality" — growth without jobs does not bank the dividend [2]. "Rwanda's Government has focused on family planning in its development agenda to accelerate its demographic transition, supported by a range of policies aimed at harnessing the demographic dividend" — transition speed itself as policy [2]. "The Government of Bangladesh has made remarkable progress in human capital formation and achieving gender equality, laying a good foundation for sustainable development" — capabilities built before they are needed [2]. The shared precondition is fiscal: human-capital investment "with a particular focus on education of girls and women," job creation drawing on "more successful experiences from East Asia and less successful stories from Latin America and the Caribbean," and governance plus "debt restructuring or debt relief [to] boost the domestic resource base" [2].

The distilled lessons are stage-specific. Early- and mid-transition governments should “continue to prioritize investments to ensure universal access to sexual and reproductive health and reproductive rights including family planning”; late-transition governments should “prioritize enhancing social protection programmes and health-care services to take care of the large number of older persons”; countries with expanding school-age populations should “invest heavily in human capital formation,” including “upskilling the workforce in anticipation of the emerging needs of the future world of work, such as digitalization and automation”; extractive economies should “redirect resources to create more jobs in the formal sector while reducing inequality”; and governments with limited fiscal space should “balance investments in youth for the future with improving social protection and health care services, in particular for increasing numbers of older persons” [2]. The crisis backdrop — the pandemic, war-driven inflation and food insecurity, mounting debt — falls hardest where coping capacity is thinnest, so “the international community is called upon to continue to provide the support needed” for the 2030 Agenda’s promise to leave no one behind [2]. The cases close the report’s global arc: East Asia showed what preparation buys, and these three show the preparation still available to countries whose ageing is mostly still ahead of them.



Age pyramid, 2023

Exhibit. Population of Angola by age and sex, in thousands, shown as age pyramids for 1975 and 2023 within a 1975–2050 span, tracing the country's youthful age structure over time. Source: S0005.

Projections, methods and the fiscal outlook

The Ageing Report: underlying assumptions and projection methodologies

Every fiscal number in this report descends from a projection, and every projection is a stack of assumptions wearing a result. The final chapter opens the machinery — principally the European Commission's 2024 Ageing Report, whose "baseline (previously named 'AWG reference scenario') is used by the European Commission in the context of fiscal surveillance and the European Semester" [4]. Understanding the baseline's five moving parts is the price of reading any of its outputs critically.

For long-term care, the parts are these. "Population projections rely on the baseline population projections by Eurostat. Dependency ratio: it is assumed that half of the projected longevity gains up to 2070 will be spent in good health and free of disability/dependency. The age-gender specific dependency rates are shifted by ½ of the change in life expectancy (e.g. if life expectancy increases by 2 years by year 2070, then the dependency rate of 80-year-old men in 2070 is that of 79-year-old men in 2022). Age-related expenditure profiles: the age-related expenditure profiles from 2022 are used, uprated in terms of the unit cost development. Public coverage: the shares of the older disabled population who receive either informal care, cash benefits, formal care at home or institutional care that is provided or financed by the public sector are kept constant over the projection period" [4]. Costs then do quiet but decisive work: "[u]nit costs are generally indexed according to GDP per hour worked in the case of in-kind benefits and to GDP per capita in case of cash benefits. This reflects the current imbalance of the care mix, with a relative deficit of formal care provision. Furthermore, the sector is highly labour-intensive and productivity gains due to technology are difficult to achieve. Therefore, public expenditure on long-term care is expected to be rather supply- than demand-driven" [4], with country-specific indexation for "France, Germany and Slovenia" where legislation dictates it, and a demand tailwind: "[a]s countries become richer, they are likely to spend a larger proportion of their GDP on long-term care. This is modelled in the baseline by including the assumption that the income elasticity starts at 1.1 in base year 2022, falling to 1 by the end of the projection period for those countries that are below the first quartile in terms of expenditure on long-term care as a proportion of GDP... expected for all countries except NL, SE, DK, BE, FI, DE, FR and NO" [4]. Note what the baseline therefore is: not a forecast of policy, but a no-policy-change measuring rod — coverage frozen at 2022 shares even where informal care is visibly unsustainable, and the health dividend set at exactly half the longevity gain, a diplomatic midpoint between the morbidity hypotheses of the health chapter.

The scenario fan around the baseline prices each assumption. The "'no healthy ageing scenario' is identical to the baseline apart from its assumption on the dependency status. It assumes that the disability rates of the population of a given age will not improve despite increases in life expectancy... Since the prevalence of dependency is kept constant over the projection period, the dependent population evolves precisely in line with the total elderly population. This implies that, in practice, none of the gains in life expectancy translate into an improvement of health" [4] — the expansion-of-morbidity bound. The opposite bound is "inspired by the so-called 'dynamic equilibrium hypothesis'... The age-gender specific dependency rates are shifted in line with changes in life expectancy (e.g. if life expectancy increases by 2 years by year 2070, then the dependency rate of 80-year-old men in 2070 is that of 78-year-old men today). This results in a gradual decrease over time in the prevalence of disability for each age cohort that goes beyond the gradual decrease in the baseline. Lower dependency rates translate in lower demand for and therefore lower expenditure on long-term care services for each age group" [4]. A third scenario changes policy rather than health: "[t]he pressure for increased public provision and financing of long-term care services may grow substantially in the coming decades, especially in Member States where the bulk of long-term care is currently provided informally. This scenario assumes that economic convergence across Member States, the exchange of best practices and growing expectations of the populations will drive an expansion of publicly financed formal care provision into the groups of the population that have not been covered by the public programmes so far... This scenario should be considered as a policy-change scenario, as it assumes a considerable shift in the current long-term care provision mix from informal care to formal in-kind care" [4].

Health care projections run the same architecture with sector-specific dials: the baseline "reflects (i) the demographic impact of population ageing, based on Eurostat projections, (ii) a 'neutral' assumption on the evolution of health status, by assuming that half of the extra years of life gained through higher life expectancy are spent in good health, (iii) unit costs assumed to evolve in line with GDP per capita and (iv) the assumption of an income effect, which is modelled through a cost sensitivity of 1.1 in the base year converging to unity by 2070" [4]. Its scenario set isolates each force: the demographic scenario estimates "the pure ageing effect" with unit costs on real GDP per

capita only — "if the age and mortality structure of the population did not change, public health expenditure as a share of GDP would remain the same over the projection period" [4]; the no-healthy-ageing scenario tests "the expansion of morbidity hypothesis," in which "all the gains in life expectancy are assumed to be spent in poor health. The number of years spent in good health remains constant" [4]; the healthy-ageing scenario tests "the relative compression of morbidity hypothesis… the number of years spent in poor health during a lifetime remains constant over the projection period. This means that all future gains in life expectancy are assumed to be spent in good health" [4]; and the labour-intensity scenario reprices the workforce: "[h]ealth care is and will remain a highly labour-intensive sector. Therefore, it is assumed that unit costs… evolve in line with changes in labour productivity (modelled by real GDP per hours worked), rather than real GDP per capita growth (as assumed in the baseline). This assumption implies that wages in the health sector grow in line with productivity in the overall economy and generally faster than GDP per capita" [4] — Baumol's cost disease, formalised as a scenario. The methodological verdict for readers of every earlier chapter: the projections are not predictions but conditional arithmetic, and the spread between their scenarios — driven overwhelmingly by the health-status assumption and the unit-cost indexation — is the honest measure of what we do not know. The next section sets out the baseline's demographic and macroeconomic assumptions themselves.

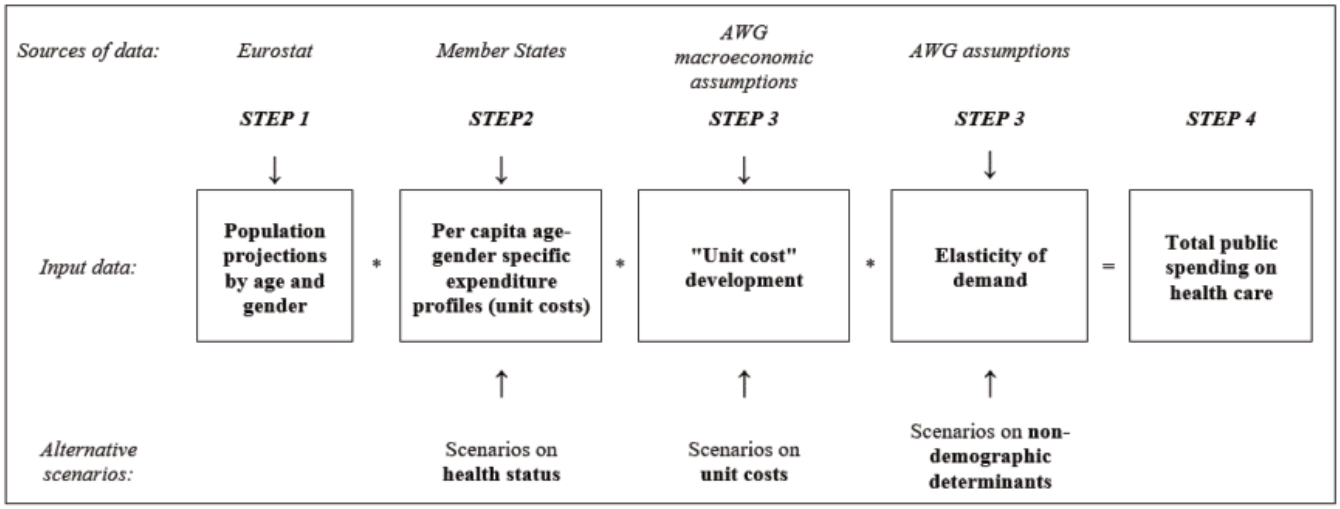


Exhibit. Graph II.2.1: Schematic overview of the health care projection methodology Source: European Commission, EPC. Source: S0015.

Main demographic and macroeconomic assumptions

Methodology becomes concrete in the country dashboards: for each Member State, the 2024 Ageing Report publishes the full assumption set — EUROPOP2023 demographics, labour-force trajectories, macro paths and the resulting dependency ratios — on a single comparable grid running from 2022 to 2070. Two worked examples and one convergence case show what the baseline actually believes about the future.

The **Netherlands** dashboard assumes gentle demographic pressure, largely offset by participation. Fertility edges up from 1.53 (2022) to 1.63 (2070); life expectancy at birth rises from 80.3 to 86.7 years for men and from 83.6 to 90.0 for women, with life expectancy at 65 reaching 23.6 years (men) and 26.4 (women) by 2070; net migration settles at about 0.2 per cent of the population per year after the exceptional 2022 inflow (1.3 per cent); the working-age (20–64) share falls from 58.7 to 52.0 per cent while the 65+ share rises from 20.1 to 29.3 per cent and the 80+ share more than doubles, from 4.9 to 10.9 per cent [4]. The old-age dependency ratio climbs from 34.3 to 56.3 per cent and the total dependency ratio from 70.3 to 92.4 per cent — but the economic old-age dependency ratio (non-workers per worker) rises far less, from 37.3 to 55.9 per cent, because the labour assumptions do heavy lifting: participation at 20–64 rises from 85.4 to 90.4 per cent, participation at 55–64 from 75.3 to 84.2 per cent, participation at 65–74 from 17.0 to 30.0 per cent (women aged 55–64 from 68.5 to 82.4 per cent), and the average exit age from 64.9 to 67.8, with unemployment held at about 3 per cent; potential GDP growth glides from 2.2 to 1.0 per cent on labour productivity growth of about 1.1 per cent a year (total factor productivity 0.7, capital deepening contributing 0.4) [4].

Belgium starts older in behaviour, if not in demography, and converges only partway. Fertility rises from 1.53 to 1.64; life expectancy at birth from 79.5 to 86.4 (men) and 84.6 to 90.5 (women); the 65+ share from 19.6 to 28.1 per cent

and the 80+ share from 5.5 to 11.3 per cent; net migration falls from about 1.0 per cent of population in 2022 toward 0.2 per cent [4]. The dependency arithmetic is similar to the Netherlands' (old-age dependency 33.7 → 53.0 per cent), but the economic version is much worse — 45.4 rising to 66.9 per cent — because Belgian participation starts lower and rises less: 20–64 participation from 76.1 to 80.3 per cent, 55–64 from 59.1 to 70.6 per cent, women's 55–64 participation from 53.9 to 68.7 per cent, and the average exit age only from 62.8 to 64.5, with unemployment around 5.5 per cent and potential growth easing from 1.8 to 1.1 per cent on the same productivity skeleton (TFP 0.7, capital deepening 0.4) [4]. Same demographic century, different fiscal experience — driven almost entirely by the labour-market assumptions, which is why those assumptions, not the demography, are where national debates concentrate.

Bulgaria displays the convergence engine at full power. Its fertility assumption rises from 1.56 to 1.69, but the striking numbers are mortality: male life expectancy at birth is assumed to gain 12.3 years, from 70.5 to 82.8, and female 10.0 years, from 77.7 to 87.7, with life expectancy at 65 gaining 7.8 years for men (13.5 → 21.3) and 7.1 for women (17.5 → 24.6) [4]. The projection thus assumes that the East–West longevity gap largely closes within two generations — the single most optimistic assumption family in the entire exercise, and the one on which Eastern Europe's dependency paths most depend.

Read across the dashboards, the common architecture is visible. Demography is taken from a single source (EUROPOP2023) with fertility recovering slightly but remaining below replacement everywhere, mortality improving with partial convergence, and migration reverting from the 2022 spike to a few tenths of a per cent of population. Labour supply is modelled, not extrapolated: participation rises through cohorts and legislated reforms, concentrated at older ages and among women, with exit ages drifting up two to three years. Macroeconomics is deliberately bland — potential growth converging toward about 1 per cent, inflation at 2 per cent, nominal interest rates toward about 4 per cent — so that differences in outcomes reflect demography and labour markets rather than macro heroics. And the dependency decomposition repeats the report's master lesson in numerical form: demographic dependency rises steeply everywhere (+20 points or so), while economic dependency rises materially less where participation assumptions are strong — the gap between the two ratios is the policy space. The next section stress-tests the edifice: sensitivity tests and alternative scenarios.

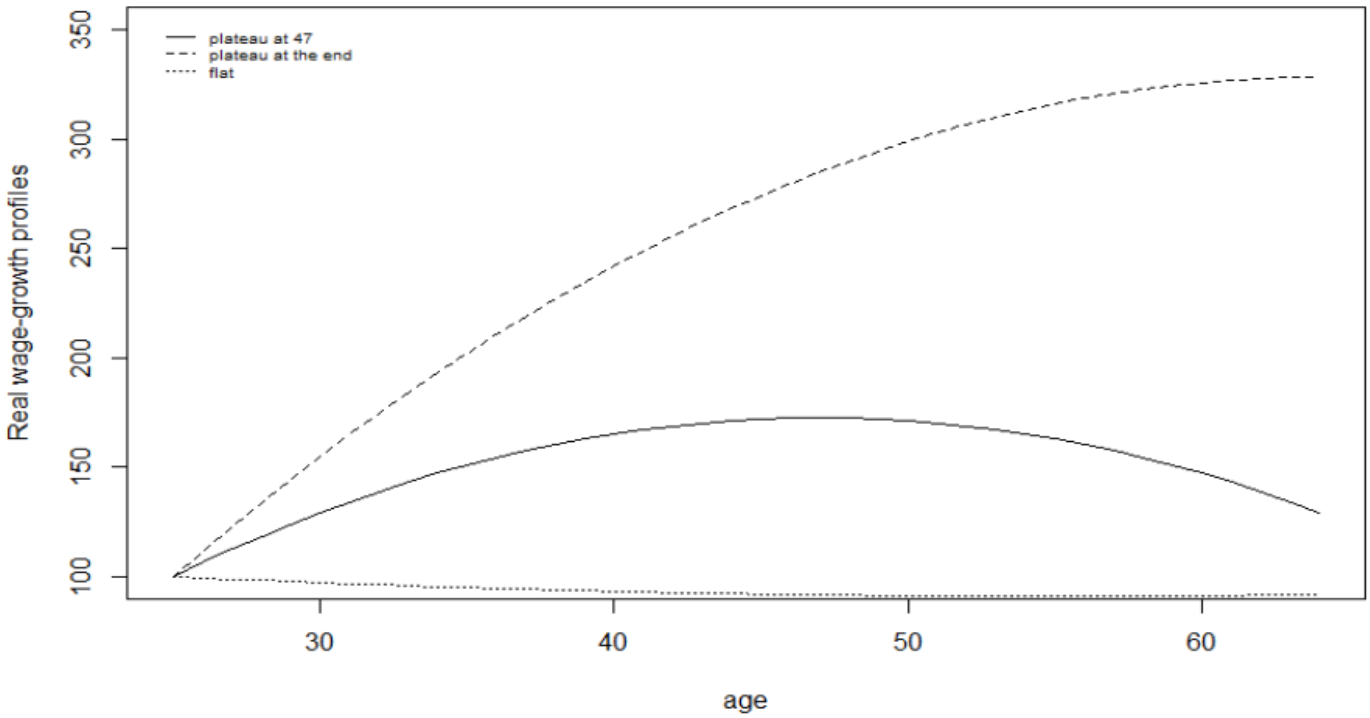


Exhibit. Real wage-growth indexes used in the modelling, under which nominal wages grow with inflation and account for the impact of unemployment. Source: S0013.

Sensitivity tests and alternative scenarios

A baseline is a single path through a wide fan of futures; the Ageing Report's sensitivity suite maps the fan's edges. The procedure is properly bottom-up: "[f]ollowing the procedure for the baseline assumptions, a bottom-up ap-

proach is followed: from population through labour input to GDP growth projections. Each sensitivity test therefore involves recalculating all assumptions and rerunning the labour force model and the production function" [4]. The suite is deliberately continuous with previous cycles — "the standard tests applied in the 2021 Ageing Report were appropriate for conducting a broad sensitivity analysis… Changes are therefore limited. They concern the design of the migration scenarios, the introduction of a higher inflation scenario, and changes in the name of some of the tests and scenarios" [4].

The demographic shocks bracket the baseline dials. In the higher-life-expectancy test, "the age-specific mortality rates are reduced linearly to achieve an additional increase in life expectancy at birth of two years by 2070 compared to the baseline" [4]. Lower fertility sets "the fertility rate… 20% lower than the baseline during the entire projection period" [4]. Migration is shocked symmetrically — "immigration is, respectively, 33% lower and 33% higher than the baseline over the entire projection period" — but with a 2024 refinement: "the shocks are applied to the non-EU immigration flows. Since some countries have negative net migration in the baseline because of intra-EU emigration, the modified approach increases comparability of the scenario across countries" [4]. Each test thereby isolates a single uncertainty: two extra years of life, a fifth fewer births, a third more or fewer arrivals.

The economic shocks target the assumptions with the widest error bars. The new higher-inflation test, prompted by "the high increases in consumer prices of recent years," slows convergence: where "[i]n the baseline these targets are reached by 2027 for all Member States… the sensitivity test assumes inflation to converge linearly from current country-specific values to market-based inflation rates by 2032. Thereafter, inflation rates would converge by 2052 to the central bank targets: 2% for euro area countries, Bulgaria, Czechia, Denmark and Sweden, 2.5% for Poland and Romania, and 3% for Hungary" [4]. Productivity is bracketed around the baseline's steady state: "total factor productivity growth (TFP) converges to a steady-state growth rate of 0.6%/1.0%, compared to a baseline assumption of 0.8%," the lower variant being 2021's "TFP risk" test renamed [4]. Finally, the one policy scenario in the suite tests this report's master lever directly: the "[h]igher employment rate of older workers" scenario, in which "through a reduction in the inactive population, the employment rate of older workers (55 to 74) is increased by 10 pps over the period 2024-2036 and remain at this higher level for the remainder of the projection period" [4].

The suite's design philosophy is the report's epistemology in miniature: change one dial at a time, rerun the whole machine, and report the spread rather than a spurious point estimate. A ten-point rise in older-worker employment is deliberately the largest single policy shock in the set — a quantitative statement that, among all the uncertainties a finance ministry faces to 2070, the one it most controls is also among the most consequential.

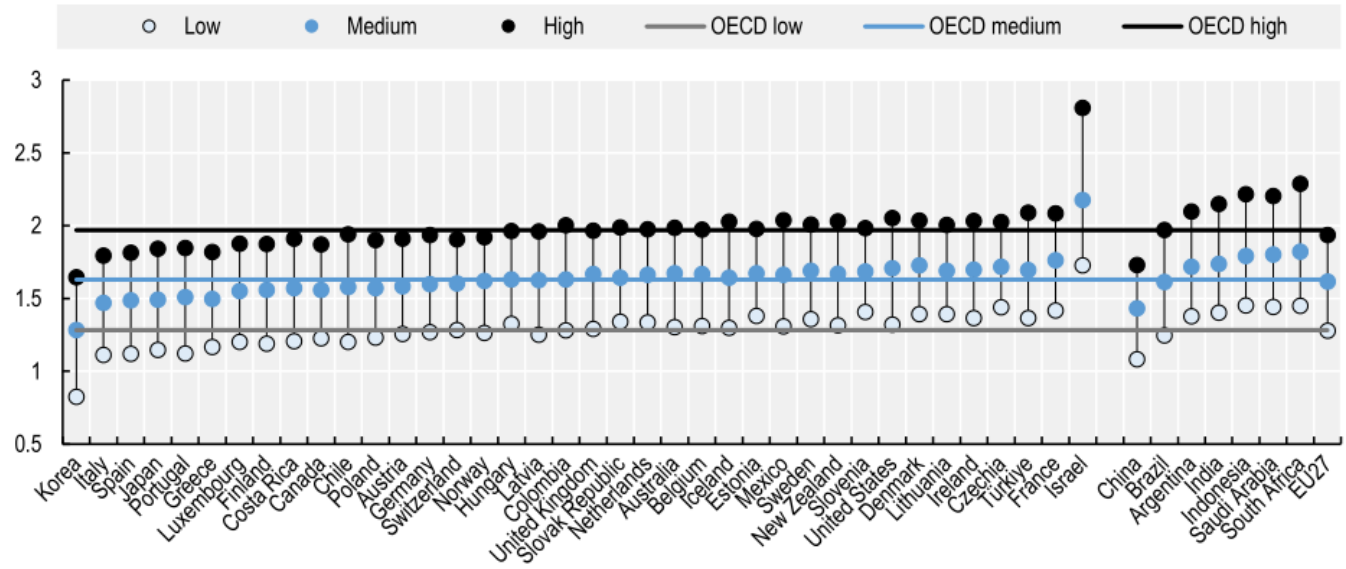


Exhibit. Low, medium and high variant projections for 2055, placing each country's projected outcome alongside the corresponding OECD low, medium and high variants. Source: S0002.

Projecting public expenditure: pensions, health, education and long-term care

The expenditure projections share one arithmetic skeleton across all four blocks: a projected population, distributed by age, sex and — for care — dependency status, multiplied by age-related expenditure profiles, with the profiles carried forward by explicit indexation rules. Pensions run through the entitlement and cohort machinery examined in

the pensions chapter; health, education and long-term care apply the same identity with sector-specific data plumbing, and it is the plumbing — what counts as public expenditure, which year anchors the base, how profiles are uprated — that decides how much of the projected increase is demography and how much is bookkeeping.

Health and long-term care rest on internationally standardised accounts: "the 2024 Ageing Report will rely whenever possible on data following common methodologies and definitions (i.e. the System of Health Accounts - SHA) agreed by international institutions (Eurostat, OECD and WHO), gathered through the joint data collection exercise (i.e. the joint OECD-Eurostat-WHO questionnaire) and reported in Eurostat (Cronos) and OECD (Health Data) databases" [4]. Coverage is uneven in an instructive place: "SHA 2011 expenditure data are available for all EU Member States for variable HC.3, but only for a limited number of Member States for the voluntary variable HC.R.1, which requires the use of the European System of integrated Social Protection Statistics (ESSPROS) database to construct a proxy for this missing variable in several countries" [4] — the social half of long-term care is, in much of Europe, a statistical estimate rather than an account. The base year carries a pandemic scar: "the COVID-19 pandemic increased the current public health care expenditure in all Member States, though to a varying degree. Given the difficulties to collect complete data on COVID-19 related expenditure in order to subtract the ones that can be considered as one-off or temporary spending, public current expenditure for the year 2019 is used instead as a reference point for the 2022 (base) year estimate" [4], with GDP denominators from the AMECO database and a German exception — "data on health care expenditure of the German compulsory private health insurances are not considered. Current public health care expenditure for Germany is therefore comprised by the government and social health insurance shares, which are only reported on the OECD database" [4]. What the international databases cannot supply, capitals must: "[w]hen data are not available in the international databases, they have to be provided by each Member State individually. This is particularly the case for the number of recipients, age-cost profiles and the breakdown of expenditure by care setting" [4].

For long-term care, the data contract is explicit. Member States provide "[t]otal number of dependent people receiving long-term care a) in institutions b) at home and c) cash benefits, by sex and five-year cohorts", the possible overlaps between cash-benefit and in-kind recipients, "[p]ublic expenditure per user (patient) on long-term care, by sex and single age or five-year cohorts (so-called 'age-related expenditure profiles')", the "[p]ublic expenditure breakdown by care setting (institutional care, home care, cash benefits)" and "[d]ata on all policy measures implemented since 2019 that have a potential impact on the long-term care expenditure path" [4]. The Commission pre-fills the rest — total public LTC spending split between nursing and social components, setting breakdowns from ESSPROS and/or SHA, and "[d]isability rates by sex and five-year cohorts (based on EU-SILC data)" [4]. The definitional boundary is the SHA's two-part sum: "[s]ervices of long-term nursing care (HC.3 in SHA 2011) (which is also called 'long-term healthcare' and includes both nursing care and medical care related to the cause of the dependency as well as assistance with ADL tasks)" plus "'[s]ocial services' of long-term care (HC.R. 1 in SHA 2011), which represent the 'assistance services' part, relating primarily to assistance with IADL tasks. Taken together these should represent the total benefits allocated to dependent people, although… these data have to be supplemented to different degrees with ESSPROS data to fulfil the projection needs" [4]. Dependency counts times profiles times indexation, run under the baseline and scenario assumptions of the previous sections — that is the whole engine, and its outputs inherit every caveat embedded in the inputs: a 2019 spending anchor, proxied social-care totals, and survey-based disability rates.

Indexation is where judgement hides. Beyond the headline rules (GDP per hour worked for in-kind care, GDP per capita for cash benefits), the health projections use a sector-specific composite built from history: ten-year average growth rates of expenditure components are compared with GDP growth, but "[d]ue to high volatility in the relative growth rates for prevention, capital formation and governance and administration, these items are excluded from the indexation. Moreover, the relative growth rates of the other three components (hospitals, outpatient care and medical goods) are capped at their respective 25th and 75th percentiles" [4]. Trimming and capping are prudence devices — they stop a volatile decade from compounding into an absurd half-century — but they also bake the recent past's spending structure into the future's.

The methodological moral generalises across pensions, health, education and care alike: the projected expenditure paths are best read as demography passed through today's systems at today's relative prices, with policy frozen and measurement gaps papered over by proxies. That is exactly what a fiscal-surveillance baseline should be — and exactly why the levels deserve less confidence than the comparisons between scenarios and countries. The next section turns to those comparisons: the country dashboards and indicator overviews.

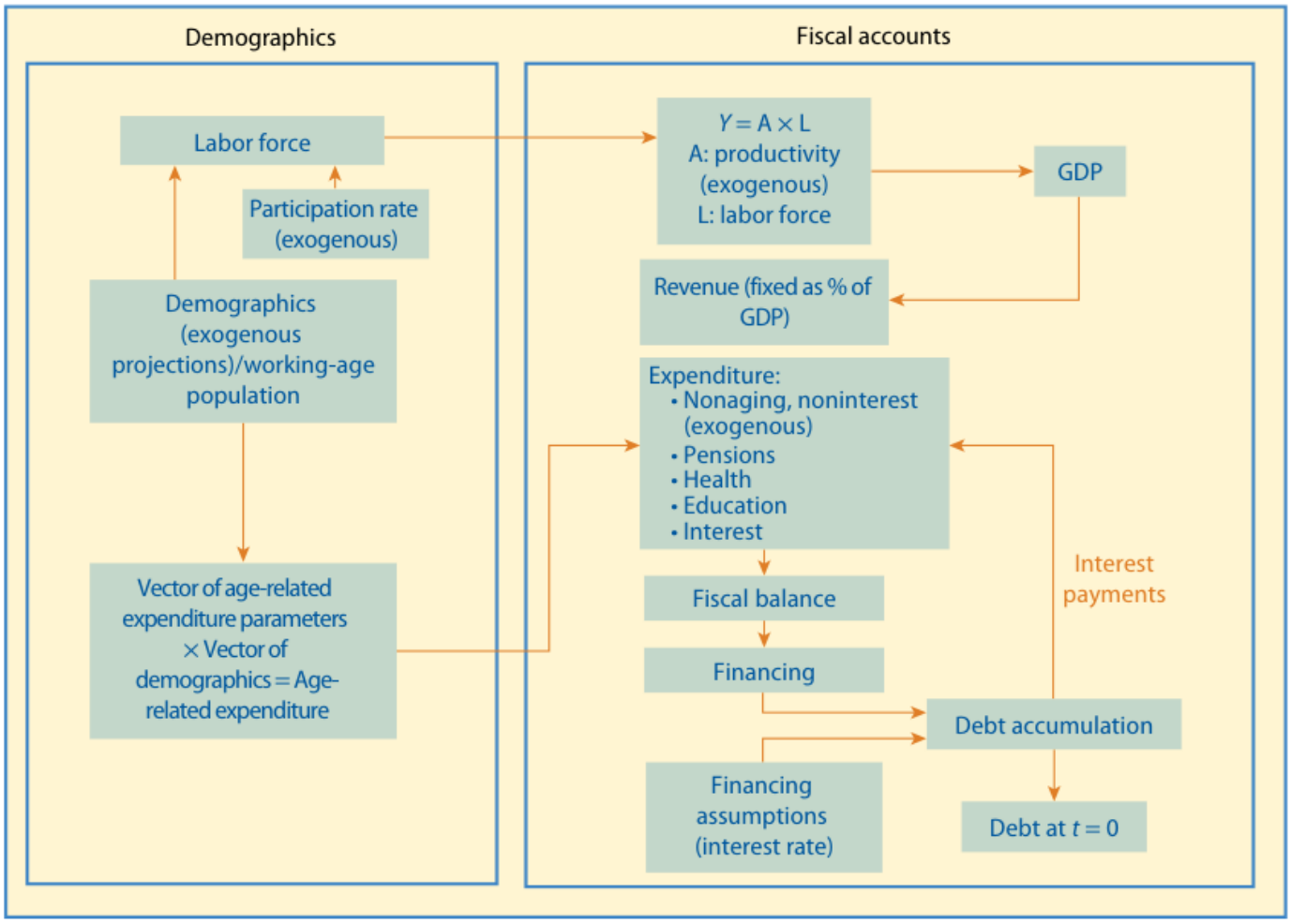


Exhibit. Schematic overview of the links between demographics and fiscal accounts, tracing how the labour force and productivity determine GDP (Y = A × L) and thereby government revenue and age-related expenditure. Source: S0029.

Country dashboards and indicator overviews

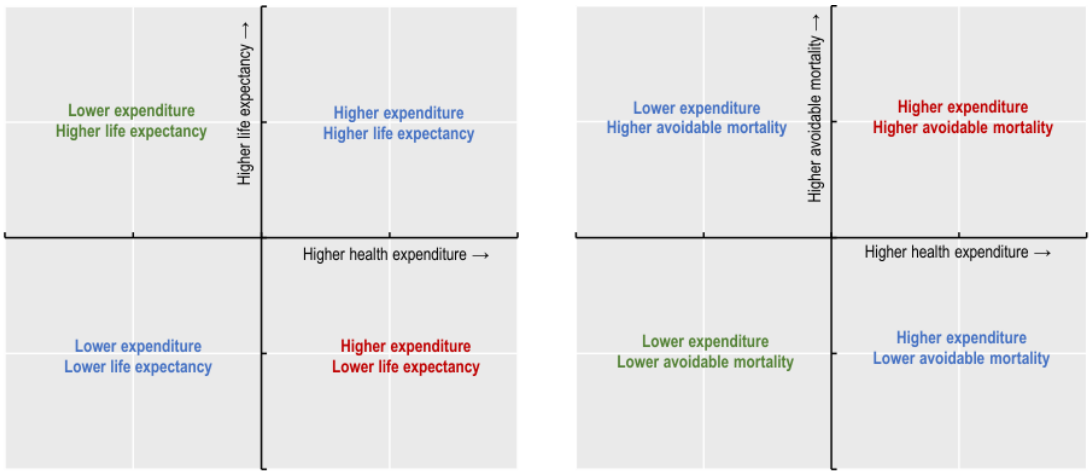
Projections look forward; dashboards look sideways. The comparative instrument this report has drawn on throughout — OECD Health at a Glance's indicator overview — exists to answer a different question from the Ageing Report's: not what will happen? but where do we stand, against whom, and is the gap closing? Its machinery deserves the same scrutiny given to the projections, because league-table colours have a way of hardening into verdicts.

The design is disciplined. "Based on these indicators, country dashboards are produced. These compare a country's performance to that of other countries and to the OECD average. Comparisons are made based on the latest year available. For most indicators this refers to 2021, or to the nearest year if 2021 data are not available for a given country. Country classification for each indicator is into one of three colour-coded groups: blue when the country's performance is close to the OECD average; green when the country's performance is considerably better than the OECD average; red when the country's performance is considerably worse than the OECD average" [6]. One panel refuses the verdict format: "[t]he exception to this grouping is the dashboard on health system capacity and resources, where indicators cannot be easily classified as showing better or worse performance. Here, lighter and darker shades of blue signal whether a country has considerably less or more of a given healthcare resource than the OECD average" [6] — more beds or doctors is a fact, not a grade. The statistics behind the colours are equally explicit: "[t]he classification of countries as being close to, better or worse than the OECD average is based on an indicator's standard deviation... Countries are classified as 'close to the OECD average' (blue) whenever the value for an indicator is within one standard deviation from the OECD average for the latest year. Particularly large outliers (larger than three standard deviations) are excluded from calculations of the standard deviation to avoid statistical distortions. For a typical indicator, about 65% of countries will be close to the OECD average, with the remaining 35% performing significantly better (green) or worse (red)" [6], and "[f]or each indicator, the OECD average, highest and lowest values are shown, as are the three countries with the largest improvements over time in terms of changes to absolute values" [6] — level and momentum side by side. The companion quadrant charts carry their own warning label: they "illustrate simple associations (not causal relationships) between how much countries spend on health and how effectively health systems function", plot each country as a percentage difference from the OECD

average, and concede that "[a] limitation is that lagged effects are not taken into account – for example, it may take some years before higher health spending translates into longer life expectancy" [6].

What the overview shows, domain by domain, is the empirical backdrop to every earlier chapter. Health status is summarised by four indicators — "life expectancy is a key indicator for the overall health of a population; avoidable mortality focuses on premature deaths that could have been prevented or treated. Diabetes prevalence shows morbidity for a major chronic condition; self-rated health offers a more holistic measure of mental and physical health" [6]. Risk factors display both progress and dispersion: "the share of people smoking daily in 2021 ranged from around 25% or more in Türkiye and France to below 10% in Iceland, Costa Rica, Norway, Mexico, Canada, the United States and Sweden. Daily smoking rates have decreased in most OECD countries over the last decade, taking the average from 20.4% in 2011 to 15.9% in 2021" [6]; alcohol peaks in the Baltics — "Latvia and Lithuania reported the highest levels of consumption in 2021 (above 12 litres of pure alcohol per person per year)... Average consumption has fallen in 23 OECD countries since 2011. Still, harmful drinking is a concern among certain population groups, and nearly one in five adults reported heavy episodic drinking at least once a month" [6]; "on average in 2021, 19.5% of the population were obese, and 54% of the population were overweight or obese"; and air pollution's toll "ranged from over 70 per 100 000 people in Poland and Hungary to less than 7 per 100 000 people in Iceland, New Zealand and Sweden in 2019", against projections that ambient pollution "may cause 6-9 million premature deaths a year worldwide by 2060" [6]. Quality of care shows the widest operational spreads in the whole dashboard set: against OECD averages of 13.1 defined daily doses of antibiotics per 1,000 people, 463 avoidable hospital admissions per 100,000, 55.1 per cent mammography coverage and 30-day mortality of 6.8 (heart attack) and 7.8 (stroke) per 100 admissions, the extremes run from 7.2–7.6 antibiotic doses in Austria and the Netherlands to 21.7 in Greece, from 195 avoidable admissions per 100,000 in Mexico to 827 in Türkiye, from 83.0 per cent screening coverage in Denmark to about 20 per cent in Türkiye and Mexico, and from 1.7 heart-attack deaths per 100 admissions in Iceland to 15.9 in Latvia and 23.7 in Mexico [6].

The dashboard lesson mirrors the projection lesson. Both instruments are honest about their construction — standard-deviation bands and outlier rules here, scenario fans there — and both are routinely read with more confidence than their construction warrants: a green cell is a position within one standard deviation of a moving average in a single year, not a certificate of system quality, just as a baseline path is conditional arithmetic, not destiny. Used as designed — to locate a country, spot a divergence, and ask why — they are the best comparative instruments the field has. The report closes by drawing the findings together into its key conclusions and policy recommendations.



Methodology, interpretation and use

Exhibit. How to interpret quadrant charts that plot health expenditure against health outcome variables, classifying countries by lower or higher expenditure combined with higher or lower life expectancy or avoidable mortality. Source: S0026.

Key findings and policy recommendations

Sixty-one sections reduce to a handful of findings, and they point the same way.

Finding 1: The demography is certain; its fiscal meaning is chosen. Population ageing is locked in by people already alive: 1.6 billion people aged 65+ by 2050, the OECD's old-age dependency ratio heading toward 59 per cent in the EU projection. But the economic dependency ratio — non-workers per worker — rises far less where participation rises, and the sensitivity suite's largest single policy shock is not a demographic variable at all but a ten-point rise in older-worker employment. The gap between the two ratios is the policy space, and it is wide.

Finding 2: Healthspan is the master variable. Every fiscal path in the methods chapter hinges on how much of the longevity gain is lived in good health: the baseline splits the difference (half), the no-healthy-ageing scenario spends far more, the dynamic-equilibrium scenario far less. The same fork governs individuals: morbidity's trajectory decides whether longer lives are working, spending years or care-consuming years. Prevention — vaccination, risk-factor control, early-life investment with a measured 13 per cent annual return by age 65 — is therefore not a health policy adjacent to the ageing agenda; it is the ageing agenda.

Finding 3: The labour market, not the pension system, is where ageing is won or lost. The hazardous-work analysis states the general principle in its sharpest form: arduousness "raise[s] first and foremost an issue for labour market policies. That is, the policy priority should be, when possible, to improve working conditions for such jobs" through "health and safety regulations to limit exposure to risky factors" and incentives for social partners, because "[s]pecial pensions for hazardous or arduous work cannot be in the long term an antidote for strong limitations in labour market policies" [13]. The verdict on the pension-led alternative is blunt: "[t]he permanent withdrawal from the labour market in countries where special pension schemes exist, sometimes at very early ages, is an inefficient solution to address the decline in the ability to perform some specific tasks at older ages. Indeed, people in this case can most often successfully perform other jobs... Not being able to continue working in one specific job should not be dealt with by early retirement" [13]. What should replace it is capability: "[r]eskilling and upskilling aim to enable workers to remain productive and adapt to new job tasks, boosting employment prospects... A professional-training and guidance framework has therefore to be put in place by governments and social partners to allow workers in hazardous or arduous jobs to make the needed extra effort into acquiring new skills, such that they can prolong their careers in different jobs" [13] — with working models to copy, from Finland's free, employment-service-purchased training system to Austria's basic-competences programme (about 50,000 participants in 2012–17) and the Netherlands' jobsearch training and vouchers for over-50s [13]. The systemic risks of doing the opposite are enumerated: overly broad early-retirement provisions "provide obvious disincentives to continue working for people who can actually work", "[o]ccupation-specific pension provisions can also reduce job mobility if changing jobs results in losing some pension entitlements", and publicly financed special provisions "lower the incentives by firms to reduce the incidence of hazardous or arduous work" [13]. Hence the standing OECD guidance, from the 2015 Council Recommendations on Ageing and Employment Policies: strengthen "incentives for workers to build up longer careers and continue working at older ages through, among others, restricting the use of publicly funded early retirement schemes and enhancing participation in training by workers throughout their working lives", while "strengthening workplace safety and physical and mental health as well as reducing the incidence of hazardous or arduous work" [13]. Even risk communication is a policy instrument: informing workers of hazards "helps workers weigh different job opportunities and ask to be compensated for the risks to be taken, which in turn contributes to internalising the negative impact of those jobs, mainly by raising their cost" [13]. And where harm has already materialised, the boundary is honest: disability benefits "do not compensate for the health impacts that may become fully visible only after having retired", and causal links from prolonged exposure "are often difficult to prove" [13] — a gap prevention must close because compensation cannot.

Finding 4: Floors, not averages, decide old-age poverty. Replacement rates average about 51 per cent, yet old-age poverty stands at 14.2 per cent, women's at 16.6 per cent, and the inflation episode showed targeted and basic pensions losing a fifth of their real value in laggard-indexation countries while wage- and price-protected floors held. Indexation rules, eligibility years (34 on average for a full contribution-based basic pension) and coverage of non-standard workers are poverty policy in technical dress.

Finding 5: Old-age inequality is manufactured across the life course. Lifetime earnings divide by education mainly through employment; care penalties, broken careers (−22 per cent of pension on average) and education gaps in life expectancy at 65 (widening in the Norwegian series) are all set decades before retirement. Pensions can only partially correct them — the prevent/mitigate/cope sequence, in that order, is the finding of the inequality chapter and of the whole report.

Recommendations. (1) Invest early, across the whole life course — childhood programmes, education, mid-life health and training — because every later intervention is more expensive and less complete. (2) Make participation

the centrepiece: link retirement ages to life expectancy with equity brakes for short-lived occupations, close the older-worker training gap, enforce age-discrimination law, and treat employer retention practice as infrastructure. (3) Protect the floor first in pension reform: honest indexation, coverage for the self-employed and informal workers, careful decumulation defaults that guarantee essentials without locking in harm. (4) Build care as a system — workforce, pay, coverage and data (the HC.R.1 proxy gap is a policy failure in itself) — before demography prices the informal default out of existence. (5) Design environments for the lives people will actually live: age-friendly cities, transport that survives driving cessation, housing wealth products with public trust anchors. (6) Measure honestly: publish scenario fans and dashboard bands with their construction visible, and judge systems by their floors and their gaps, not their averages.

The hundred-year life is not a problem to be solved but a dividend to be collected — in health, work, income and care — by societies willing to rebuild institutions designed for shorter lives. The evidence assembled here says the dividend is real, unevenly shared by default, and still, for the cohorts now young, almost entirely a matter of choice.

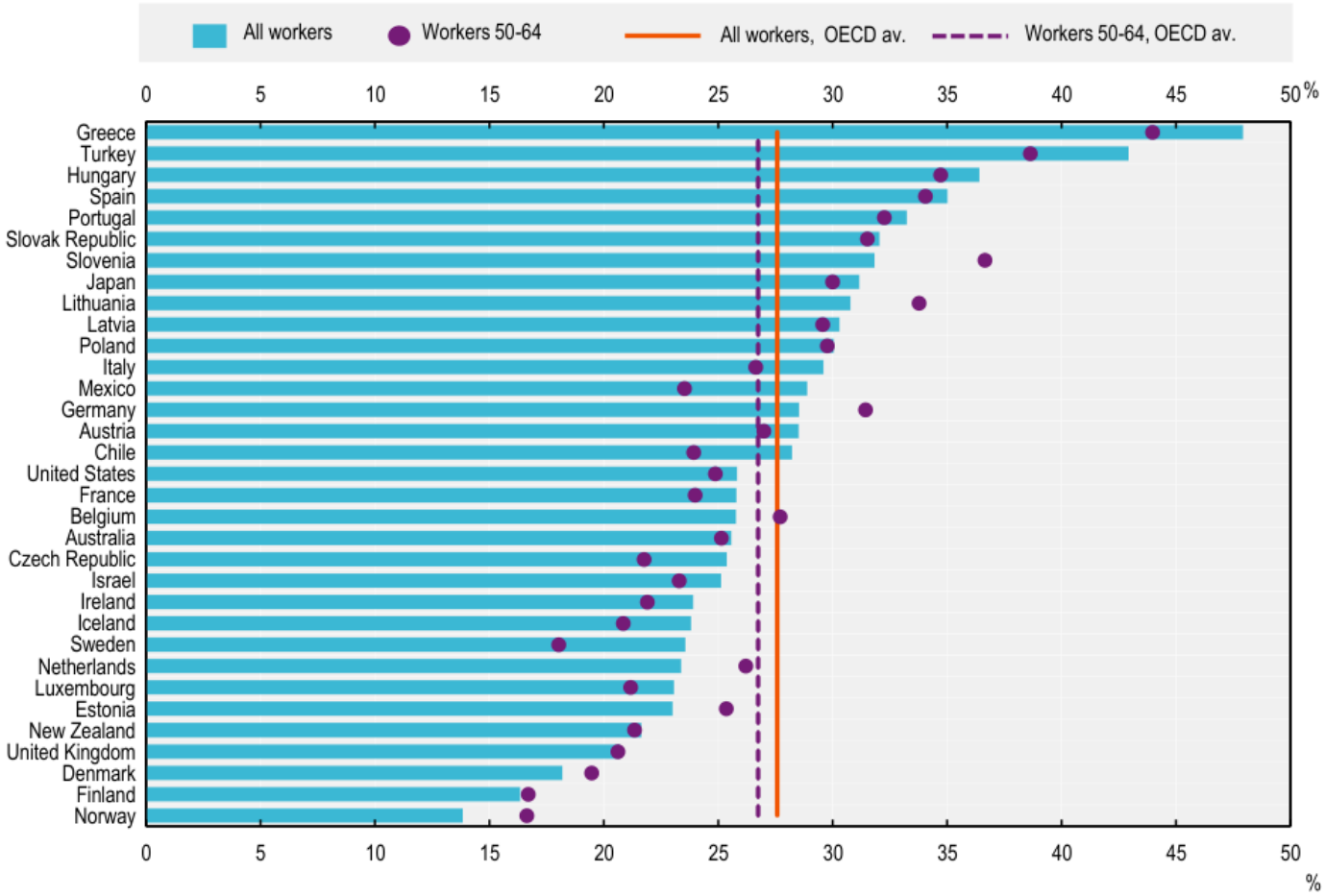


Exhibit. Job strain — high job demands relative to job resources — in OECD countries in 2015, comparing all workers with workers aged 50–64 against the OECD averages for both groups. Source: S0019.

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